Faculty of Medicine, General Sir John Kotelawala Defence University

Application for Ethics Review (Part I) - Basic Information

For official use		Date Received:	
Application No:		Date Informed:	
REC Meeting Date:		Decision:	
Reviewer I:		Decision:	
Reviewer II:		Decision:	
Reviewer (ICF):			

1. Title of Project

Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

2. Investigators & Supervisors

Applications from investigators based overseas will only be considered if the project is done in collaboration with investigators based in institutions in Sri Lanka who take equal responsibility for the conduct of the study and who will appear as co-authors in any publication arising out of the study.

Title, Name and Designation of Investigators and Supervisors	**Role**
KH Viraj Sameera, PhD Candidate, Department of Computer Science, Faculty of Applied Sciences, University of Sri Jayewardenepura (USJ)	Principal Investigator
Dr. WMKS Ilmini, Department of Computer Science, Faculty of Applied Sciences, University of Sri Jayewardenepura	Principal Supervisor
Dr. WGCW Kumara, Department of Computer Engineering, Faculty of Engineering, University of Sri Jayewardenepura	Co-Supervisor
Dr. AM Deepal K Attanayake, Consultant Neurosurgeon, Neurosurgery Wards 9, 10 & 12, National Hospital of Sri Lanka (NHSL), Colombo 10	Co-Supervisor (Clinical) / Clinical Investigator

**Please note that signed and dated short curriculum vitae of all investigators and supervisors must be attached to the application.*

3. Contact Details of the Principal Investigator

Address:	39/3, Hills Street, Dehiwala
Telephone numbers:	0775816789
Fax number:	-
Email address:	virajofficialinfo@gmail.com

4. Funding

Name and Address of Funding Source(s)	Amount
Self-funded (PhD candidate). No external sponsor or commercial funding.	N/A

5. Proposed starting and ending dates: *‡

Start Date: 01.04.2027 End Date: 30.09.2029

**From initial recruitment of participants until completion of all data collection.*

‡Retrospective approval will not be given for projects already started or completed.

No part of this study involving patient data or human participants has started. Data collection will begin only after approval from this committee and from the ethics review committee of each participating hospital.

6. Has ethics approval for this study been requested earlier from KDU/REC or another similar committee?

☐ Yes ☒ No

If yes, give details (names of committees and outcome of review)

Not previously requested. In line with the study protocol, approval will also be obtained from the Ethics Review Committee of each participating hospital (including the National Hospital of Sri Lanka) before any data are accessed at that site, and copies of those approvals will be forwarded to this committee.

Please note that for studies sponsored by foreign funding agencies or sponsors ethics review and approval is required from the country of the funding agency or the sponsor.

7. Scientific Review

Has this research proposal been subjected to scientific review by any other committee?

☒ Yes ☐ No

If yes, give details (names of committees and outcome of review)

Name of the committee: Faculty of Graduate Studies (FGS), University of Sri Jayewardenepura. The full PhD research proposal was submitted to the FGS on 01.10.2026 as part of PhD registration, endorsed by the supervisory panel named above. Outcome of review: awaited; the FGS decision will be forwarded to this committee as soon as it is received.

8. Clinical Trials

8.1 What phase clinical trial is being conducted?

☐ Phase I ☐ Phase II ☐ Phase III ☐ Phase IV (Post Marketing) ☒ Other

If OTHER specify: Not a clinical trial. This is a multi-centre observational study that uses routinely collected, de-identified brain-tumour data (MRI scans, digitised biopsy slides, clinical, pathology and operative reports and follow-up records) to develop and evaluate an AI decision-support framework, together with a clinician evaluation (reader study,

questionnaires and interviews). No intervention is given to patients, no additional tests, samples or visits are required, and AI outputs are not used to manage any patient during the study.

8.2 Is it a multi-centre trial?

☒ Yes ☐ No (multi-centre observational study, not a trial)

If yes, list the other trial sites

Sites: Neurosurgery Units (Wards 9, 10 & 12), National Hospital of Sri Lanka, Colombo; with the radiology and histopathology departments that hold the corresponding MRI scans and biopsy slides. Other tertiary neurosurgical units in Sri Lanka may join; each will take part only after its own ethics approval and written institutional permission. At the time of submission the confirmed site is the National Hospital of Sri Lanka; any additional site will be notified to this committee as an amendment before data collection begins there.

Please attach ethics approval from the sponsoring country or country of the overseas principal investigator (if any). Not applicable: no foreign sponsor and no overseas investigator.

8.3 Is the clinical trial registered with a clinical trials registry?

☐ Yes ☒ No

If yes, give details (name of register and registration number)

Not applicable (not a clinical trial).

8.4 Data Safety Monitoring Board (only if available)

Name and Designation of Members	Role
Not applicable	Not applicable

Please attach the curriculum vitae of all members of the DSMB. Not applicable.

8.5 Details of Indemnity and Insurance coverage for participants, investigators and ethics committee

Not applicable. The study involves no intervention and no physical risk to participants.

8.6 Evidence of GCP training of ALL investigators - Attached / not attached

Not applicable - not attached (not a clinical trial).

9. Conflict of Interest

9.1 Do you believe this project has a Conflict of Interest?

Commercially	No
Financially	No
Intellectually	No
Other (explain)	No

9.2 Does any member of the research team have any affiliation with the provider(s) of funding/support, or a financial interest in the outcome of the research?

☐ Yes ☒ No

If yes, please explain: Not applicable.

9.3 If there is a duality of interest identified above describe the interest and state whether it constitutes a potential conflict of interest.

Not applicable. No duality of interest or potential conflict of interest has been identified among the research team.

9.4 Declaration of Conflict of Interest and Consent by Investigators

We, the undersigned investigators involved in this research project, hereby:

1. Declare whether or not we have any conflict of interest (COI) related to the study (e.g., financial gain, affiliations, intellectual property, or personal relationships).

2. Confirm that we have read and understood the study protocol, agree to participate in this research project as described, and will comply with all applicable national and international ethical guidelines.

Name of Investigator	Affiliation	COI Declared (Yes/No)	If Yes, Nature of COI	Signature	Date
KH Viraj Sameera	PhD Candidate, USJ	No	N/A		
Dr. WMKS Ilmini	Principal Supervisor, USJ	No	N/A		
Dr. WGCW Kumara	Co-Supervisor, USJ	No	N/A		
Dr. AM Deepal K Attanayake	Co-Supervisor (Clinical), NHSL	No	N/A		

Note: All investigators must sign individually. Unsigned forms will be considered incomplete. If a conflict of interest is declared, the REC may request further clarification or documentation.

Faculty of Medicine, General Sir John Kotelawala Defence University

Application for Ethics Review (Part II) - Protocol Checklist

For official use - Application No:	

1. Title of Protocol

Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

2. Name of Principal Investigator

K.H. Viraj Sameera

3. A List of Documents Submitted for Review

Title of Document	Version	Date
Application for Ethics Review (Part I) - Basic Information	V1	09.11.2026
Application for Ethics Review (Part II) - Protocol Checklist	V1	09.11.2026
Application for Ethics Review (Part III) - Document Checklist	V1	09.11.2026
PhD Research Proposal (full study protocol)	V2	01.10.2026
Ethics Considerations Annex (sections A1-A12)	V1	09.11.2026
Appendix 1: Patient Information Sheet and Consent Form (English; Sinhala and Tamil translations)	V1	09.11.2026
Appendix 2: Clinician Information Sheet and Consent Form (English; Sinhala and Tamil translations)	V1	09.11.2026
Appendix 3: Data collection forms - case report form (English); clinician background form, reader-study case record, post-session questionnaire (System Usability Scale, NASA Task Load Index, trust) and interview guide (English, Sinhala and Tamil)	V1	09.11.2026
Signed and dated short CVs of all investigators and supervisors	-	09.11.2026

4. Protocol Checklist

Section numbers refer to the attached Ethics Considerations Annex (A1-A12).

Please indicate the following:	Applicable (Yes/No)	Protocol Section Number
Collaborative partnership		
1. The collaborations you have established with institutions where the study is to be conducted	Yes	A2
2. The collaborations you have established with the community where the study is to be conducted	No	N/A

Please indicate the following:	Applicable (Yes/No)	Protocol Section Number
3. The benefits to institutions, communities, and participants in your research	Yes	A3
Social value		
1. The beneficiaries of your research and the benefit to them	Yes	A3
2. The plan for dissemination of study findings	Yes	A3
Scientific validity		
1. The scientific importance of your study in relation to improving healthcare and/or knowledge on the subject	Yes	A1, A4
2. The justification for a replication study, if your study is a replication study	No	N/A
3. How the sample size was calculated	Yes	A4
Assessment of risks/benefits		
1. The risks to research subjects	Yes	A6
2. Benefits to research subjects	Yes	A6
3. Steps taken to minimize risks	Yes	A6, A8
4. Steps taken to enhance benefits	Yes	A6
5. Justification of the potential benefits against the risks	Yes	A6
6. Support provided to the research participants (medical, psychological and other)	Yes	A6
Consent		
1. The procedure for initial contact of participants	Yes	A7
2. The procedure for obtaining informed consent	Yes	A7
3. The information (written/oral) provided to participants	Yes	A7, Appendix 1 & 2
4. The procedure for ensuring that subjects have understood the information provided	Yes	A7
5. The procedure for obtaining proxy consent	Yes	A7, A11
6. The procedure for withdrawing consent	Yes	A9
7. Incentives/rewards/compensation provided to participants	Yes	A7
8. The procedure for re-consenting if the research protocol changes during the course of research	Yes	A7
Confidentiality		
1. How the data and samples will be obtained	Yes	A8

Please indicate the following:	Applicable (Yes/No)	Protocol Section Number
2. How long data and samples will be kept	Yes	A8
3. Justification for collection of personal identification data	Yes	A8
4. Who will have access to the personal data of the research participants	Yes	A8
5. How the confidentiality of participants will be ensured	Yes	A8
6. The procedure for data and sample storage	Yes	A8
7. The procedure for data and sample disposal	Yes	A8
Rights of the participants		
1. Procedure for subjects to withdraw from the research at any time	Yes	A9
2. Procedure for subjects to ask questions and register complaints	Yes	A9
3. The contact person for research subjects	Yes	A9
4. Provisions for participants to be informed of results	Yes	A9
5. Provision to make the study product available to the study participants after research	No	N/A
Fair participant selection		
1. The justification for the selection of the study population	Yes	A5
2. The inclusion and exclusion criteria	Yes	A5
Responsibilities of the researcher		
1. The provision of medical services to research participants	No	N/A
2. The provisions for continuation of care after the research is completed	No	N/A
3. Declaration of conflicts of interests and how the investigators plan to manage the conflicts	Yes	A10
4. The ethical / legal / social and financial issues relevant to the study	Yes	A10
Vulnerable populations		
1. Justification for conducting the study in this population	Yes	A11
Research funded by foreign agencies / companies (not applicable - self-funded)		
1. Justification for conducting the study in Sri Lanka	No	N/A
2. Relevance of the study to Sri Lanka	No	N/A
3. Post research benefits to Sri Lanka	No	N/A
4. The steps taken to take into account cultural and social customs, practices, and taboos in Sri Lanka	No	N/A

Please indicate the following:	Applicable (Yes/No)	Protocol Section Number
5. The sharing of rights to intellectual property	No	N/A
6. The fate of data and biological samples including whether they will be transferred abroad and what will happen to them after the conclusion of the study	No	N/A
7. How the results of research will be conveyed to relevant authorities in Sri Lanka	No	N/A
8. The agreement between the sponsor/funding agency and the investigator	No	N/A
9. The materials transfer agreement, if biological material is to be transferred abroad	No	N/A
Community based research		
1. The impact and relevance of the research on the community in which it is to be carried out	No	N/A
2. The steps taken to consult with the concerned community during the design of the research	No	N/A
3. The procedure used to obtain community consent	No	N/A
4. The contribution to capacity building of the community	No	N/A
5. The procedure for making available results of research to the community	No	N/A
Clinical trials		
1. Justification for withdrawing any therapy from participants to prepare them for the trial	No	N/A
2. Justification for withholding standard therapy from trial participants (e.g. control group)	No	N/A
3. Justification for providing care which is not the standard of care	No	N/A
4. Procedure for dealing with adverse events	No	N/A
5. Procedure for reporting adverse events	No	N/A
6. Provisions for safety monitoring	No	N/A
7. Provisions/criteria for termination of the trial	No	N/A
8. Provisions for making the trial drug available to participants after the trial if found to be effective	No	N/A

Information Sheet (IFS) / Informed Consent Form (ICF) Check List

Sections in the information sheets (Appendix 1 for patients, Appendix 2 for clinicians) where each item is dealt with:

Item	Section in IFS/ICF
1. Purpose of the study	Section 1
2. Voluntary participation	Section 2
3. Duration, procedures of the study and participant's responsibilities	Section 3
4. Potential benefits	Section 4
5. Risks, hazards and discomforts	Section 5
6. Reimbursements	Section 6
7. Confidentiality	Section 7
8. Termination of study participation	Section 8

Are the investigator's qualifications and experience appropriate to conduct the study?

☒ Yes ☐ No

To be filled by REC Office:	
Processing charge of Rs 1000/- / Rs 2000/-	Receipt no:
Serial No: REC ..... /	Signature REC Administrative Assistant

Faculty of Medicine, General Sir John Kotelawala Defence University

Application for Ethics Review (Part III) - Document Checklist

For official use - Application No:	

Application Checklist

I declare that I have attached the following documents (Please tick the check box and confirm). If any of the following sections are not relevant, please indicate N/A in the given space.

Document	Attached
1. Application Form: Part I [1 copy]	☒
2. Application Form: Part II [1 copy]	☒
3. The complete research protocol including a section on ethics considerations [1 copy] (PhD Research Proposal + Ethics Considerations Annex)	☒
4. Information sheet for research participants (Should be provided in all three languages - Sinhala, Tamil, and English - if the participant is being interviewed or is filling up the form) [1 copy each] - patients and clinicians	☒
5. Consent forms (Should be provided in all three languages: Sinhala, Tamil, and English) [1 copy each] - patients and clinicians	☒
6. Data collection booklets/forms/questionnaires (Should be provided in all three languages - Sinhala, Tamil, and English) [1 copy] - clinician forms 3B-3E in all three languages; case report form 3A in English only, as it is completed by hospital staff from English-language medical records and is never given to patients	☒
7. Clinical Trials Contract (required for clinical trials)	N/A
8. Materials Transfer Agreement (required for all research involving transfer of biological samples abroad)	N/A
9. Indemnity/Insurance coverage (required for clinical trials)	N/A
10. Ethics approval from sponsoring country or country of the overseas investigator (if any)	N/A
11. Brief curriculum vitae of all investigators [1 copy]	☒
12. Curriculum vitae of all DSMB members [1 copy]	N/A
13. The downloaded payment slip for REC processing charges	☒
14. Soft copies of all documents (The documents should be in a compressed folder (zip/rar)) have been submitted via email to ethicsreviewcommittee.fom@kdu.ac.lk	☒

**The above documents should be handed over to Secretary, Research Ethics Committee, Faculty of Medicine, KDU. Deadline for submission is 10th of each month for a new protocol and re-submission.*

I understand that the application for ethics clearance will not be accepted unless all documents are submitted. I declare that I am not seeking approval for a study that has already commenced or has already been completed. I understand that at least two months are required for ethics review and granting ethics clearance.

Signature of Principal Investigator	Date

Ethics Considerations Annex

Supplement to the PhD Research Proposal - Version 1

Study: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

Principal Investigator: KH Viraj Sameera, PhD Candidate, Department of Computer Science, University of Sri Jayewardenepura

This annex sets out the ethical aspects of the study. Section numbers A1-A12 are referenced in the Protocol Checklist (Part II). Section numbers of the form 3.x refer to the attached PhD Research Proposal.

A1. Study summary

Background. Sri Lanka records about 700 new brain and nervous-system tumours and 587 related deaths each year, with a five-year prevalence of about 2,500, but has no registry that links imaging and tissue findings. Artificial intelligence (AI) can analyse MRI scans and biopsy slides, but a clinically safe system must also know when it is not confident and must then refer the decision to a clinician (Proposal sections 1.1-1.2).

Aim. To develop and evaluate a multi-agent AI decision-support framework for the brain-tumour care pathway in which each AI component reports a calibrated confidence, gives a decision-support output only when that confidence is sufficient, and refers all other cases to a clinician, who keeps final authority at every step.

Objectives.

- O1. Design a multi-agent AI architecture for the brain-tumour pathway that passes uncertainty between components and keeps an auditable record of every AI output and clinician validation.
- O2. Develop and evaluate a calibrated, uncertainty-aware escalation mechanism that decides when a case must be referred to a clinician.
- O3. Develop data-efficient models for MRI and biopsy-slide analysis on small Sri Lankan datasets, including privacy-preserving federated training across hospitals without sharing raw patient data.
- O4. Evaluate the human-in-the-loop framework with clinicians: diagnostic performance, appropriateness of escalation, workload, usability, trust and safety, with surgical planning as a proof of concept.

Design. A multi-centre observational study with three components: (a) collection of routinely acquired clinical data (MRI scans, digitised biopsy slides, clinical, pathology and operative reports and follow-up records) into a de-identified research registry, retrospectively and prospectively; (b) computational development and evaluation of AI models on these data and on public datasets; and (c) evaluation by clinicians through a reader study, questionnaires and interviews. The study involves no intervention: no additional tests, scans, biopsies, samples or hospital visits are required, and AI outputs will not be used to diagnose or manage any patient during the study.

A2. Study sites and collaborations

- Academic host: Department of Computer Science, Faculty of Applied Sciences, University of Sri Jayewardenepura, with the Department of Computer Engineering, Faculty of Engineering, USJ.
- Clinical site: Neurosurgery Units (Wards 9, 10 & 12), National Hospital of Sri Lanka, Colombo, under the clinical co-supervisor, together with the radiology and histopathology departments that hold the corresponding MRI scans and biopsy slides.
- Further tertiary neurosurgical units in Sri Lanka may join. Each site will take part only after approval from its own ethics review committee and written permission from the head of the institution, and under a data-sharing agreement.

Where federated learning is used, raw clinical, imaging and tissue data stay at each hospital and only model parameters are exchanged (Proposal section 3.10).

A3. Social value, benefits and dissemination

Beneficiaries and benefits.

- Participating units: the first imaging- and tissue-linked brain-tumour registry in Sri Lanka, supporting follow-up, clinical audit and research, which remains with the units after the study.
- Future patients: AI support that could make diagnosis faster and more consistent, while referring uncertain cases to specialists.
- Clinicians: reduced review workload, measured directly in the study.
- Research community: an open, de-identified Sri Lankan brain-tumour benchmark (released only with the approvals described in A8), addressing the near-absence of South Asian data in public datasets; methods transferable to other low-resource settings.

Individual participants are not expected to benefit directly.

Dissemination. Findings will be reported in the PhD thesis, in at least four peer-reviewed publications and at conferences, and in a summary report to each participating hospital and its ethics committee, with a presentation to clinical staff. Results will be reported only in aggregate, with no information that could identify a patient or clinician. Negative results will be reported in the same detail as positive ones.

A4. Scientific validity and sample size

Scientific importance. Existing medical AI is mostly evaluated on accuracy alone; it does not specify when a system should act and when it must defer to a clinician, and it is trained on large foreign datasets that do not represent Sri Lankan patients or scanners. This study makes clinician oversight a measurable part of the system and evaluates it on local data (Proposal sections 1.5 and 2.12). It is not a replication study.

Sample size - patients. Patients are recruited by consecutive sampling of all eligible patients at participating units during the study period. Because the number is limited by national incidence, a formal power calculation for the patient cohort is not used; the expected cohort is several hundred patients, and fewer will have complete linked imaging, tissue and follow-up data

(Proposal section 3.5). The study is designed for this: models are pre-trained on public datasets and unlabelled local scans, and results are reported with patient-level bootstrap confidence intervals so the precision of every estimate is visible.

Sample size - clinicians. Clinicians are recruited by purposive sampling. The reader study uses a multi-reader, multi-case design in which each clinician reviews a common set of cases with and without AI support, with counterbalanced order and a washout period, which uses scarce specialist time efficiently. The final number of readers and cases will be set with a statistician before the reader study begins and notified to the committee. The indicative target is at least six clinician readers.

A5. Study populations and selection

Population 1: Patients (data only)

Inclusion: patients of any age with a suspected or confirmed brain or nervous-system tumour, managed at a participating unit during the study period, with at least one clinically relevant MRI examination in usable DICOM format and enough clinical information to establish the reference diagnosis.

Exclusion: records where imaging is unusable and cannot be recovered; where the reference diagnosis cannot be established; where de-identification cannot be completed; where data-governance requirements are not met; or where the patient (or proxy) has declined or withdrawn.

Patients are not excluded only because tissue, laboratory-marker or follow-up data are missing; they contribute to the parts of the study for which data exist.

Population 2: Clinicians (active participants)

Inclusion: consultant neurosurgeons, radiologists, histopathologists and experienced senior trainees at participating institutions who are involved in the diagnosis or management of brain tumours and who give written informed consent.

Exclusion: members of the research team (excluded from the reader study and usability evaluation).

Population 3: Public datasets

Public de-identified datasets such as BraTS are used for pre-training and benchmarking under their published licences. They are not used as the sole basis for any clinical claim about Sri Lankan patients.

Justification. Brain tumours are diagnosed and treated at tertiary neurosurgical units, so these are the appropriate sites. All eligible patients are included consecutively to avoid selection bias and so that the AI is evaluated on the real mix of cases, including incomplete records.

A6. Risks, benefits and risk minimisation

Risks to patients. There is no physical risk: no procedure, test, sample or visit is added, and treatment is decided by the treating team exactly as usual. The main risk is loss of confidentiality

or re-identification from medical images; the safeguards are described in A8. The AI system is a research decision-support tool only. It will not issue or implement any clinical decision, its outputs are not used in patient care during the study, and the surgical-planning component is an offline proof of concept that never performs or controls any part of an operation.

Risks to clinicians. The main burdens are time and possible discomfort at having their reading performance measured. Sessions are scheduled at the clinician's convenience and can be paused or stopped at any time. Individual clinicians' performance is never reported by name, is never shared with employers or colleagues, and is analysed only in aggregate.

Benefits. There is no direct benefit to individual participants. Clinicians gain early experience with AI decision support. Wider benefits are described in A3. To increase benefit, the registry remains with the participating units and summary results are fed back to them.

Justification. The study is minimal risk, adds nothing to patient care, and has substantial potential benefit for brain-tumour services in Sri Lanka.

Support. Because there is no intervention, no medical or psychological support is expected to be required. Participants may contact the Principal Investigator or the clinical co-supervisor with any concern (A9). If a participating clinician raises a patient-safety concern during the study, it will be referred immediately to the clinical co-supervisor and the treating team.

A7. Consent

Retrospective patient records - request for waiver of consent

For records created during routine care before the study starts, a waiver of individual consent is requested because: (i) the data already exist and no contact with the patient is needed; (ii) data are de-identified at the hospital before research use; (iii) tracing patients is impracticable, as many will have been discharged, lost to follow-up or have died, and approaching bereaved families could cause distress; (iv) the study is minimal risk; and (v) use will take place only with written permission from the head of each institution and approval from its ethics committee.

Prospective patients - written informed consent

Initial contact. Patients presenting during the study period are approached by a member of their treating team, not by the Principal Investigator, at a time that does not interfere with their care.

Information and consent. The patient receives the Patient Information Sheet (Appendix 1) in Sinhala, Tamil or English as preferred, has time to read it and ask questions, and gives written consent to a trained member of the clinical team. Declining has no effect on care.

Understanding. Staff explain the study orally in the patient's language and ask the patient to describe in their own words what taking part means before consent is signed.

Proxy consent. For patients under 18 years, a parent or legal guardian gives consent, and assent is sought from children old enough to understand. For adults who temporarily or permanently lack capacity, which is common with brain tumours, consent is obtained from the legally authorised representative or next of kin. If the patient later regains capacity, their own consent is sought; if they decline, their data are withdrawn.

Clinicians - written informed consent

Clinicians are invited by letter or email from the Principal Investigator with the Clinician Information Sheet (Appendix 2), are given at least 48 hours to decide, have the opportunity to ask questions, and sign the consent form before taking part. The information sheet, consent form, questionnaires and interview are available in Sinhala, Tamil or English, as the clinician prefers.

Incentives. No payment, gift or other incentive is offered to any participant, and taking part costs participants nothing.

Re-consent. If the protocol changes in a way that affects participants, an amendment will be submitted to the committee, and participants will be given the revised information sheet and asked to consent again.

A8. Confidentiality and data management

How data are obtained. MRI scans are exported from the hospital imaging system by authorised hospital staff. Existing archived biopsy slides are digitised within the hospital pathology department under its own custody procedures; glass slides are not removed from the hospital and no new tissue is taken. Clinical information is abstracted from hospital records into a case report form. No biological samples are collected.

De-identification at source. Before any data leave the hospital system: direct identifiers (name, NIC number, hospital/BHT and clinic numbers, address, telephone) are removed; DICOM headers are scrubbed; structural MRI scans are defaced; and slide labels and identifying metadata are removed. Each patient is given a study code.

Justification for personal identification data. A linkage key connecting the study code to the hospital number is kept only at the hospital, by the clinical co-supervisor or designated hospital staff. It is needed only to add follow-up information and to honour withdrawal requests. No personal identifiers are held in the research dataset.

Access. The research team and trained annotators, who sign a confidentiality agreement, access de-identified data only. Only the clinical co-supervisor and designated hospital staff can access the linkage key.

Storage. Data are stored on encrypted, password-protected servers within the participating institution or the University of Sri Jayewardenepura, with role-based access control and audit logging of all access. Data are not stored on personal devices or on external cloud services. In federated experiments raw data never leave the hospital and only model parameters are shared. Clinician consent forms are kept in a locked cabinet in the Department of Computer Science, USJ, separately from questionnaire data, which are stored under a code.

Retention and disposal. Research data are kept for five years after the end of the study, or longer if required by the participating institution, and then securely deleted; paper records are shredded. The linkage key is destroyed at the end of follow-up data collection.

Open benchmark. Any public release of a de-identified dataset will happen only after a further de-identification review and with the explicit approval of the participating hospitals and their ethics committees. No data or biological material will be transferred abroad.

Legal compliance. Data will be handled in accordance with the Personal Data Protection Act, No. 9 of 2022, of Sri Lanka, and the data-governance requirements of each participating institution.

A9. Rights of participants

Withdrawal. Participants (or their proxies) may withdraw at any time without giving a reason, and withdrawal does not affect their medical care or employment. Patients may withdraw through their treating team or by contacting the Principal Investigator; clinicians by contacting the Principal Investigator. On withdrawal, the participant's data are removed from the registry and from further analysis. Data already included in trained models or in published aggregate results cannot be removed; this is explained in the information sheets.

Questions and complaints. Participants may contact the Principal Investigator or the clinical co-supervisor with questions, and the Secretary of the Research Ethics Committee with complaints or concerns about their rights.

Principal Investigator	KH Viraj Sameera - 0775816789 - virajofficialinfo@gmail.com
Clinical co-supervisor	Dr. AM Deepal K Attanayake, Consultant Neurosurgeon, NHSL - +94 71 421 9049 - drdeepal@yahoo.com
Principal Supervisor	Dr. WMKS Ilmini, Department of Computer Science, USJ - wmksilmini@sjp.ac.lk
Ethics Committee	Secretary, Research Ethics Committee, Faculty of Medicine, KDU - ethicsreviewcommittee.fom@kdu.ac.lk

Results. A plain-language summary of findings will be provided to participating units and is available to any participant on request. Individual AI results are not returned to patients, because the system is a research tool and is not used for clinical decisions.

Study product. The AI framework is a research prototype, not a certified medical device, and will not be made available to participants for clinical use after the study. The registry platform remains with the participating units.

A10. Responsibilities of the researcher and ethical, legal and social issues

Medical care. The researchers provide no medical services within the study. Patients continue to receive standard care from their treating teams during and after the study, unaffected by participation.

Conflicts of interest. No investigator has a commercial, financial or intellectual conflict of interest (Part I, section 9). Any conflict that arises during the study will be declared to the committee immediately.

AI safety. The system is decision support only; every AI output requires validation by a qualified clinician; uncertain cases are referred to clinicians rather than acted on; and every AI output and clinician decision is recorded in an audit log.

Fairness and bias. Model performance will be reported by subgroup (tumour type, site, age group and sex) so that differences in performance are visible and are not hidden in an overall figure.

Financial. Participants incur no cost. The study is self-funded.

A11. Vulnerable populations

Two groups that may be considered vulnerable are included: children with brain tumours, and adults whose decision-making capacity is impaired by their tumour.

Justification. Brain tumours occur in children, and impaired capacity is common in exactly the most serious cases. Excluding these groups would bias the data towards milder cases and would deny future patients in these groups the benefit of a system validated on people like them.

Protections. The study adds no procedure and carries minimal risk; proxy consent and age-appropriate assent are used (A7); and all data are de-identified at source (A8).

A12. Intellectual property and data ownership

Clinical data remain the property of the participating hospitals. Intellectual property in software and models developed in the study is governed by the intellectual-property policy of the University of Sri Jayewardenepura. Public datasets are used under their published licences. There is no foreign sponsor, and no data or biological material will be transferred outside Sri Lanka.

Appendix 1: Patient Information Sheet

Study: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

You are invited to take part in a research study. Please read this information carefully and ask us about anything that is not clear before you decide. If you are reading this on behalf of a child or a family member who cannot decide for themselves, "you" means the person you are responsible for.

1. Purpose of the study

We are developing a computer system that uses artificial intelligence (AI) to help doctors read brain MRI scans and biopsy slides of brain tumours. The system is designed to say when it is not sure, so that a doctor always checks those cases. This study uses information from patients' existing scans and records to build and test the system. It is part of a PhD at the University of Sri Jayewardenepura, carried out with the Neurosurgery Units of the National Hospital of Sri Lanka.

2. Voluntary participation

Taking part is entirely your choice. If you decide not to take part, or later change your mind, your treatment and care will not change in any way.

3. Duration, procedures and your responsibilities

If you agree, copies of the MRI scans, biopsy slides and medical records made during your normal care, and later follow-up information, will be used for the study for about the next two to three years. You do not need to have any extra test, scan, blood sample, biopsy or hospital visit, and you have nothing else to do. Your doctors will decide your treatment exactly as usual; the computer system is not used to make decisions about you.

4. Potential benefits

There is no direct benefit to you. The study may help future patients in Sri Lanka by supporting faster and more consistent diagnosis.

5. Risks, hazards and discomforts

There is no physical risk. The main risk is that your information could be seen by someone who should not see it. We protect against this as described in Section 7.

6. Reimbursements

You will not be paid for taking part, and it will not cost you anything.

7. Confidentiality

Your name, NIC number, hospital number, address and telephone number are removed before your information is used, and your scans are processed so that your face cannot be recognised. Your information is given a code. Only your hospital team keeps the list linking the code to you. Data are kept on secure, password-protected computers within the hospital or university. Results are published only as group totals, and you will never be named.

8. Termination of participation

You may stop taking part at any time without giving a reason, by telling your doctor or the researcher. Your information will then be removed from the study. Information that has already been used in published group results or in a trained computer model cannot be removed. The researchers may also stop the study early, for example if the ethics committee asks.

Contacts

Principal Investigator: KH Viraj Sameera, PhD Candidate, University of Sri Jayewardenepura - 0775816789 - virajofficialinfo@gmail.com

Clinical co-supervisor: Dr. AM Deepal K Attanayake, Consultant Neurosurgeon, NHSL - +94 71 421 9049

For concerns about your rights as a participant: Secretary, Research Ethics Committee, Faculty of Medicine, KDU - ethicsreviewcommittee.fom@kdu.ac.lk

Appendix 1: Patient Consent Form

Study: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

Please initial each box:

Statement	Initials
I have read (or had read to me) the Patient Information Sheet, Version 1, and have had the chance to ask questions.	
I understand that taking part is voluntary and that I can withdraw at any time without my medical care being affected.	
I agree that copies of my MRI scans, biopsy slides, medical records and follow-up information may be used for this study after my personal details are removed.	
I understand that my information will be kept confidential and that I will not be named in any report or publication.	
I agree to take part in this study.	

Name	Signature	Date
Patient		
Parent / guardian / legal representative (if applicable) - relationship:		
Person taking consent		

Assent (children old enough to understand): "The doctor explained this study to me and I am happy for my scans to be used." Child's name / signature: ..... Date: .....

Appendix 2: Clinician Information Sheet

Study: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

You are invited to take part in a research study. Please read this information carefully and ask us about anything that is not clear before you decide. You are being invited because you are involved in the diagnosis or management of patients with brain tumours.

1. Purpose of the study

We are developing and evaluating an AI decision-support framework for brain-tumour diagnosis and surgical-planning support that reports its own confidence and refers uncertain cases to clinicians. We want to measure whether it helps clinicians, how it affects workload, and whether clinicians find it usable and trustworthy.

2. Voluntary participation

Taking part is entirely voluntary. Your decision will not affect your employment, training or relationship with any participating institution.

3. Duration, procedures and your responsibilities

You may be asked to: (a) review a set of de-identified cases with and without AI support, in several sessions spread over time (estimated total 4-6 hours); (b) complete short questionnaires on usability, workload and trust (about 15 minutes each time); and (c) take part in an optional interview of about 30-45 minutes, which will be audio-recorded only with your separate permission. All sessions are arranged at times convenient to you.

4. Potential benefits

There is no direct benefit to you, although you will gain early experience with AI decision-support tools. The findings may improve brain-tumour services in Sri Lanka.

5. Risks, hazards and discomforts

The main burden is your time. You may feel uncomfortable having your reading performance measured; your individual results will never be reported by name or shared with your employer or colleagues.

6. Reimbursements

No payment is offered, and taking part costs you nothing.

7. Confidentiality

Your responses are stored under a code, separately from your consent form, on secure systems at the University of Sri Jayewardenepura. Results are reported only in aggregate. Interview quotations, if used, will be anonymised. Audio recordings are deleted after transcription.

8. Termination of participation

You may withdraw at any time without giving a reason by contacting the Principal Investigator; your data will then be excluded from further analysis. Data already included in published aggregate results cannot be removed.

Contacts

Principal Investigator: KH Viraj Sameera, PhD Candidate, University of Sri Jayewardenepura - 0775816789 - virajofficialinfo@gmail.com

Clinical co-supervisor: Dr. AM Deepal K Attanayake, Consultant Neurosurgeon, NHSL - +94 71 421 9049

For concerns about your rights as a participant: Secretary, Research Ethics Committee, Faculty of Medicine, KDU - ethicsreviewcommittee.fom@kdu.ac.lk

Appendix 2: Clinician Consent Form

Study: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

Please initial each box:

Statement	Initials
I have read the Clinician Information Sheet, Version 1, and have had the chance to ask questions.	
I understand that taking part is voluntary and that I can withdraw at any time without giving a reason.	
I understand that my individual performance will not be reported by name or shared with my employer.	
I agree to take part in the case-review sessions and questionnaires.	
(Optional) I agree to take part in an interview and to it being audio-recorded.	

Name	Signature	Date
Clinician		
Researcher taking consent		

ඇමුණුම 1: රෝගී තොරතුරු පත්‍රිකාව

1 වන අනුවාදය, *09.11.2026 (සිංහල)*

අධ්‍යයනය: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

(මොළයේ ගෙඩි හඳුනා ගැනීමට සහ ශල්‍යකර්ම සැලසුම් කිරීමට වෛද්‍යවරුන්ට උදව් වන කෘත්‍රිම බුද්ධි පද්ධතියක් පිළිබඳ පර්යේෂණය)

පර්යේෂණ අධ්‍යයනයකට සහභාගී වන ලෙස ඔබට ආරාධනා කරනු ලැබේ. තීරණයක් ගැනීමට පෙර කරුණාකර මෙම තොරතුරු හොඳින් කියවා, පැහැදිලි නැති ඕනෑම දෙයක් ගැන අපෙන් විමසන්න. ඔබ මෙය කියවන්නේ දරුවෙකු වෙනුවෙන් හෝ තමන්ටම තීරණය කළ නොහැකි පවුලේ සාමාජිකයෙකු වෙනුවෙන් නම්, "ඔබ" යන්නෙන් අදහස් වන්නේ ඔබ භාරකාරත්වය දරන පුද්ගලයාය.

1. අධ්‍යයනයේ අරමුණ

මොළයේ MRI ස්කෑන් පරීක්ෂණ සහ මොළයේ ගෙඩිවල බයොප්සි (පටක සාම්පල) කදා කියවීමට වෛද්‍යවරුන්ට උදව් කිරීම සඳහා කෘත්‍රිම බුද්ධිය (AI) භාවිත කරන පරිගණක පද්ධතියක් අපි නිර්මාණය කරමින් සිටිමු. මෙම පද්ධතිය තමන්ට විශ්වාස නැති අවස්ථා පෙන්වා දෙන ලෙස සකස් කර ඇති අතර, එවැනි අවස්ථා සෑම විටම වෛද්‍යවරයෙකු විසින් පරීක්ෂා කරනු ලැබේ. මෙම පද්ධතිය නිර්මාණය කිරීමට සහ පරීක්ෂා කිරීමට, රෝගීන්ගේ දැනටමත් පවතින ස්කෑන් සහ වාර්තා මෙම අධ්‍යයනයේදී භාවිත කෙරේ. මෙය ශ්‍රී ලංකා ජාතික රෝහලේ ස්නායු ශල්‍ය ඒකක සමඟ එක්ව, ශ්‍රී ජයවර්ධනපුර විශ්වවිද්‍යාලයේ සිදු කෙරෙන PhD උපාධියක කොටසකි.

2. ස්වේච්ඡා සහභාගීත්වය

සහභාගී වීම සම්පූර්ණයෙන්ම ඔබේ කැමැත්ත මතය. ඔබ සහභාගී නොවීමට තීරණය කළත්, පසුව ඔබේ අදහස වෙනස් කළත්, ඔබට ලැබෙන ප්‍රතිකාර හා සත්කාර කිසිසේත් වෙනස් නොවේ.

3. කාලසීමාව, ක්‍රියාපටිපාටි සහ ඔබේ වගකීම්

ඔබ එකඟ වන්නේ නම්, ඔබේ සාමාන්‍ය ප්‍රතිකාර අතරතුර ලබාගත් MRI ස්කෑන්, බයොප්සි කදා සහ වෛද්‍ය වාර්තාවල පිටපත් ද, පසුව ලැබෙන පසු විපරම් තොරතුරු ද, ඉදිරි වසර දෙක තුනක පමණ කාලයක් අධ්‍යයනය සඳහා භාවිත කෙරේ. ඔබට අමතර පරීක්ෂණයක්, ස්කෑන් එකක්, රුධිර සාම්පලයක්, බයොප්සියක් හෝ රෝහලට පැමිණීමක් අවශ්‍ය නොවන අතර, ඔබට වෙනත් කිසිවක් කිරීමට නැත. ඔබේ වෛද්‍යවරුන් සුපුරුදු පරිදිම ඔබේ ප්‍රතිකාර තීරණය කරනු ඇත; ඔබ පිළිබඳ තීරණ ගැනීමට මෙම පරිගණක පද්ධතිය භාවිත නොකෙරේ.

4. ඇති විය හැකි ප්‍රතිලාභ

ඔබට සෘජු ප්‍රතිලාභයක් නැත. වේගවත් හා වඩාත් ස්ථාවර රෝග විනිශ්චයකට සහාය වීමෙන්, මෙම අධ්‍යයනය ශ්‍රී ලංකාවේ අනාගත රෝගීන්ට උපකාරී විය හැකිය.

5. අවදානම්, අනතුරු සහ අපහසුතා

ශාරීරික අවදානමක් නැත. ප්‍රධාන අවදානම වන්නේ ඔබේ තොරතුරු නොබැලිය යුතු අයෙකු දැකීමේ හැකියාවයි. 7 වන කොටසේ විස්තර කර ඇති පරිදි අපි මෙයින් ආරක්ෂා කරන්නෙමු.

6. ප්‍රතිපූරණ

සහභාගී වීම සඳහා ඔබට මුදල් නොගෙවන අතර, ඔබට එයින් කිසිදු වියදමක් ද නොවේ.

7. රහස්‍යභාවය

ඔබේ තොරතුරු භාවිත කිරීමට පෙර ඔබේ නම, ජාතික හැඳුනුම්පත් අංකය, රෝහල් අංකය, ලිපිනය සහ දුරකථන අංකය ඉවත් කරනු ලබන අතර, ඔබේ මුහුණ හඳුනාගත නොහැකි වන පරිදි ඔබේ ඡායාරූප සකසනු ලැබේ. ඔබේ තොරතුරුවලට කේත අංකයක් ලබා දේ. එම කේතය ඔබට සම්බන්ධ කරන ලැයිස්තුව තබා ගන්නේ ඔබේ රෝහල් කණ්ඩායම පමණි. දත්ත රෝහල හෝ විශ්වවිද්‍යාලය තුළ ඇති, මුරපදයකින් ආරක්ෂිත පරිගණකවල තබා ගැනේ. ප්‍රතිඵල ප්‍රකාශයට පත් කරන්නේ සමූහ එකතු ලෙස පමණක් වන අතර, ඔබේ නම කිසි විටෙක සඳහන් නොවේ.

8. සහභාගීත්වය අවසන් කිරීම

ඔබේ වෛද්‍යවරයාට හෝ පර්යේෂකයාට දැනුම් දී, හේතුවක් දැක්වීමකින් තොරව ඕනෑම වේලාවක ඔබට සහභාගීත්වය නතර කළ හැකිය. එවිට ඔබේ තොරතුරු අධ්‍යයනයෙන් ඉවත් කරනු ලැබේ. දැනටමත් ප්‍රකාශිත සමූහ ප්‍රතිඵලවල හෝ පුහුණු කළ පරිගණක ආකෘතියක භාවිත කර ඇති තොරතුරු ඉවත් කළ නොහැක. උදාහරණයක් ලෙස ආචාර ධර්ම කමිටුව ඉල්ලා සිටියහොත්, පර්යේෂකයන්ට ද අධ්‍යයනය කලින් නතර කළ හැකිය.

සම්බන්ධතා

ප්‍රධාන පර්යේෂක: කේ.එච්. විරාජ් සමීර, PhD අපේක්ෂක, ශ්‍රී ජයවර්ධනපුර විශ්වවිද්‍යාලය - 0775816789 - virajofficialinfo@gmail.com

සායනික සම-අධීක්ෂක: වෛද්‍ය ඒ.එම්. දීපාල් කේ. අත්තනායක, විශේෂඥ ස්නායු ශල්‍ය වෛද්‍ය, ශ්‍රී ලංකා ජාතික රෝහල - +94 71 421 9049

සහභාගී වන්නෙකු ලෙස ඔබේ අයිතිවාසිකම් පිළිබඳ ගැටලු සඳහා: ලේකම්, පර්යේෂණ ආචාර ධර්ම කමිටුව, වෛද්‍ය පීඨය, ජෙනරාල් ශ්‍රීමත් ජෝන් කොතලාවල ආරක්ෂක විශ්වවිද්‍යාලය - ethicsreviewcommittee.fom@kdu.ac.lk

ඇමුණුම 1: රෝගී කැමැත්ත පත්‍රය

1 වන අනුවාදය, 09.11.2026 (සිංහල)

අධ්‍යයනය: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

(මොළයේ ගෙඩි හඳුනා ගැනීමට සහ ශල්‍යකර්ම සැලසුම් කිරීමට වෛද්‍යවරුන්ට උදව් වන කෘත්‍රිම බුද්ධි පද්ධතියක් පිළිබඳ පර්යේෂණය)

කරුණාකර සෑම කොටුවකම ඔබේ මුලකුරු යොදන්න:

ප්‍රකාශය	**මුලකුරු**
1. මම රෝගී තොරතුරු පත්‍රිකාව (1 වන අනුවාදය) කියවා ඇත (හෝ මට කියවා දී ඇත), ප්‍රශ්න ඇසීමට අවස්ථාව ලැබී ඇත.	
2. සහභාගී වීම ස්වේච්ඡාවෙන් බවත්, මගේ වෛද්‍ය සත්කාරයට බලපෑමක් නොමැතිව ඕනෑම වේලාවක ඉවත් විය හැකි බවත් මම තේරුම් ගනිමි.	
3. මගේ පෞද්ගලික තොරතුරු ඉවත් කිරීමෙන් පසු, මගේ MRI ස්කෑන්, බයොප්සි කදා, වෛද්‍ය වාර්තා සහ පසු විපරම් තොරතුරුවල පිටපත් මෙම අධ්‍යයනය සඳහා භාවිත කිරීමට මම එකඟ වෙමි.	
4. මගේ තොරතුරු රහසිගතව තබා ගන්නා බවත්, කිසිදු වාර්තාවක හෝ ප්‍රකාශනයක මගේ නම සඳහන් නොවන බවත් මම තේරුම් ගනිමි.	
5. මෙම අධ්‍යයනයට සහභාගී වීමට මම එකඟ වෙමි.	

	නම	**අත්සන**	**දිනය**
රෝගියා			
දෙමාපිය / භාරකරු / නීත්‍යානුකූල නියෝජිත (අදාළ නම්) - සම්බන්ධතාවය:			
කැමැත්ත ලබා ගන්නා පුද්ගලයා			

අනුමැතිය (තේරුම් ගැනීමට තරම් වයස ඇති දරුවන් සඳහා): "වෛද්‍යවරයා මෙම අධ්‍යයනය මට පැහැදිලි කළා. මගේ ස්කෑන් භාවිත කිරීමට මම කැමතියි." දරුවාගේ නම / අත්සන: ..... දිනය: .....

ඇමුණුම 2: සායනික වෛද්‍යවරුන් සඳහා තොරතුරු පත්‍රිකාව

1 වන අනුවාදය, 09.11.2026 (සිංහල)

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පර්යේෂණ අධ්‍යයනයකට සහභාගී වන ලෙස ඔබට ආරාධනා කරනු ලැබේ. තීරණයක් ගැනීමට පෙර කරුණාකර මෙම තොරතුරු හොඳින් කියවා, පැහැදිලි නැති ඕනෑම දෙයක් ගැන අපෙන් විමසන්න. මොළයේ ගෙඩි සහිත රෝගීන්ගේ රෝග විනිශ්චයට හෝ ප්‍රතිකාරවලට ඔබ සම්බන්ධ වන බැවින් ඔබට ආරාධනා කෙරේ.

1. අධ්‍යයනයේ අරමුණ

තමන්ගේ විශ්වාසනීයත්ව මට්ටම වාර්තා කරන සහ අවිනිශ්චිත අවස්ථා සායනික වෛද්‍යවරුන් වෙත යොමු කරන, මොළයේ ගෙඩි රෝග විනිශ්චය හා ශල්‍යකර්ම සැලසුම් කිරීම සඳහා වූ AI තීරණ-සහාය රාමුවක් අපි නිර්මාණය කර ඇගයීමට ලක් කරමින් සිටිමු. එය සායනික වෛද්‍යවරුන්ට උපකාරී වේද, වැඩ බරට බලපාන්නේ කෙසේද, සහ වෛද්‍යවරුන් එය භාවිතයට පහසු හා විශ්වාසදායක ලෙස සලකන්නේද යන්න මැනීමට අපට අවශ්‍යය.

2. ස්වේච්ඡා සහභාගීත්වය

සහභාගී වීම සම්පූර්ණයෙන්ම ස්වේච්ඡාවෙනි. ඔබේ තීරණය ඔබේ රැකියාවට, පුහුණුවට හෝ සහභාගී වන කිසිදු ආයතනයක් සමඟ ඔබට ඇති සම්බන්ධතාවයට බලපාන්නේ නැත.

3. කාලසීමාව, ක්‍රියාපටිපාටි සහ ඔබේ වගකීම්

ඔබෙන් පහත දෑ ඉල්ලා සිටිය හැකිය: (අ) හඳුනාගත නොහැකි ලෙස සකස් කළ රෝගී අවස්ථා සමූහයක්, AI සහාය සහිතව සහ රහිතව, කාලයත් සමඟ බෙදී ගිය සැසි කිහිපයකදී සමාලෝචනය කිරීම (ඇස්තමේන්තුගත මුළු කාලය පැය 4-6); (ආ) භාවිතයේ පහසුව, වැඩ බර සහ විශ්වාසය පිළිබඳ කෙටි ප්‍රශ්නාවලි සම්පූර්ණ කිරීම (වරකට මිනිත්තු 15 ක් පමණ); සහ (ඇ) මිනිත්තු 30-45 ක් පමණ වන, ඔබගේ කැමැත්ත අනුව සහභාගී විය හැකි සම්මුඛ සාකච්ඡාවක්; එය ශ්‍රව්‍ය පටිගත කරනු ලබන්නේ ඔබේ වෙනම අවසරය ඇත්නම් පමණි. සියලු සැසි ඔබට පහසු වේලාවන්හි සංවිධානය කෙරේ.

4. ඇති විය හැකි ප්‍රතිලාභ

ඔබට සෘජු ප්‍රතිලාභයක් නැත, නමුත් AI තීරණ-සහාය මෙවලම් පිළිබඳ මුල් අත්දැකීමක් ඔබට ලැබෙනු ඇත. සොයාගැනීම් ශ්‍රී ලංකාවේ මොළයේ ගෙඩි සම්බන්ධ සේවා වැඩිදියුණු කිරීමට උපකාරී විය හැකිය.

5. අවදානම්, අනතුරු සහ අපහසුතා

ප්‍රධාන බර වන්නේ ඔබේ කාලයයි. ඔබේ කියවීමේ කාර්ය සාධනය මැනීම ගැන ඔබට අපහසුතාවයක් දැනිය හැකිය; ඔබේ තනි ප්‍රතිඵල කිසි විටෙක නමින් වාර්තා නොකරන අතර, ඔබේ සේවා යෝජකයා හෝ සගයන් සමඟ බෙදා නොගනී.

6. ප්‍රතිපූරණ

කිසිදු ගෙවීමක් පිරිනමනු නොලබන අතර, සහභාගී වීමෙන් ඔබට කිසිදු වියදමක් නොවේ.

7. රහස්‍යභාවය

ඔබේ ප්‍රතිචාර ඔබේ කැමැත්ත පත්‍රයෙන් වෙන්ව, කේතයක් යටතේ, ශ්‍රී ජයවර්ධනපුර විශ්වවිද්‍යාලයේ ආරක්ෂිත පද්ධතිවල ගබඩා කෙරේ. ප්‍රතිඵල වාර්තා කරන්නේ සමූහ වශයෙන් පමණි. සම්මුඛ සාකච්ඡාවලින් උපුටා දැක්වීම් භාවිත කරන්නේ නම්, ඒවා නිර්නාමික කෙරේ. ශ්‍රව්‍ය පටිගත කිරීම් ලිඛිත පිටපත් කිරීමෙන් පසු මකා දමනු ලැබේ.

8. සහභාගීත්වය අවසන් කිරීම

ප්‍රධාන පර්යේෂක අමතා, හේතුවක් දැක්වීමකින් තොරව ඕනෑම වේලාවක ඔබට ඉවත් විය හැකිය; එවිට ඔබේ දත්ත තවදුරටත් විශ්ලේෂණයෙන් බැහැර කෙරේ. දැනටමත් ප්‍රකාශිත සමූහ ප්‍රතිඵලවලට ඇතුළත් දත්ත ඉවත් කළ නොහැක.

සම්බන්ධතා

ප්‍රධාන පර්යේෂක: කේ.එච්. විරාජ් සමීර, PhD අපේක්ෂක, ශ්‍රී ජයවර්ධනපුර විශ්වවිද්‍යාලය - 0775816789 - virajofficialinfo@gmail.com

සායනික සම-අධීක්ෂක: වෛද්‍ය ඒ.එම්. දීපාල් කේ. අත්තනායක, විශේෂඥ ස්නායු ශල්‍ය වෛද්‍ය, ශ්‍රී ලංකා ජාතික රෝහල - +94 71 421 9049

සහභාගී වන්නෙකු ලෙස ඔබේ අයිතිවාසිකම් පිළිබඳ ගැටලු සඳහා: ලේකම්, පර්යේෂණ ආචාර ධර්ම කමිටුව, වෛද්‍ය පීඨය, ජෙනරාල් ශ්‍රීමත් ජෝන් කොතලාවල ආරක්ෂක විශ්වවිද්‍යාලය - ethicsreviewcommittee.fom@kdu.ac.lk

ඇමුණුම 2: සායනික වෛද්‍යවරුන් සඳහා කැමැත්ත පත්‍රය

1 වන අනුවාදය, 09.11.2026 (සිංහල)

අධ්‍යයනය: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

(මොළයේ ගෙඩි හඳුනා ගැනීමට සහ ශල්‍යකර්ම සැලසුම් කිරීමට වෛද්‍යවරුන්ට උදව් වන කෘත්‍රිම බුද්ධි පද්ධතියක් පිළිබඳ පර්යේෂණය)

කරුණාකර සෑම කොටුවකම ඔබේ මුලකුරු යොදන්න:

ප්‍රකාශය	**මුලකුරු**
1. මම සායනික වෛද්‍යවරුන් සඳහා තොරතුරු පත්‍රිකාව (1 වන අනුවාදය) කියවා ඇති අතර, ප්‍රශ්න ඇසීමට අවස්ථාව ලැබී ඇත.	
2. සහභාගී වීම ස්වේච්ඡාවෙන් බවත්, හේතුවක් දැක්වීමකින් තොරව ඕනෑම වේලාවක ඉවත් විය හැකි බවත් මම තේරුම් ගනිමි.	
3. මගේ තනි කාර්ය සාධනය නමින් වාර්තා නොකරන බවත්, මගේ සේවා යෝජකයා සමඟ බෙදා නොගන්නා බවත් මම තේරුම් ගනිමි.	
4. රෝගී අවස්ථා සමාලෝචන සැසිවලට සහ ප්‍රශ්නාවලිවලට සහභාගී වීමට මම එකඟ වෙමි.	
5. (අත්‍යවශ්‍ය නොවේ) සම්මුඛ සාකච්ඡාවකට සහභාගී වීමට සහ එය ශ්‍රව්‍ය පටිගත කිරීමට මම එකඟ වෙමි.	

	නම	**අත්සන**	**දිනය**
සායනික වෛද්‍යවරයා			
කැමැත්ත ලබා ගන්නා පර්යේෂකයා			

பின்னிணைப்பு 1: நோயாளர் தகவல் தாள்

பதிப்பு *1, 09.11.2026 (தமிழ்)*

ஆய்வு: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

(மூளைக் கட்டிகளைக் கண்டறிவதற்கும் அறுவை சிகிச்சையைத் திட்டமிடுவதற்கும் மருத்துவர்களுக்கு உதவும் செயற்கை நுண்ணறிவு முறைமை பற்றிய ஆய்வு)

ஓர் ஆய்வில் பங்குபெற நீங்கள் அழைக்கப்படுகிறீர்கள். முடிவெடுப்பதற்கு முன் இந்தத் தகவலைக் கவனமாகப் படித்து, தெளிவில்லாத எதைப் பற்றியும் எங்களிடம் கேளுங்கள். நீங்கள் ஒரு பிள்ளைக்காகவோ அல்லது தாமாக முடிவெடுக்க முடியாத குடும்ப உறுப்பினர் ஒருவருக்காகவோ இதனைப் படிக்கிறீர்கள் என்றால், "நீங்கள்" என்பது நீங்கள் பொறுப்பேற்றுள்ள நபரைக் குறிக்கும்.

1. ஆய்வின் நோக்கம்

மூளை MRI ஸ்கேன்களையும் மூளைக் கட்டிகளின் திசுப் பரிசோதனை (பயாப்ஸி) வில்லைகளையும் மருத்துவர்கள் வாசிப்பதற்கு உதவுவதற்காக, செயற்கை நுண்ணறிவைப் (AI) பயன்படுத்தும் கணினி முறைமை ஒன்றை நாம் உருவாக்கி வருகிறோம். இந்த முறைமை தனக்கு உறுதியில்லாத சந்தர்ப்பங்களைத் தெரிவிக்கும் வகையில் வடிவமைக்கப்பட்டுள்ளது; அத்தகைய சந்தர்ப்பங்களை ஒரு மருத்துவர் எப்போதும் பரிசோதிப்பார். இம்முறைமையை உருவாக்கவும் பரிசோதிக்கவும், நோயாளர்களின் ஏற்கனவே உள்ள ஸ்கேன்களும் பதிவுகளும் இந்த ஆய்வில் பயன்படுத்தப்படும். இது இலங்கை தேசிய வைத்தியசாலையின் நரம்பியல் அறுவை சிகிச்சைப் பிரிவுகளுடன் இணைந்து, ஸ்ரீ ஜயவர்தனபுர பல்கலைக்கழகத்தில் மேற்கொள்ளப்படும் PhD பட்டப்படிப்பின் ஒரு பகுதியாகும்.

2. தன்னார்வப் பங்கேற்பு

பங்குபெறுவது முற்றிலும் உங்கள் விருப்பம். நீங்கள் பங்குபெற வேண்டாம் என முடிவெடுத்தாலும், பின்னர் உங்கள் மனதை மாற்றிக்கொண்டாலும், உங்கள் சிகிச்சையும் பராமரிப்பும் எவ்வகையிலும் மாறாது.

3. காலம், நடைமுறைகள் மற்றும் உங்கள் பொறுப்புகள்

நீங்கள் சம்மதித்தால், உங்கள் வழமையான சிகிச்சையின் போது எடுக்கப்பட்ட MRI ஸ்கேன்கள், பயாப்ஸி வில்லைகள், மருத்துவப் பதிவுகள் ஆகியவற்றின் பிரதிகளும், பின்னர் கிடைக்கும் தொடர் கண்காணிப்புத் தகவல்களும் அடுத்த இரண்டு முதல் மூன்று ஆண்டுகளுக்கு ஆய்வுக்காகப் பயன்படுத்தப்படும். உங்களுக்கு மேலதிகப் பரிசோதனை, ஸ்கேன், இரத்த மாதிரி, பயாப்ஸி அல்லது வைத்தியசாலை வருகை எதுவும் தேவையில்லை; நீங்கள் வேறு எதுவும் செய்ய வேண்டியதில்லை. உங்கள் மருத்துவர்கள் வழமை போலவே உங்கள் சிகிச்சையைத் தீர்மானிப்பார்கள்; உங்களைப் பற்றிய முடிவுகளை எடுக்க இந்தக் கணினி முறைமை பயன்படுத்தப்படமாட்டாது.

4. சாத்தியமான நன்மைகள்

உங்களுக்கு நேரடி நன்மை எதுவும் இல்லை. விரைவான, மேலும் சீரான நோயறிதலுக்கு உதவுவதன் மூலம் இந்த ஆய்வு இலங்கையின் எதிர்கால நோயாளர்களுக்குப் பயனளிக்கக்கூடும்.

5. அபாயங்கள், ஆபத்துகள் மற்றும் அசௌகரியங்கள்

உடல் ரீதியான அபாயம் எதுவும் இல்லை. பார்க்கக்கூடாத ஒருவர் உங்கள் தகவலைப் பார்க்கக்கூடும் என்பதே பிரதான அபாயம். பிரிவு 7 இல் விவரிக்கப்பட்டுள்ளவாறு இதிலிருந்து நாம் பாதுகாப்போம்.

6. இழப்பீடுகள்

பங்குபெறுவதற்காக உங்களுக்குப் பணம் வழங்கப்படமாட்டாது; இதனால் உங்களுக்கு எந்தச் செலவும் ஏற்படாது.

7. இரகசியத்தன்மை

உங்கள் தகவல் பயன்படுத்தப்படுவதற்கு முன், உங்கள் பெயர், தேசிய அடையாள அட்டை இலக்கம், வைத்தியசாலை இலக்கம், முகவரி, தொலைபேசி இலக்கம் ஆகியவை நீக்கப்படும்; உங்கள் முகத்தை அடையாளம் காண முடியாதவாறு உங்கள் ஸ்கேன்கள் செயல்முறைப்படுத்தப்படும். உங்கள் தகவலுக்கு ஒரு குறியீடு வழங்கப்படும். அந்தக் குறியீட்டை உங்களுடன் இணைக்கும் பட்டியலை உங்கள் வைத்தியசாலைக் குழு மட்டுமே வைத்திருக்கும். தரவுகள் வைத்தியசாலை அல்லது பல்கலைக்கழகத்தினுள் உள்ள, கடவுச்சொல்லால் பாதுகாக்கப்பட்ட பாதுகாப்பான கணினிகளில் சேமிக்கப்படும். முடிவுகள் குழு மொத்தங்களாக மட்டுமே வெளியிடப்படும்; உங்கள் பெயர் ஒருபோதும் குறிப்பிடப்படமாட்டாது.

8. பங்கேற்பை நிறுத்துதல்

உங்கள் மருத்துவரிடமோ ஆய்வாளரிடமோ தெரிவித்து, காரணம் கூறாமல் எந்நேரத்திலும் நீங்கள் பங்கேற்பதை நிறுத்தலாம். அப்போது உங்கள் தகவல் ஆய்விலிருந்து நீக்கப்படும். ஏற்கனவே வெளியிடப்பட்ட குழு முடிவுகளிலோ பயிற்றுவிக்கப்பட்ட கணினி மாதிரியிலோ பயன்படுத்தப்பட்ட தகவலை நீக்க முடியாது. உதாரணமாக, நெறிமுறைக் குழு கோரினால், ஆய்வாளர்களும் ஆய்வை முன்கூட்டியே நிறுத்தலாம்.

தொடர்புகளுக்கு

பிரதான ஆய்வாளர்: கே.எச். விராஜ் சமீர, PhD மாணவர், ஸ்ரீ ஜயவர்தனபுர பல்கலைக்கழகம் - 0775816789 - virajofficialinfo@gmail.com

மருத்துவ இணை மேற்பார்வையாளர்: வைத்தியர் ஏ.எம். தீபால் கே. அத்தநாயக்க, விசேட நரம்பியல் அறுவை சிகிச்சை நிபுணர், இலங்கை தேசிய வைத்தியசாலை - +94 71 421 9049

பங்கேற்பாளராக உங்கள் உரிமைகள் தொடர்பான கவலைகளுக்கு: செயலாளர், ஆய்வு நெறிமுறைக் குழு, மருத்துவ பீடம், ஜெனரல் சேர் ஜோன் கொத்தலாவல பாதுகாப்புப் பல்கலைக்கழகம் - ethicsreviewcommittee.fom@kdu.ac.lk

பின்னிணைப்பு 1: நோயாளர் சம்மதப் படிவம்

பதிப்பு *1, 09.11.2026 (தமிழ்)*

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ஒவ்வொரு பெட்டியிலும் உங்கள் முதலெழுத்துகளை இடவும்:

கூற்று	**முதலெழுத்து**
1. நோயாளர் தகவல் தாளை (பதிப்பு 1) நான் படித்துள்ளேன் (அல்லது எனக்கு வாசித்துக் காட்டப்பட்டது); கேள்விகள் கேட்க எனக்கு வாய்ப்பு வழங்கப்பட்டது.	
2. பங்கேற்பது தன்னார்வமானது என்பதையும், எனது மருத்துவப் பராமரிப்பு பாதிக்கப்படாமல் எந்நேரத்திலும் நான் விலகலாம் என்பதையும் புரிந்துகொள்கிறேன்.	
3. எனது தனிப்பட்ட விவரங்கள் நீக்கப்பட்ட பின்னர், எனது MRI ஸ்கேன்கள், பயாப்ஸி வில்லைகள், மருத்துவப் பதிவுகள், தொடர் கண்காணிப்புத் தகவல்கள் ஆகியவற்றின் பிரதிகளை இந்த ஆய்வுக்குப் பயன்படுத்த நான் சம்மதிக்கிறேன்.	
4. எனது தகவல் இரகசியமாக வைக்கப்படும் என்பதையும், எந்த அறிக்கையிலும் வெளியீட்டிலும் எனது பெயர் குறிப்பிடப்படாது என்பதையும் புரிந்துகொள்கிறேன்.	
5. இந்த ஆய்வில் பங்குபெற நான் சம்மதிக்கிறேன்.	

	பெயர்	**கையொப்பம்**	**திகதி**
நோயாளர்			
பெற்றோர் / பாதுகாவலர் / சட்டப் பிரதிநிதி (பொருந்துமாயின்) - உறவுமுறை:			
சம்மதம் பெறுபவர்			

இணக்கம் (புரிந்துகொள்ளக்கூடிய வயதுடைய பிள்ளைகளுக்கு): "மருத்துவர் இந்த ஆய்வை எனக்கு விளக்கினார். எனது ஸ்கேன்களைப் பயன்படுத்துவதில் எனக்கு மகிழ்ச்சி." பிள்ளையின் பெயர் / கையொப்பம்: ..... திகதி: .....

பின்னிணைப்பு 2: மருத்துவர்களுக்கான தகவல் தாள்

பதிப்பு *1, 09.11.2026 (தமிழ்)*

ஆய்வு: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

(மூளைக் கட்டிகளைக் கண்டறிவதற்கும் அறுவை சிகிச்சையைத் திட்டமிடுவதற்கும் மருத்துவர்களுக்கு உதவும் செயற்கை நுண்ணறிவு முறைமை பற்றிய ஆய்வு)

ஓர் ஆய்வில் பங்குபெற நீங்கள் அழைக்கப்படுகிறீர்கள். முடிவெடுப்பதற்கு முன் இந்தத் தகவலைக் கவனமாகப் படித்து, தெளிவில்லாத எதைப் பற்றியும் எங்களிடம் கேளுங்கள். மூளைக் கட்டிகள் உள்ள நோயாளர்களின் நோயறிதல் அல்லது சிகிச்சையில் நீங்கள் ஈடுபட்டுள்ளதால் அழைக்கப்படுகிறீர்கள்.

1. ஆய்வின் நோக்கம்

தனது நம்பகத்தன்மை அளவைத் தானே தெரிவித்து, உறுதியற்ற சந்தர்ப்பங்களை மருத்துவர்களிடம் பரிந்துரைக்கும், மூளைக் கட்டி நோயறிதலுக்கும் அறுவை சிகிச்சைத் திட்டமிடலுக்குமான AI முடிவு-ஆதரவுக் கட்டமைப்பொன்றை நாம் உருவாக்கி மதிப்பீடு செய்கிறோம். அது மருத்துவர்களுக்கு உதவுகிறதா, வேலைப்பளுவை எவ்வாறு பாதிக்கிறது, மருத்துவர்கள் அதனைப் பயன்படுத்த எளிதானதாகவும் நம்பகமானதாகவும் கருதுகிறார்களா என்பதை அளவிட விரும்புகிறோம்.

2. தன்னார்வப் பங்கேற்பு

பங்குபெறுவது முற்றிலும் தன்னார்வமானது. உங்கள் முடிவு உங்கள் வேலைவாய்ப்பு, பயிற்சி அல்லது பங்குபெறும் எந்த நிறுவனத்துடனுமான உங்கள் உறவைப் பாதிக்காது.

3. காலம், நடைமுறைகள் மற்றும் உங்கள் பொறுப்புகள்

உங்களிடம் பின்வருவன கோரப்படலாம்: (அ) அடையாளம் நீக்கப்பட்ட நோயாளர் சந்தர்ப்பங்களின் தொகுப்பொன்றை AI ஆதரவுடனும் ஆதரவின்றியும், காலப்போக்கில் பகிரப்பட்ட பல அமர்வுகளில் மீளாய்வு செய்தல் (மதிப்பிடப்பட்ட மொத்த நேரம் 4-6 மணித்தியாலங்கள்); (ஆ) பயன்பாட்டுத்தன்மை, வேலைப்பளு, நம்பிக்கை பற்றிய சுருக்கமான வினாக்கொத்துகளைப் பூர்த்தி செய்தல் (ஒவ்வொரு முறையும் சுமார் 15 நிமிடங்கள்); (இ) சுமார் 30-45 நிமிடங்கள் நீடிக்கும், விருப்பத்தேர்வான நேர்காணலில் பங்குபெறுதல்; உங்கள் தனியான அனுமதியுடன் மட்டுமே அது ஒலிப்பதிவு செய்யப்படும். அனைத்து அமர்வுகளும் உங்களுக்கு வசதியான நேரங்களில் ஏற்பாடு செய்யப்படும்.

4. சாத்தியமான நன்மைகள்

உங்களுக்கு நேரடி நன்மை எதுவும் இல்லை; எனினும் AI முடிவு-ஆதரவுக் கருவிகள் பற்றிய ஆரம்ப அனுபவம் உங்களுக்குக் கிடைக்கும். கண்டறிவுகள் இலங்கையின் மூளைக் கட்டிச் சேவைகளை மேம்படுத்தக்கூடும்.

5. அபாயங்கள், ஆபத்துகள் மற்றும் அசௌகரியங்கள்

பிரதான சுமை உங்கள் நேரமாகும். உங்கள் வாசிப்புச் செயல்திறன் அளவிடப்படுவது குறித்து நீங்கள் அசௌகரியமாக உணரலாம்; உங்கள் தனிப்பட்ட

முடிவுகள் ஒருபோதும் பெயருடன் அறிவிக்கப்படவோ, உங்கள் தொழில் வழங்குநருடனோ சக ஊழியர்களுடனோ பகிரப்படவோ மாட்டாது.

6. இழப்பீடுகள்

எந்தக் கொடுப்பனவும் வழங்கப்படமாட்டாது; பங்குபெறுவதால் உங்களுக்கு எந்தச் செலவும் ஏற்படாது.

7. இரகசியத்தன்மை

உங்கள் பதில்கள் உங்கள் சம்மதப் படிவத்திலிருந்து வேறாக, ஒரு குறியீட்டின் கீழ், ஸ்ரீ ஜயவர்தனபுர பல்கலைக்கழகத்தின் பாதுகாப்பான முறைமைகளில் சேமிக்கப்படும். முடிவுகள் தொகுப்பாக மட்டுமே அறிவிக்கப்படும். நேர்காணல் மேற்கோள்கள் பயன்படுத்தப்பட்டால், அவை அநாமதேயமாக்கப்படும். ஒலிப்பதிவுகள் எழுத்துப்படியாக்கப்பட்ட பின் அழிக்கப்படும்.

8. பங்கேற்பை நிறுத்துதல்

பிரதான ஆய்வாளரைத் தொடர்புகொண்டு, காரணம் கூறாமல் எந்நேரத்திலும் நீங்கள் விலகலாம்; அப்போது உங்கள் தரவுகள் மேலதிகப் பகுப்பாய்விலிருந்து நீக்கப்படும். ஏற்கனவே வெளியிடப்பட்ட தொகுப்பு முடிவுகளில் உள்ள தரவுகளை நீக்க முடியாது.

தொடர்புகளுக்கு

பிரதான ஆய்வாளர்: கே.எச். விராஜ் சமீர, PhD மாணவர், ஸ்ரீ ஜயவர்தனபுர பல்கலைக்கழகம் - 0775816789 - virajofficialinfo@gmail.com

மருத்துவ இணை மேற்பார்வையாளர்: வைத்தியர் ஏ.எம். தீபால் கே. அத்தநாயக்க, விசேட நரம்பியல் அறுவை சிகிச்சை நிபுணர், இலங்கை தேசிய வைத்தியசாலை - +94 71 421 9049

பங்கேற்பாளராக உங்கள் உரிமைகள் தொடர்பான கவலைகளுக்கு: செயலாளர், ஆய்வு நெறிமுறைக் குழு, மருத்துவ பீடம், ஜெனரல் சேர் ஜோன் கொத்தலாவல பாதுகாப்புப் பல்கலைக்கழகம் - ethicsreviewcommittee.fom@kdu.ac.lk

பின்னிணைப்பு 2: மருத்துவர்களுக்கான சம்மதப் படிவம்

பதிப்பு *1, 09.11.2026* (தமிழ்)

ஆய்வு: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

(மூளைக் கட்டிகளைக் கண்டறிவதற்கும் அறுவை சிகிச்சையைத் திட்டமிடுவதற்கும் மருத்துவர்களுக்கு உதவும் செயற்கை நுண்ணறிவு முறைமை பற்றிய ஆய்வு)

ஒவ்வொரு பெட்டியிலும் உங்கள் முதலெழுத்துகளை இடவும்:

கூற்று	**முதலெழுத்து**
1. மருத்துவர்களுக்கான தகவல் தாளை (பதிப்பு 1) நான் படித்துள்ளேன்; கேள்விகள் கேட்க எனக்கு வாய்ப்பு வழங்கப்பட்டது.	
2. பங்கேற்பது தன்னார்வமானது என்பதையும், காரணம் கூறாமல் எந்நேரத்திலும் நான் விலகலாம் என்பதையும் புரிந்துகொள்கிறேன்.	
3. எனது தனிப்பட்ட செயல்திறன் பெயருடன் அறிவிக்கப்படாது என்பதையும், எனது தொழில் வழங்குநருடன் பகிரப்படாது என்பதையும் புரிந்துகொள்கிறேன்.	
4. நோயாளர் சந்தர்ப்ப மீளாய்வு அமர்வுகளிலும் வினாக்கொத்துகளிலும் பங்குபெற நான் சம்மதிக்கிறேன்.	
5. (விருப்பத்தேர்வு) நேர்காணலில் பங்குபெறவும் அது ஒலிப்பதிவு செய்யப்படவும் நான் சம்மதிக்கிறேன்.	

	பெயர்	**கையொப்பம்**	**திகதி**
மருத்துவர்			
சம்மதம் பெறும் ஆய்வாளர்			

Appendix 3: Data Collection Forms

Version 1, 09.11.2026

Study: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

This appendix contains every form used to collect data in the study. Forms 3B-3E, which clinicians complete or take part in, are also provided in Sinhala and Tamil (Appendix 3, Sinhala and Tamil versions); the clinician uses the language they prefer. Form 3A is in English only, because it is completed by hospital staff from English-language medical records and is never given to patients. No form records a patient's name, NIC number, hospital/BHT or clinic number, address, telephone number or any full date. Clinicians are identified only by a reader code.

Form	Purpose	Completed by	When
3A Case Report Form (CRF)	Abstract routinely recorded clinical, imaging, pathology, treatment and follow-up data for the de-identified registry	Designated hospital staff / trained abstractor under the clinical co-supervisor	Once per patient, updated at each follow-up
3B Clinician Background Form	Describe the reader panel (aggregate reporting only)	Participating clinician	Once, after consent
3C Reader-Study Case Record	Record each reader's decision on each case, with and without AI support	Participating clinician (time is also logged by the study software)	Every case, every session
3D Post-Session Questionnaire	System Usability Scale, NASA Task Load Index and trust questionnaire	Participating clinician	End of each reading session
3E Interview Guide	Semi-structured interview on trust, escalation, workload and use in practice	Principal Investigator (interviewer)	Once, optional, separate consent

Appendix 3A: Case Report Form (CRF)

Version 1, 09.11.2026

Study: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

Instructions. Complete from the hospital record. Write the study code only; do not write any identifier on this form. Record time as the number of days from the first diagnostic MRI (Day 0). Tick one box unless stated otherwise. Write NR if not recorded. Completed CRFs are kept at the hospital and entered into the registry under the study code.

Section 1. Administration

Study code	BT-____-_____	Site code	_____
Data source	☐ Retrospective record (waiver) ☐ Prospective (consent)	Consent given by	☐ Patient ☐ Parent/guardian ☐ Legal representative ☐ N/A (waiver)
Abstractor code	_____	CRF entry (Day)	_____

Section 2. Patient characteristics

Age at diagnosis (years)	_____	Sex	☐ Male ☐ Female
Age group	☐ <18 ☐ 18-39 ☐ 40-59 ☐ 60+	Handedness	☐ Right ☐ Left ☐ NR
Relevant history	☐ Previous cancer ☐ Previous cranial radiotherapy ☐ Genetic syndrome ☐ None ☐ NR		

Section 3. Presentation

Presenting features (tick all)	☐ Headache ☐ Seizure ☐ Focal weakness ☐ Speech disturbance ☐ Visual disturbance ☐ Cognitive / behavioural change ☐ Raised ICP features ☐ Incidental ☐ Other: _____
Duration of symptoms (weeks)	_____
Performance status at presentation	KPS _____ / Lansky (child) _____ / ECOG _____ GCS _____ /15

Section 4. Imaging (one row per MRI study; Day 0 = first diagnostic MRI)

Study	Day	Scanner make / field strength	Sequences available (tick)	Quality	De-identified & defaced
Pre-op	0	_____ / ___ T	☐T1 ☐T1c ☐T2 ☐FLAIR ☐DWI ☐SWI ☐Perfusion ☐DTI ☐MRS	☐Good ☐Usable ☐Poor	☐Yes ☐No
Post-op		_____ / ___ T	☐T1 ☐T1c ☐T2 ☐FLAIR ☐DWI ☐Other	☐Good ☐Usable ☐Poor	☐Yes ☐No
Other		_____ / ___ T	☐T1 ☐T1c ☐T2 ☐FLAIR ☐DWI ☐Other	☐Good ☐Usable ☐Poor	☐Yes ☐No

Lesion location (from report)	☐ Frontal ☐ Temporal ☐ Parietal ☐ Occipital ☐ Insular ☐ Deep/thalamic ☐ Ventricular ☐ Sellar/suprasellar ☐ Cerebellar ☐ Brainstem ☐ Extra-axial ☐

	Spinal
Side / number of lesions	☐ Left ☐ Right ☐ Bilateral ☐ Midline Number of lesions: _____
Radiological impression	_____

Section 5. Surgery and tissue

Procedure	☐ Stereotactic / needle biopsy ☐ Open biopsy ☐ Subtotal resection ☐ Gross total resection ☐ No surgery
Day of procedure	_____
Extent of resection (post-op MRI)	☐ Gross total ☐ Near total ☐ Subtotal ☐ Partial ☐ Biopsy only ☐ NR
Peri-operative complication	☐ None ☐ New neurological deficit ☐ Haemorrhage ☐ Infection ☐ Other: _____
Archived slides	Number of H&E slides _____ Number of IHC slides _____ Digitised: ☐ Yes ☐ No Slide quality: ☐ Good ☐ Usable ☐ Poor

Section 6. Pathology and reference diagnosis

Histological diagnosis (report)	_____
Integrated diagnosis (WHO CNS5, 2021)	_____
CNS WHO grade	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ Not applicable ☐ NR
Markers (where tested)	IDH ☐Mut ☐WT ☐NT 1p/19q ☐Codel ☐Intact ☐NT MGMT ☐Methyl ☐Unmeth ☐NT ATRX ☐Lost ☐Retained ☐NT p53 ☐Pos ☐Neg ☐NT H3 K27M ☐Pos ☐Neg ☐NT Ki-67 _____ %
Reference standard confirmed by	☐ Consultant histopathologist ☐ Multidisciplinary team ☐ Imaging + clinical only (no tissue)

Section 7. Treatment

Radiotherapy	☐ Yes ☐ No ☐ NR Day started _____ Dose _____ Gy / _____ fractions
Chemotherapy	☐ Temozolomide ☐ Other: _____ ☐ None ☐ NR
Other treatment	☐ Steroids ☐ Anti-epileptic drugs ☐ Shunt / drain ☐ Palliative care only ☐ Other: _____

Section 8. Follow-up (one row per visit)

Visit	Day	MRI done	Response (RANO)	Clinical status	KPS
1		☐Y ☐N	☐CR ☐PR ☐SD ☐PD ☐NA	☐Stable ☐Improved ☐Worse	
2		☐Y ☐N	☐CR ☐PR ☐SD ☐PD ☐NA	☐Stable ☐Improved ☐Worse	
3		☐Y ☐N	☐CR ☐PR ☐SD ☐PD ☐NA	☐Stable ☐Improved ☐Worse	
4		☐Y ☐N	☐CR ☐PR ☐SD ☐PD ☐NA	☐Stable ☐Improved ☐Worse	

Status at last contact	☐ Alive, no progression ☐ Alive with progression ☐ Died ☐ Lost to follow-up Day of last contact / death: _____
Recurrence	☐ No ☐ Yes Day _____

Section 9. Completeness and sign-off

Data available	☐ MRI ☐ Digitised slides ☐ Markers ☐ Treatment ☐ Follow-up
Identifier check	☐ I confirm that no name, NIC, hospital/BHT/clinic number, address, telephone number or full date appears on this form or in the exported data.
Abstractor code / signature / date	
Checked by (clinical co-supervisor or delegate)	

Appendix 3B: Clinician Background Form

Version 1, 09.11.2026

Study: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

Please complete once, after signing the consent form. Your answers are stored under your reader code and are reported only as group totals.

Reader code	R-____
Specialty	☐ Neurosurgery ☐ Radiology ☐ Histopathology ☐ Other: _____
Current grade	☐ Consultant ☐ Senior registrar ☐ Registrar ☐ Other: _____
Years in specialty (since starting specialty training)	☐ <5 ☐ 5-9 ☐ 10-19 ☐ 20+
Brain-tumour cases you see or report per month (approx.)	☐ <5 ☐ 5-14 ☐ 15-29 ☐ 30+
Previous use of AI tools in clinical work	☐ Never ☐ Tried once or twice ☐ Occasionally ☐ Regularly
General attitude to AI in your specialty	☐ Very negative ☐ Negative ☐ Neutral ☐ Positive ☐ Very positive

Appendix 3C: Reader-Study Case Record

Version 1, 09.11.2026

Study: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

How the reader study works. Each clinician reviews the same set of de-identified cases twice: once without AI support and once with AI support, in separate sessions at least four weeks apart (washout). Half the readers start with the unassisted condition and half with the assisted condition. Case order is randomised within each session. Reading time is recorded automatically by the study software; this record captures the reader's decision. One row per case.

Reader code	R-____	Session no.	____	Condition	☐ Unassisted ☐ AI-assisted

Columns: Dx = most likely tumour type (WHO CNS5 family); Gr = CNS WHO grade 1-4; Conf = your confidence 1 (very unsure) to 5 (very sure). AI-assisted sessions only: AI = what you did with the AI output (A = accepted, M = amended, R = rejected); Esc = for cases the AI referred to you, was referral appropriate? (Y/N); for cases it did not refer, should it have? (Y/N).

#	Case ID	Dx (type)	Gr	Conf	AI (A/M/R)	Esc (Y/N)	Comment / reason for change
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							

Appendix 3D: Post-Session Questionnaire

Version 1, 09.11.2026

Study: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

Reader code R-____ Session ____ Condition: ☐ Unassisted ☐ AI-assisted Date _____

Part 1 (workload) is completed after every session. Parts 2 and 3 are completed after AI-assisted sessions only. About 10-15 minutes.

Part 1. NASA Task Load Index (raw TLX)

Hart & Staveland (1988). For each scale, mark a cross on one of the 21 positions to show how the session you have just finished felt.

Scale	Question	Very low ← → Very high
Mental demand	How mentally demanding was the task?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
Physical demand	How physically demanding was the task?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
Temporal demand	How hurried or rushed was the pace of the task?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
Performance	How successful were you in accomplishing what you were asked to do? (left = perfect, right = failure)	Perfect Failure |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
Effort	How hard did you have to work to accomplish your level of performance?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
Frustration	How insecure, discouraged, irritated, stressed and annoyed were you?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|

Part 2. System Usability Scale (SUS)

Brooke (1996). Tick one box per statement. 1 = strongly disagree, 5 = strongly agree. "System" means the AI decision-support system you used in this session.

#	Statement	1	2	3	4	5
1	I think that I would like to use this system frequently.	☐	☐	☐	☐	☐
2	I found the system unnecessarily complex.	☐	☐	☐	☐	☐
3	I thought the system was easy to use.	☐	☐	☐	☐	☐
4	I think that I would need the support of a technical person to be able to use this system.	☐	☐	☐	☐	☐
5	I found the various functions in this system were well integrated.	☐	☐	☐	☐	☐
6	I thought there was too much inconsistency in this system.	☐	☐	☐	☐	☐
7	I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☐
8	I found the system very cumbersome to use.	☐	☐	☐	☐	☐
9	I felt very confident using the system.	☐	☐	☐	☐	☐
10	I needed to learn a lot of things before I could get going with this system.	☐	☐	☐	☐	☐

Part 3. Trust in the AI decision-support system

Items adapted from Jian, Bisantz & Drury (2000) and written for this study. 1 = strongly disagree, 5 = strongly agree.

#	Statement	1	2	3	4	5
1	I could rely on the system's outputs when it did not refer a case to me.	☐	☐	☐	☐	☐
2	The confidence the system reported matched how correct its outputs were.	☐	☐	☐	☐	☐
3	The system referred the right cases to me for full review.	☐	☐	☐	☐	☐
4	The system missed cases that I think it should have referred to me.	☐	☐	☐	☐	☐
5	The explanations (outlines, heat-maps, evidence breakdown) helped me judge whether the output was correct.	☐	☐	☐	☐	☐
6	I was suspicious of the system's outputs.	☐	☐	☐	☐	☐
7	Using the system did not reduce my own responsibility for the decision.	☐	☐	☐	☐	☐
8	I would be comfortable using this system as a second reader in routine practice.	☐	☐	☐	☐	☐
9	The system could lead a less experienced clinician into error.	☐	☐	☐	☐	☐
10	Overall, I trust this system.	☐	☐	☐	☐	☐

Any comments on this session: _____

_____

Appendix 3E: Semi-Structured Interview Guide

Version 1, 09.11.2026

Study: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

Optional interview with clinicians who completed the reader study and gave separate consent for interview and audio recording. About 30-45 minutes, in person or online, at a time chosen by the clinician, in Sinhala, Tamil or English as the clinician prefers. The questions below are a guide; follow-up questions depend on the answers. The interviewer does not comment on the clinician's own reading performance.

Opening (about 3 minutes)

- Thank the participant; confirm they have read the information sheet and signed the interview consent item.
- Confirm consent to audio recording and start the recorder only after verbal agreement. Remind them that they can skip any question or stop at any time.
- Explain that quotations will be anonymised and that the recording will be deleted after transcription.

1. Current practice

- Can you walk me through how a new brain-tumour case reaches you and how you reach a decision?
- Where in that process do delays or uncertainty usually occur?

2. Experience of the system

- What was it like to review cases with the AI support compared with without it?
- Which outputs did you find most and least useful (outlines, type and grade, evidence breakdown, surgical route)?

3. Confidence and referral (escalation)

- How did you interpret the confidence the system showed?
- Did it refer the right cases to you? Can you recall a case where it referred something it should not have, or did not refer something it should have?
- Where would you set the line between quick sign-off and full review?

4. Trust and responsibility

- What made you trust or distrust a particular output?
- Who do you feel is responsible for a decision made with AI support?
- How do you think the system would affect less experienced clinicians?

5. Workload

- Did the system change how much time or effort the cases took? In what way?
- Were there cases where checking the AI added work?

6. Surgical-planning proof of concept (neurosurgeons)

- How useful was the 3D model and candidate route for thinking about an approach?
- What would it need to show before you would consider it in planning?

7. Use in practice in Sri Lanka

- What would stop or help this kind of system being used in your unit?
- What safeguards would you want in place (training, audit, governance, data protection)?

Closing (about 2 minutes)

- Is there anything we have not discussed that you think is important?
- Stop the recording. Thank the participant and explain how they can obtain the summary of findings.

ඇමුණුම 3B: සායනික වෛද්‍ය පසුබිම් ආකෘති පත්‍රය

1 වන අනුවාදය, 09.11.2026 (සිංහල)

අධ්‍යයනය: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

සටහන: මෙය ඉංග්‍රීසි මුල් පිටපතේ (ඇමුණුම 3) පරිවර්තනයකි. අර්ථයේ වෙනසක් ඇත්නම්, ඉංග්‍රීසි පිටපත වලංගු වේ.

කැමැත්ත පත්‍රය අත්සන් කිරීමෙන් පසු එක් වරක් සම්පූර්ණ කරන්න. ඔබේ පිළිතුරු ඔබේ පාඨක කේතය යටතේ ගබඩා කරනු ලබන අතර, සමූහ එකතු ලෙස පමණක් වාර්තා කෙරේ.

පාඨක කේතය	R-____
විශේෂඥ ක්ෂේත්‍රය	☐ ස්නායු ශල්‍යකර්ම ☐ විකිරණවේදය ☐ පටක රෝගවේදය (Histopathology) ☐ වෙනත්: _____
වර්තමාන තනතුර	☐ විශේෂඥ වෛද්‍ය ☐ ජ්‍යෙෂ්ඨ රෙජිස්ට්‍රාර් ☐ රෙජිස්ට්‍රාර් ☐ වෙනත්: _____
විශේෂඥ ක්ෂේත්‍රයේ වසර ගණන (විශේෂඥ පුහුණුව ආරම්භයේ සිට)	☐ <5 ☐ 5-9 ☐ 10-19 ☐ 20+
මසකට ඔබ බලන හෝ වාර්තා කරන මොළයේ ගෙඩි රෝගී අවස්ථා (දළ වශයෙන්)	☐ <5 ☐ 5-14 ☐ 15-29 ☐ 30+
සායනික කටයුතුවලදී AI මෙවලම් පෙර භාවිතය	☐ කිසි විටෙක නැත ☐ එක් හෝ දෙවරක් ☐ ඉදහිට ☐ නිතිපතා
ඔබේ ක්ෂේත්‍රයේ AI පිළිබඳ සාමාන්‍ය ආකල්පය	☐ ඉතා ඍණාත්මක ☐ ඍණාත්මක ☐ මධ්‍යස්ථ ☐ ධනාත්මක ☐ ඉතා ධනාත්මක

ඇමුණුම 3C: පාඨක අධ්‍යයන රෝගී අවස්ථා වාර්තාව

1 වන අනුවාදය, 09.11.2026 (සිංහල)

අධ්‍යයනය: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

සටහන: මෙය ඉංග්‍රීසි මුල් පිටපතේ (ඇමුණුම 3) පරිවර්තනයකි. අර්ථයේ වෙනසක් ඇත්නම්, ඉංග්‍රීසි පිටපත වලංගු වේ.

පාඨක අධ්‍යයනය ක්‍රියාත්මක වන ආකාරය. සෑම සායනික වෛද්‍යවරයෙකුම හඳුනාගත නොහැකි ලෙස සකස් කළ එකම රෝගී අවස්ථා සමූහය දෙවරක් සමාලෝචනය කරයි: වරක් AI සහාය නොමැතිව සහ වරක් AI සහාය සහිතව, අවම වශයෙන් සති හතරක පරතරයක් (washout) සහිත වෙනම සැසිවලදී. පාඨකයන්ගෙන් අඩක් සහාය රහිත තත්ත්වයෙන් ද, අනෙක් අඩ සහාය සහිත තත්ත්වයෙන් ද ආරම්භ කරයි. සෑම සැසියකදීම රෝගී අවස්ථා අනුපිළිවෙල අහඹු ලෙස සකසනු ලැබේ. කියවීමට ගතවන කාලය අධ්‍යයන මෘදුකාංගය මගින් ස්වයංක්‍රීයව සටහන් වේ; මෙම වාර්තාවේ පාඨකයාගේ තීරණය සටහන් කෙරේ. එක් රෝගී අවස්ථාවකට එක් පේළියක්.

පාඨක කේතය	R-____	සැසි අංකය	____	තත්ත්වය	☐ AI සහාය රහිත ☐ AI සහාය සහිත

තීරු: Dx = වඩාත් ඉඩ ඇති ගෙඩි වර්ගය (WHO CNS5 කාණ්ඩය); Gr = CNS WHO ශ්‍රේණිය 1-4; Conf = ඔබේ විශ්වාසය, 1 (ඉතා අවිනිශ්චිත) සිට 5 (ඉතා නිශ්චිත) දක්වා. AI සහාය සහිත සැසි සඳහා පමණි: AI = AI ප්‍රතිදානය සමඟ ඔබ කළ දේ (A = පිළිගත්තා, M = සංශෝධනය කළා, R = ප්‍රතික්ෂේප කළා); Esc = AI ඔබ වෙත යොමු කළ අවස්ථා සඳහා, එම යොමු කිරීම සුදුසු වූයේද? (Y/N); යොමු නොකළ අවස්ථා සඳහා, එය යොමු කළ යුතුව තිබුණේද? (Y/N).

#	**රෝගී අවස්ථා අංකය**	**Dx (වර්ගය)**	**Gr**	**Conf**	**AI (A/M/R)**	**Esc (Y/N)**	**අදහස් / වෙනස් කිරීමට හේතුව**
1							
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ඇමුණුම 3D: සැසියෙන් පසු ප්‍රශ්නාවලිය

1 වන අනුවාදය, 09.11.2026 (සිංහල)

අධ්‍යයනය: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

සටහන: මෙය ඉංග්‍රීසි මුල් පිටපතේ (ඇමුණුම 3) පරිවර්තනයකි. අර්ථයේ වෙනසක් ඇත්නම්, ඉංග්‍රීසි පිටපත වලංගු වේ.

පාඨක කේතය R-____ සැසිය ____ තත්ත්වය: ☐ AI සහාය රහිත ☐ AI සහාය සහිත දිනය _____

1 වන කොටස (වැඩ බර) සෑම සැසියකින් පසුම සම්පූර්ණ කරන්න. 2 සහ 3 කොටස් AI සහාය සහිත සැසිවලින් පසු පමණක් සම්පූර්ණ කරන්න. මිනිත්තු 10-15 ක් පමණ.

1 වන කොටස. NASA කාර්ය බර දර්ශකය (NASA-TLX, raw)

Hart සහ Staveland (1988). සෑම පරිමාණයක් සඳහාම, ඔබ දැන් අවසන් කළ සැසිය ඔබට දැනුණු ආකාරය පෙන්වීමට ස්ථාන 21 න් එකක් මත කතිරයක් (X) ලකුණු කරන්න.

පරිමාණය	ප්‍රශ්නය	ඉතා අඩු ← → ඉතා ඉහළ
මානසික ඉල්ලුම	කාර්යය මානසිකව කොතරම් ඉල්ලුම් සහිත වූයේද?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
ශාරීරික ඉල්ලුම	කාර්යය ශාරීරිකව කොතරම් ඉල්ලුම් සහිත වූයේද?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
කාලීන ඉල්ලුම	කාර්යයේ වේගය කොතරම් ඉක්මන් හෝ කලබලකාරී වූයේද?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
කාර්ය සාධනය	ඔබෙන් ඉල්ලූ දේ ඉටු කිරීමට ඔබ කොතරම් සාර්ථක වූයේද? (වම = පරිපූර්ණ, දකුණ = අසාර්ථක)	පරිපූර්ණ අසාර්ථක |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
උත්සාහය	ඔබේ කාර්ය සාධන මට්ටම ළඟා කර ගැනීමට ඔබට කොතරම් වෙහෙස වී වැඩ කිරීමට සිදු වූයේද?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
කලකිරීම	ඔබට කොතරම් අනාරක්ෂිත බවක්, අධෛර්යයක්, කෝපයක්, ආතතියක් සහ කරදරයක් දැනුණේද?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|

2 වන කොටස. පද්ධති භාවිත පහසුව පරිමාණය (System Usability Scale - SUS)

Brooke (1996). සෑම ප්‍රකාශයකටම එක් කොටුවක් ලකුණු කරන්න. 1 = දැඩි ලෙස එකඟ නොවේ, 5 = දැඩි ලෙස එකඟ වේ. "පද්ධතිය" යනු මෙම සැසියේදී ඔබ භාවිත කළ AI තීරණ-සහාය පද්ධතියයි.

#	ප්‍රකාශය	1	2	3	4	5
1	මෙම පද්ධතිය නිතර භාවිත කිරීමට මා කැමති වනු ඇතැයි මම සිතමි.	☐	☐	☐	☐	☐
2	මෙම පද්ධතිය අනවශ්‍ය ලෙස සංකීර්ණ බව මට පෙනී ගියේය.	☐	☐	☐	☐	☐
3	මෙම පද්ධතිය භාවිතයට පහසු යැයි මම සිතුවෙමි.	☐	☐	☐	☐	☐
4	මෙම පද්ධතිය භාවිත කිරීමට හැකි වීමට තාක්ෂණික පුද්ගලයෙකුගේ සහාය මට අවශ්‍ය වනු ඇතැයි මම සිතමි.	☐	☐	☐	☐	☐
5	මෙම පද්ධතියේ විවිධ කාර්යයන් හොඳින් ඒකාබද්ධ කර ඇති බව මට පෙනී ගියේය.	☐	☐	☐	☐	☐
6	මෙම පද්ධතියේ අනනුකූලතා ඕනෑවට වඩා ඇතැයි මම සිතුවෙමි.	☐	☐	☐	☐	☐
7	බොහෝ දෙනා මෙම පද්ධතිය භාවිත කිරීමට ඉතා ඉක්මනින් ඉගෙන ගනු ඇතැයි මම සිතමි.	☐	☐	☐	☐	☐
8	මෙම පද්ධතිය භාවිත කිරීම ඉතා අපහසු සහ අවුල් සහගත බව මට පෙනී	☐	☐	☐	☐	☐

#	**ප්‍රකාශය**	**1**	**2**	**3**	**4**	**5**
	ගියේය.					
9	මෙම පද්ධතිය භාවිත කිරීමේදී මට ඉතා විශ්වාසයක් දැනුණි.	☐	☐	☐	☐	☐
10	මෙම පද්ධතිය භාවිත කිරීම ආරම්භ කිරීමට පෙර මට බොහෝ දේ ඉගෙන ගැනීමට සිදු විය.	☐	☐	☐	☐	☐

3 වන කොටස. AI තීරණ-සහාය පද්ධතිය කෙරෙහි විශ්වාසය

Jian, Bisantz සහ Drury (2000) ඇසුරින් මෙම අධ්‍යයනය සඳහා සකස් කළ ප්‍රකාශ. 1 = දැඩි ලෙස එකඟ නොවේ, 5 = දැඩි ලෙස එකඟ වේ.

#	**ප්‍රකාශය**	**1**	**2**	**3**	**4**	**5**
1	පද්ධතිය මා වෙත රෝගී අවස්ථාවක් යොමු නොකළ විට, එහි ප්‍රතිදාන මත මට විශ්වාසය තැබිය හැකි විය.	☐	☐	☐	☐	☐
2	පද්ධතිය වාර්තා කළ විශ්වාස මට්ටම එහි ප්‍රතිදාන කොතරම් නිවැරදිද යන්නට ගැළපුණි.	☐	☐	☐	☐	☐
3	සම්පූර්ණ සමාලෝචනය සඳහා පද්ධතිය මා වෙත යොමු කළේ නිවැරදි රෝගී අවස්ථාය.	☐	☐	☐	☐	☐
4	මා වෙත යොමු කළ යුතු යැයි මා සිතන රෝගී අවස්ථා පද්ධතිය මග හැරියේය.	☐	☐	☐	☐	☐
5	ප්‍රතිදානය නිවැරදිද යන්න විනිශ්චය කිරීමට පැහැදිලි කිරීම් (ගෙඩි මායිම්, තාප සිතියම්, සාක්ෂි විශ්ලේෂණය) මට උපකාරී විය.	☐	☐	☐	☐	☐
6	පද්ධතියේ ප්‍රතිදාන පිළිබඳව මට සැකයක් තිබුණි.	☐	☐	☐	☐	☐
7	පද්ධතිය භාවිත කිරීමෙන් තීරණය සඳහා මගේම වගාකීම අඩු නොවීය.	☐	☐	☐	☐	☐
8	සාමාන්‍ය සායනික කටයුතුවලදී දෙවන පාඨකයෙකු ලෙස මෙම පද්ධතිය භාවිත කිරීමට මට පහසුවක් දැනේ.	☐	☐	☐	☐	☐
9	අඩු අත්දැකීම් ඇති සායනික වෛද්‍යවරයෙකු වැරදි තීරණයකට යොමු කිරීමට මෙම පද්ධතියට හැකිය.	☐	☐	☐	☐	☐
10	සමස්තයක් ලෙස, මම මෙම පද්ධතිය විශ්වාස කරමි.	☐	☐	☐	☐	☐

මෙම සැසිය පිළිබඳ ඕනෑම අදහසක්: _____

_____

ඇමුණුම 3E: අර්ධ-ව්‍යුහගත සම්මුඛ සාකච්ඡා මාර්ගෝපදේශය

1 වන අනුවාදය, 09.11.2026 (සිංහල)

අධ්‍යයනය: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

සටහන: මෙය ඉංග්‍රීසි මුල් පිටපතේ (ඇමුණුම 3) පරිවර්තනයකි. අර්ථයේ වෙනසක් ඇත්නම්, ඉංග්‍රීසි පිටපත වලංගු වේ.

පාඨක අධ්‍යයනය සම්පූර්ණ කළ, සම්මුඛ සාකච්ඡාව සහ ශ්‍රව්‍ය පටිගත කිරීම සඳහා වෙනම කැමැත්ත ලබා දුන් සායනික වෛද්‍යවරුන් සමඟ පවත්වන, අනිවාර්ය නොවන සම්මුඛ සාකච්ඡාවකි. සායනික වෛද්‍යවරයා තෝරා ගන්නා වේලාවක, පෞද්ගලිකව හෝ මාර්ගගතව, මිනිත්තු 30-45 ක් පමණ. සම්මුඛ සාකච්ඡාව සායනික වෛද්‍යවරයා කැමති භාෂාවෙන් (සිංහල, දෙමළ හෝ ඉංග්‍රීසි) පවත්වනු ලැබේ. පහත ප්‍රශ්න මාර්ගෝපදේශයක් පමණි; අමතර ප්‍රශ්න පිළිතුරු මත රඳා පවතී. සම්මුඛ සාකච්ඡාකරු සායනික වෛද්‍යවරයාගේ කියවීමේ කාර්ය සාධනය ගැන අදහස් දක්වන්නේ නැත.

ආරම්භය (මිනිත්තු 3 ක් පමණ)

- සහභාගිවන්නාට ස්තුති කරන්න; ඔවුන් තොරතුරු පත්‍රිකාව කියවා සම්මුඛ සාකච්ඡා කැමැත්ත අයිතමය අත්සන් කර ඇති බව තහවුරු කරන්න.
- ශ්‍රව්‍ය පටිගත කිරීමට කැමැත්ත තහවුරු කර, වාචික එකඟතාවයෙන් පසුව පමණක් පටිගත කිරීම ආරම්භ කරන්න. ඕනෑම ප්‍රශ්නයක් මඟ හැරීමට හෝ ඕනෑම වේලාවක නතර කිරීමට හැකි බව මතක් කරන්න.
- උපුටා දැක්වීම් නිර්නාමික කරන බවත්, ලිඛිත පිටපත් කිරීමෙන් පසු පටිගත කිරීම මකා දමන බවත් පැහැදිලි කරන්න.

1. වර්තමාන භාවිතය

- නව මොළයේ ගෙඩි රෝගී අවස්ථාවක් ඔබ වෙත පැමිණෙන ආකාරය සහ ඔබ තීරණයකට එළඹෙන ආකාරය විස්තර කළ හැකිද?
- එම ක්‍රියාවලියේ ප්‍රමාද හෝ අවිනිශ්චිතතා සාමාන්‍යයෙන් ඇති වන්නේ කොතැනද?

2. පද්ධතිය පිළිබඳ අත්දැකීම

- AI සහාය නොමැතිව සමාලෝචනය කිරීමට සාපේක්ෂව, AI සහාය සහිතව රෝගී අවස්ථා සමාලෝචනය කිරීම කෙසේ වීද?
- වඩාත්ම හා අවමව ප්‍රයෝජනවත් වූ ප්‍රතිදාන මොනවාද (ගෙඩි මායිම්, වර්ගය සහ ශ්‍රේණිය, සාක්ෂි විශ්ලේෂණය, ශල්‍ය මාර්ගය)?

3. විශ්වාස මට්ටම සහ යොමු කිරීම (escalation)

- පද්ධතිය පෙන්වූ විශ්වාස මට්ටම ඔබ අර්ථකථනය කළේ කෙසේද?
- එය නිවැරදි රෝගී අවස්ථා ඔබ වෙත යොමු කළාද? යොමු නොකළ යුතු දෙයක් යොමු කළ, හෝ යොමු කළ යුතු දෙයක් යොමු නොකළ අවස්ථාවක් ඔබට මතකද?
- ඉක්මන් අනුමැතිය සහ සම්පූර්ණ සමාලෝචනය අතර සීමාව ඔබ තබන්නේ කොතැනද?

4. විශ්වාසය සහ වගකීම

- කිසියම් ප්‍රතිදානයක් විශ්වාස කිරීමට හෝ නොකිරීමට ඔබට හේතු වූයේ කුමක්ද?
- AI සහාය ඇතිව ගන්නා තීරණයක් සඳහා වගකිව යුත්තේ කවුරුන්දැයි ඔබට හැඟේද?
- අඩු අත්දැකීම් ඇති සායනික වෛද්‍යවරුන්ට මෙම පද්ධතිය බලපානු ඇත්තේ කෙසේදැයි ඔබ සිතන්නේද?

5. වැඩ බර

- රෝගී අවස්ථා සඳහා ගත වූ කාලය හෝ උත්සාහය පද්ධතිය වෙනස් කළාද? කුමන ආකාරයෙන්ද?

– AI පරීක්ෂා කිරීම නිසා වැඩ වැඩි වූ අවස්ථා තිබුණාද?

6. ශල්‍යකර්ම සැලසුම් සංකල්ප සාධනය (ස්නායු ශල්‍ය වෛද්‍යවරුන්)

– ශල්‍ය ප්‍රවේශයක් ගැන සිතීමට ත්‍රිමාණ ආකෘතිය සහ යෝජිත මාර්ගය කොතරම් ප්‍රයෝජනවත් වූයේද?

– සැලසුම් කිරීමේදී එය සලකා බැලීමට පෙර එය පෙන්විය යුත්තේ කුමක්ද?

7. ශ්‍රී ලංකාවේ ප්‍රායෝගික භාවිතය

– ඔබේ ඒකකයේ මෙවැනි පද්ධතියක් භාවිත කිරීමට බාධා වන හෝ උපකාරී වන්නේ කුමක්ද?

– ඔබට අවශ්‍ය ආරක්ෂණ පියවර මොනවාද (පුහුණුව, විගණනය, පාලනය, දත්ත ආරක්ෂාව)?

අවසානය (මිනිත්තු 2 ක් පමණ)

– අප සාකච්ඡා නොකළ, නමුත් වැදගත් යැයි ඔබ සිතන කිසිවක් තිබේද?

– පටිගත කිරීම නවත්වන්න. සහභාගිවන්නාට ස්තුති කර, සොයාගැනීම්වල සාරාංශය ලබා ගත හැකි ආකාරය පැහැදිලි කරන්න.

பின்னிணைப்பு 3B: மருத்துவர் பின்னணிப் படிவம்

பதிப்பு *1, 09.11.2026 (தமிழ்)*

ஆய்வு: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

குறிப்பு: இது ஆங்கில மூலப் பிரதியின் (பின்னிணைப்பு 3) மொழிபெயர்ப்பாகும். பொருளில் வேறுபாடு இருப்பின், ஆங்கிலப் பிரதியே செல்லுபடியாகும்.

சம்மதப் படிவத்தில் கையொப்பமிட்ட பின்னர் ஒருமுறை பூர்த்தி செய்யவும். உங்கள் பதில்கள் உங்கள் வாசிப்பாளர் குறியீட்டின் கீழ் சேமிக்கப்பட்டு, குழு மொத்தங்களாக மட்டுமே அறிவிக்கப்படும்.

வாசிப்பாளர் குறியீடு	R-____
சிறப்புத் துறை	☐ நரம்பியல் அறுவை சிகிச்சை ☐ கதிரியக்கவியல் ☐ திசு நோயியல் (Histopathology) ☐ வேறு: _____
தற்போதைய பதவி	☐ ஆலோசகர் (Consultant) ☐ சிரேஷ்ட பதிவாளர் ☐ பதிவாளர் ☐ வேறு: _____
சிறப்புத் துறையில் ஆண்டுகள் (சிறப்புப் பயிற்சி தொடங்கியதிலிருந்து)	☐ <5 ☐ 5-9 ☐ 10-19 ☐ 20+
மாதமொன்றுக்கு நீங்கள் பார்க்கும் அல்லது அறிக்கையிடும் மூளைக் கட்டி நோயாளர்கள் (தோராயமாக)	☐ <5 ☐ 5-14 ☐ 15-29 ☐ 30+
மருத்துவப் பணியில் AI கருவிகளின் முன் பயன்பாடு	☐ ஒருபோதும் இல்லை ☐ ஓரிரு முறை ☐ அவ்வப்போது ☐ தொடர்ச்சியாக
உங்கள் துறையில் AI பற்றிய பொதுவான மனப்பாங்கு	☐ மிக எதிர்மறை ☐ எதிர்மறை ☐ நடுநிலை ☐ நேர்மறை ☐ மிக நேர்மறை

பின்னிணைப்பு 3C: வாசிப்பாளர் ஆய்வு - நோயாளர் சந்தர்ப்பப் பதிவு

பதிப்பு *1, 09.11.2026 (தமிழ்)*

ஆய்வு: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

குறிப்பு: இது ஆங்கில மூலப் பிரதியின் (பின்னிணைப்பு 3) மொழிபெயர்ப்பாகும். பொருளில் வேறுபாடு இருப்பின், ஆங்கிலப் பிரதியே செல்லுபடியாகும்.

வாசிப்பாளர் ஆய்வு நடைபெறும் முறை. ஒவ்வொரு மருத்துவரும் அடையாளம் நீக்கப்பட்ட அதே நோயாளர் சந்தர்ப்பத் தொகுப்பை இருமுறை மீளாய்வு செய்வார்: ஒருமுறை AI ஆதரவின்றியும் ஒருமுறை AI ஆதரவுடனும், குறைந்தது நான்கு வார இடைவெளியுடன் (washout) தனித்தனி அமர்வுகளில். வாசிப்பாளர்களில் பாதிப் பேர் ஆதரவற்ற நிலையிலும் மீதிப் பேர் ஆதரவுள்ள நிலையிலும் தொடங்குவர். ஒவ்வொரு அமர்விலும் சந்தர்ப்பங்களின் வரிசை சீரற்ற முறையில் அமைக்கப்படும். வாசிப்பு நேரம் ஆய்வு மென்பொருளால் தானாகப் பதிவுசெய்யப்படும்; இப்பதிவு வாசிப்பாளரின் முடிவைப் பதிவுசெய்கிறது. ஒரு சந்தர்ப்பத்திற்கு ஒரு வரிசை.

வாசிப்பாளர் குறியீடு	R-____	அமர்வு இல.	____	நிலை	☐ AI ஆதரவின்றி ☐ AI ஆதரவுடன்

நிரல்கள்: Dx = மிகவும் சாத்தியமான கட்டி வகை (WHO CNS5 குடும்பம்); Gr = CNS WHO தரம் 1-4; Conf = உங்கள் நம்பிக்கை, 1 (மிகவும் உறுதியில்லை) முதல் 5 (மிகவும் உறுதி) வரை. AI ஆதரவுடனான அமர்வுகளுக்கு மட்டும்: AI = AI வெளியீட்டை நீங்கள் என்ன செய்தீர்கள் (A = ஏற்றுக்கொண்டேன், M = திருத்தினேன், R = நிராகரித்தேன்); Esc = AI உங்களிடம் பரிந்துரைத்த சந்தர்ப்பங்களுக்கு, அப்பரிந்துரை பொருத்தமானதா? (Y/N); பரிந்துரைக்காத சந்தர்ப்பங்களுக்கு, அது பரிந்துரைத்திருக்க வேண்டுமா? (Y/N).

#	சந்தர்ப்ப இல.	Dx (வகை)	Gr	Conf	AI (A/M/R)	Esc (Y/N)	குறிப்பு / மாற்றத்திற்கான காரணம்
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							

#	சந்தர்ப்ப இல.	Dx (வகை)	Gr	Conf	AI (A/M/R)	Esc (Y/N)	குறிப்பு / மாற்றத்திற்கான காரணம்
15							

பின்னிணைப்பு 3D: அமர்வுக்குப் பிந்தைய வினாக்கொத்து

பதிப்பு *1, 09.11.2026 (தமிழ்)*

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குறிப்பு: இது ஆங்கில மூலப் பிரதியின் (பின்னிணைப்பு 3) மொழிபெயர்ப்பாகும். பொருளில் வேறுபாடு இருப்பின், ஆங்கிலப் பிரதியே செல்லுபடியாகும்.

வாசிப்பாளர் குறியீடு R-____ அமர்வு ____ நிலை: ☐ AI ஆதரவின்றி ☐ AI ஆதரவுடன் திகதி _____

பகுதி 1 (வேலைப்பளு) ஒவ்வொரு அமர்வின் பின்னரும் பூர்த்தி செய்யப்படும். பகுதிகள் 2 மற்றும் 3 AI ஆதரவுடனான அமர்வுகளின் பின்னர் மட்டுமே பூர்த்தி செய்யப்படும். சுமார் 10-15 நிமிடங்கள்.

பகுதி 1. NASA பணிச்சுமைச் சுட்டெண் (NASA-TLX, raw)

Hart மற்றும் Staveland (1988). ஒவ்வொரு அளவுகோலுக்கும், நீங்கள் இப்போது முடித்த அமர்வு எவ்வாறு இருந்தது என்பதைக் காட்ட 21 நிலைகளில் ஒன்றில் பெருக்கல் குறி (X) இடவும்.

அளவுகோல்	கேள்வி	மிகக் குறைவு ← → மிக அதிகம்
மனக் கோரிக்கை	பணி மனரீதியாக எவ்வளவு கோருவதாக இருந்தது?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
உடல் கோரிக்கை	பணி உடல்ரீதியாக எவ்வளவு கோருவதாக இருந்தது?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
நேரக் கோரிக்கை	பணியின் வேகம் எவ்வளவு அவசரமாக அல்லது விரைவாக இருந்தது?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
செயல்திறன்	உங்களிடம் கேட்கப்பட்டதை நிறைவேற்றுவதில் நீங்கள் எவ்வளவு வெற்றிகரமாக இருந்தீர்கள்? (இடம் = மிகச் சரியானது, வலம் = தோல்வி)	மிகச் சரியானது தோல்வி |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
முயற்சி	உங்கள் செயல்திறன் மட்டத்தை அடைய நீங்கள் எவ்வளவு கடினமாக உழைக்க வேண்டியிருந்தது?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
விரக்தி	நீங்கள் எவ்வளவு பாதுகாப்பின்மை, மனச்சோர்வு, எரிச்சல், மன அழுத்தம், சலிப்பு அடைந்தீர்கள்?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|

பகுதி 2. முறைமைப் பயன்பாட்டுத்தன்மை அளவுகோல் (System Usability Scale - SUS)

Brooke (1996). ஒவ்வொரு கூற்றுக்கும் ஒரு பெட்டியைக் குறிக்கவும். 1 = கடுமையாக உடன்படவில்லை, 5 = முழுமையாக உடன்படுகிறேன். "முறைமை" என்பது இந்த அமர்வில் நீங்கள் பயன்படுத்திய AI முடிவு-ஆதரவு முறைமையைக் குறிக்கும்.

#	கூற்று	1	2	3	4	5
1	இந்த முறைமையை அடிக்கடி பயன்படுத்த விரும்புவேன் என நினைக்கிறேன்.	☐	☐	☐	☐	☐
2	இந்த முறைமை தேவையற்ற அளவில் சிக்கலானதாக	☐	☐	☐	☐	☐

#	கூற்று	1	2	3	4	5
	இருப்பதைக் கண்டேன்.					
3	இந்த முறைமை பயன்படுத்த எளிதானது என நினைத்தேன்.	☐	☐	☐	☐	☐
4	இந்த முறைமையைப் பயன்படுத்த ஒரு தொழில்நுட்ப நபரின் உதவி எனக்குத் தேவைப்படும் என நினைக்கிறேன்.	☐	☐	☐	☐	☐
5	இந்த முறைமையின் பல்வேறு செயற்பாடுகள் நன்கு ஒருங்கிணைக்கப்பட்டிருப்பதைக் கண்டேன்.	☐	☐	☐	☐	☐
6	இந்த முறைமையில் அதிகளவு முரண்பாடுகள் இருப்பதாக நினைத்தேன்.	☐	☐	☐	☐	☐
7	பெரும்பாலானோர் இந்த முறைமையைப் பயன்படுத்த மிக விரைவாகக் கற்றுக்கொள்வார்கள் என நினைக்கிறேன்.	☐	☐	☐	☐	☐
8	இந்த முறைமையைப் பயன்படுத்துவது மிகவும் சிரமமானதாக இருப்பதைக் கண்டேன்.	☐	☐	☐	☐	☐
9	இந்த முறைமையைப் பயன்படுத்தும்போது மிகுந்த நம்பிக்கையுடன் உணர்ந்தேன்.	☐	☐	☐	☐	☐
10	இந்த முறைமையைப் பயன்படுத்தத் தொடங்குவதற்கு முன் நான் நிறைய விடயங்களைக் கற்றுக்கொள்ள வேண்டியிருந்தது.	☐	☐	☐	☐	☐

பகுதி 3. AI முடிவு-ஆதரவு முறைமை மீதான நம்பிக்கை

Jian, Bisantz மற்றும் Drury (2000) அடிப்படையில் இந்த ஆய்வுக்காக அமைக்கப்பட்ட கூற்றுகள். 1 = கடுமையாக உடன்படவில்லை, 5 = முழுமையாக உடன்படுகிறேன்.

#	கூற்று	1	2	3	4	5
1	முறைமை ஒரு சந்தர்ப்பத்தை என்னிடம் பரிந்துரைக்காதபோது, அதன் வெளியீடுகளில் நான் நம்பிக்கை வைக்க முடிந்தது.	☐	☐	☐	☐	☐
2	முறைமை தெரிவித்த நம்பிக்கை அளவு, அதன் வெளியீடுகள் எவ்வளவு சரியானவை என்பதுடன் பொருந்தியது.	☐	☐	☐	☐	☐
3	முழு மீளாய்வுக்காக முறைமை சரியான சந்தர்ப்பங்களையே என்னிடம் பரிந்துரைத்தது.	☐	☐	☐	☐	☐
4	என்னிடம் பரிந்துரைத்திருக்க வேண்டும் என நான் கருதும் சந்தர்ப்பங்களை முறைமை தவறவிட்டது.	☐	☐	☐	☐	☐
5	வெளியீடு சரியானதா என மதிப்பிட விளக்கங்கள் (கட்டி எல்லைகள், வெப்ப வரைபடங்கள், சான்றுப் பகுப்பாய்வு) எனக்கு உதவின.	☐	☐	☐	☐	☐
6	முறைமையின் வெளியீடுகள் குறித்து எனக்குச் சந்தேகம் இருந்தது.	☐	☐	☐	☐	☐
7	முறைமையைப் பயன்படுத்தியமை, முடிவுக்கான எனது சொந்தப் பொறுப்பைக் குறைக்கவில்லை.	☐	☐	☐	☐	☐
8	வழமையான மருத்துவப் பணியில் இரண்டாவது வாசிப்பாளராக இந்த முறைமையைப் பயன்படுத்துவதில் எனக்குத் தயக்கமில்லை.	☐	☐	☐	☐	☐
9	அனுபவம் குறைந்த மருத்துவர் ஒருவரை இந்த முறைமை தவறான முடிவுக்கு இட்டுச் செல்லக்கூடும்.	☐	☐	☐	☐	☐
10	ஒட்டுமொத்தமாக, நான் இந்த முறைமையை நம்புகிறேன்.	☐	☐	☐	☐	☐

இந்த அமர்வு பற்றிய ஏதேனும் குறிப்புகள்:

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_____

பின்னிணைப்பு 3E: பகுதி-கட்டமைக்கப்பட்ட நேர்காணல் வழிகாட்டி

பதிப்பு *1, 09.11.2026 (*தமிழ்*)*

ஆய்வு: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

குறிப்பு: இது ஆங்கில மூலப் பிரதியின் (பின்னிணைப்பு 3) மொழிபெயர்ப்பாகும். பொருளில் வேறுபாடு இருப்பின், ஆங்கிலப் பிரதியே செல்லுபடியாகும்.

வாசிப்பாளர் ஆய்வை நிறைவுசெய்து, நேர்காணலுக்கும் ஒலிப்பதிவுக்கும் தனியான சம்மதம் வழங்கிய மருத்துவர்களுடனான விருப்பத்தேர்வு நேர்காணல். மருத்துவர் தெரிவுசெய்யும் நேரத்தில், நேரடியாகவோ இணையவழியாகவோ, சுமார் 30-45 நிமிடங்கள். நேர்காணல் மருத்துவர் விரும்பும் மொழியில் (சிங்களம், தமிழ் அல்லது ஆங்கிலம்) நடத்தப்படும். கீழுள்ள கேள்விகள் ஒரு வழிகாட்டி மட்டுமே; தொடர் கேள்விகள் பதில்களைப் பொறுத்தவை. நேர்காணல் செய்பவர் மருத்துவரின் சொந்த வாசிப்புச் செயல்திறன் பற்றிக் கருத்துத் தெரிவிக்கமாட்டார்.

ஆரம்பம் (சுமார் 3 நிமிடங்கள்)

- பங்கேற்பாளருக்கு நன்றி கூறவும்; அவர் தகவல் தாளைப் படித்து, நேர்காணல் சம்மத உருப்படியில் கையொப்பமிட்டுள்ளார் என்பதை உறுதிப்படுத்தவும்.
- ஒலிப்பதிவுக்கான சம்மதத்தை உறுதிப்படுத்தி, வாய்மொழி இணக்கத்தின் பின்னரே பதிவைத் தொடங்கவும். எந்தக் கேள்வியையும் தவிர்க்கலாம் அல்லது எந்நேரத்திலும் நிறுத்தலாம் என நினைவூட்டவும்.
- மேற்கோள்கள் அநாமதேயமாக்கப்படும் என்பதையும், எழுத்துப்படியாக்கத்தின் பின் ஒலிப்பதிவு அழிக்கப்படும் என்பதையும் விளக்கவும்.

1. தற்போதைய நடைமுறை

- புதிய மூளைக் கட்டி நோயாளர் ஒருவர் உங்களிடம் எவ்வாறு வருகிறார், நீங்கள் எவ்வாறு ஒரு முடிவுக்கு வருகிறீர்கள் என்பதை விளக்க முடியுமா?
- அச்செயல்முறையில் தாமதங்கள் அல்லது உறுதியின்மை பொதுவாக எங்கு ஏற்படுகின்றன?

2. முறைமை பற்றிய அனுபவம்

- AI ஆதரவின்றி மீளாய்வு செய்ததுடன் ஒப்பிடுகையில், AI ஆதரவுடன் சந்தர்ப்பங்களை மீளாய்வு செய்தது எவ்வாறு இருந்தது?
- எந்த வெளியீடுகள் மிகவும் பயனுள்ளதாகவும் குறைவாகப் பயனுள்ளதாகவும் இருந்தன (கட்டி எல்லைகள், வகை மற்றும் தரம், சான்றுப் பகுப்பாய்வு, அறுவை சிகிச்சைப் பாதை)?

3. நம்பிக்கை அளவும் பரிந்துரையும் (escalation)

- முறைமை காட்டிய நம்பிக்கை அளவை நீங்கள் எவ்வாறு விளங்கிக்கொண்டீர்கள்?
- அது சரியான சந்தர்ப்பங்களை உங்களிடம் பரிந்துரைத்ததா? பரிந்துரைக்கக் கூடாததைப் பரிந்துரைத்த, அல்லது பரிந்துரைக்க வேண்டியதைப் பரிந்துரைக்காத சந்தர்ப்பம் ஏதேனும் நினைவிருக்கிறதா?
- விரைவான அங்கீகாரத்திற்கும் முழு மீளாய்வுக்கும் இடையிலான எல்லையை நீங்கள் எங்கு வைப்பீர்கள்?

4. நம்பிக்கையும் பொறுப்பும்

- ஒரு குறிப்பிட்ட வெளியீட்டை நம்பவோ நம்பாமலோ செய்தது எது?

- AI ஆதரவுடன் எடுக்கப்படும் முடிவுக்கு யார் பொறுப்பு என நீங்கள் கருதுகிறீர்கள்?
- அனுபவம் குறைந்த மருத்துவர்களை இந்த முறைமை எவ்வாறு பாதிக்கும் என நினைக்கிறீர்கள்?

5. வேலைப்பளு

- சந்தர்ப்பங்களுக்கு எடுத்த நேரத்தையோ முயற்சியையோ முறைமை மாற்றியதா? எவ்வகையில்?
- AI-ஐச் சரிபார்ப்பது வேலையை அதிகரித்த சந்தர்ப்பங்கள் இருந்தனவா?

6. அறுவை சிகிச்சைத் திட்டமிடல் கருத்துச் சான்று (நரம்பியல் அறுவை சிகிச்சை நிபுணர்கள்)

- ஓர் அணுகுமுறை பற்றிச் சிந்திப்பதற்கு முப்பரிமாண மாதிரியும் முன்மொழியப்பட்ட பாதையும் எவ்வளவு பயனுள்ளதாக இருந்தன?
- திட்டமிடலில் அதைக் கருத்திற்கொள்வதற்கு முன் அது எதைக் காட்ட வேண்டும்?

7. இலங்கையில் நடைமுறைப் பயன்பாடு

- உங்கள் பிரிவில் இத்தகைய முறைமை பயன்படுத்தப்படுவதைத் தடுப்பது அல்லது உதவுவது எது?
- நீங்கள் எதிர்பார்க்கும் பாதுகாப்பு ஏற்பாடுகள் யாவை (பயிற்சி, கணக்காய்வு, நிர்வாகம், தரவுப் பாதுகாப்பு)?

நிறைவு (சுமார் 2 நிமிடங்கள்)

- நாம் கலந்துரையாடாத, ஆனால் முக்கியமானது என நீங்கள் கருதும் ஏதேனும் உள்ளதா?
- பதிவை நிறுத்தவும். பங்கேற்பாளருக்கு நன்றி கூறி, கண்டறிவுகளின் சுருக்கத்தை எவ்வாறு பெறலாம் என்பதை விளக்கவும்.

Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

Doctor of Philosophy (PhD) in Computer Science
Department of Computer Science
University of Sri Jayewardenepura, Sri Lanka

PhD Research Proposal

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Field	Computer Science, Medical Image Analysis & Applied AI Systems
Mode / Duration	Full-time
Date	1 October 2026

PhD Application – Declaration and Approval

Declaration and Approval

Student Declaration

This is my research proposal. I declare that the work presented in this proposal is my own and that it has been prepared under the supervision of the supervisors named below.

Name	KHV Sameera
Signature	
Date	2026.10.01

Supervisor Approval

Each supervisor named below confirms that they agree to supervise this research, that they are aware of this proposal, and that they have verified and approved its content.

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Abstract

Brain tumours are among the most demanding cancers to manage. Diagnosis depends on expert reading of magnetic resonance imaging (MRI) and of tissue taken at biopsy and examined under a microscope; treatment is time-critical; surgery must remove the tumour while sparing the parts of the brain that control speech, movement and vision; and follow-up runs for years. In Sri Lanka this work is carried by a small number of specialist units, and, unlike breast, lung or oral cancer, there is no national brain-tumour registry linking scans, laboratory results, clinical history, treatment and outcomes in one maintained resource.

The primary scientific contribution is an uncertainty-aware, human-in-the-loop multi-agent artificial-intelligence (AI) framework for the clinical decision workflow. Instead of one large model, the system is a team of specialised software agents coordinated by an orchestrator. The central research idea is that every agent must measure how confident it is, in a form that can be trusted numerically, and on that basis decide whether to act on its own or escalate to a clinician, who always remains the final validator. This turns the familiar principle that AI should carry routine load while doctors keep authority into a mechanism that can be specified, tuned and measured.

The framework is built and validated inside Sri Lanka's first National Brain Tumour Patient Registry. Its agents span the pathway: data-efficient MRI analysis for small and inconsistent local datasets; an agent that reads digitised microscope slides of biopsy tissue; an agent that combines imaging and tissue evidence to support tumour type, severity grade and treatment recommendation; and a pre-operative surgical-planning agent, built as a proof of concept, that proposes a candidate low-damage approach with an explicit map of structures at risk. A privacy-preserving federated-learning protocol trains these models across hospitals without moving raw patient data. Throughout, the AI plans, predicts and flags; clinicians decide and act, and the surgeon always performs and validates the operation.

1. Introduction

1.1 Background of the Research

Artificial intelligence (AI) now supports diagnosis, treatment planning and clinical documentation at a scale no human reader could match [26,82], but the same capability makes unsupervised deployment hazardous: opacity, dataset bias, unclear accountability and the difficulty of detecting silent failure have all constrained adoption [56,58]. Two findings dominate. Performance degrades under dataset shift: a pneumonia-detection model generalised poorly outside its training hospitals because it had latched onto site-specific artefacts rather than disease [54,55]. And a model that reports a confident answer gives the clinician no signal that it has left the region in which it is competent [28,31].

This has made human-in-the-loop (HITL) AI the dominant design philosophy for safety-critical medical AI. HITL interleaves human judgement and machine computation rather than substituting one for the other, inserting control points at consequential moments so a clinician can validate, refine or override a recommendation using findings, comorbidity and patient preference no model can see. HITL configurations outperform both fully automated AI and clinician-only workflows on accuracy, safety, efficiency and satisfaction, because the two participants fail in uncorrelated ways [25]. Crucially, HITL is usually asserted as a principle rather than implemented as a mechanism; a rare exception coupled retinal analysis to an explicit referral decision [27]. The open engineering question, and the one this research addresses, is not whether a human should be in the loop, but when, on which cases, and on what calibrated evidence.

1.1.1 The brain-tumour pathway and the state of computational support

The pathway is long, information-dense and unforgiving. It runs from non-specific neurological symptoms and a scan, through multi-sequence MRI, biopsy and examination of the tissue under a microscope, laboratory tests on that tissue, a multidisciplinary decision, surgery and follow-on therapy, to years of surveillance scanning. Since 2021 the World Health Organization classification has been integrated, so the diagnosis combines tissue appearance with laboratory results [36], making the task inherently multi-source. Brain tumours carry disproportionate mortality and affect working-age adults and children [37]. Surgery must balance removing as much tumour as possible against preserving the regions controlling speech, movement and vision: how much is removed is repeatedly associated with survival [38,39], while removing too much healthy tissue causes permanent loss of function. The pathway is also dependent: an error in outlining propagates into grading, grading into treatment selection, and treatment selection into outcome.

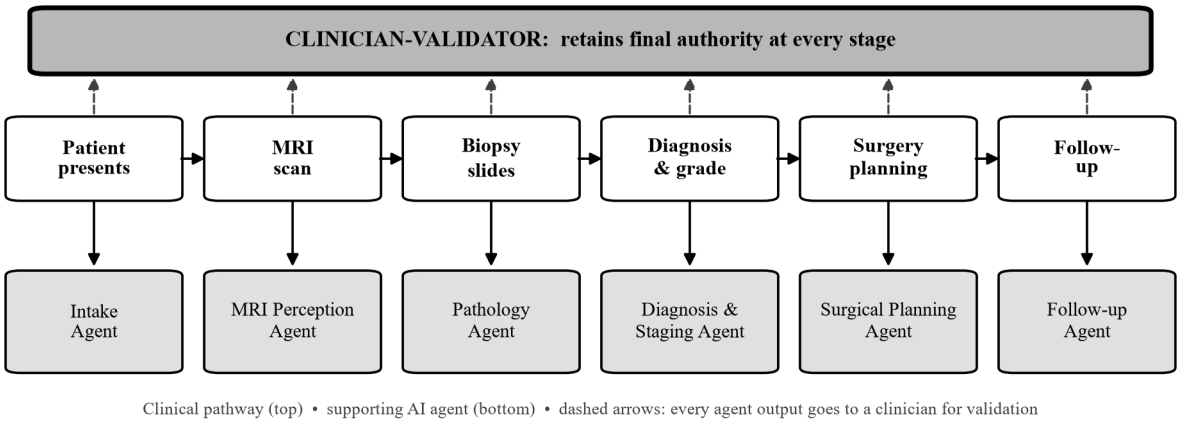


Figure 1.1: The brain-tumour care pathway, the specialised AI agent that supports each stage, and the clinician who retains final authority throughout.

Computational support has advanced rapidly at the level of individual sub-tasks. Deep learning dominates detection, outlining and classification on MRI, benchmarked on BraTS [40,41,42], with reviews documenting substantial gains [1,2,11], nnU-Net establishing strong baselines [43] and good outlining attainable with fewer sequences [12]. Computational pathology reaches clinical-grade analysis of whole digitised slides [44,45,46], extended to brain-tumour classification and grading [18,19,20,21]; combining tissue appearance with laboratory markers outperforms either alone [47], and near-real-time diagnosis during surgery has been demonstrated [48]. Yet individual models solve individual tasks: no accepted architecture assembles them into a supervised workflow with defined accountability. Two further shifts frame this work: agentic AI, whose safety, evaluation, traceability and oversight remain unresolved [3,4,5,60], and federated learning [49], which lets institutions train a shared model without disclosing raw data [50,51,52], proven across 71 sites on six continents [53] and where site data differ markedly [6,7,8,13].

1.1.2 The Sri Lankan context and the case for a national registry

GLOBOCAN 2024 estimates approximately 700 new brain and nervous-system cancer cases and around 587 deaths annually in Sri Lanka, with a five-year prevalence of roughly 2,500 [9]. Deaths are high relative to new cases and care is concentrated in a few tertiary units served by a limited pool of specialists. A National Cancer Registry exists for incidence and mortality reporting [10], but there is no imaging- and tissue-linked brain-tumour registry; patients are tracked in fragmented, institution-specific records. Clinically this impedes continuity of care, multidisciplinary review and outcome auditing. Scientifically it makes locally valid AI almost impossible to build, because models trained on foreign cohorts do not transfer reliably to Sri Lankan scanners, protocols, laboratory practice or case mix [2,54,55]. The project therefore separates a platform layer, the registry itself, which is the real-world deliverable giving the research authentic data and users; and a research layer, the focus of this PhD, which is the design, formalisation and validation of the AI framework that makes such a platform safe, data-efficient and privacy-preserving.

1.2 Problem Statement and Research Gap

1.2.1 Why accuracy alone is not enough

A model that is 95% accurate is not automatically safe: the danger lies in the 5% of confident mistakes, and in the absence of a mechanism by which the system can recognise the limits of its own competence. Modern deep networks are systematically over-confident, and their raw probabilities are poor estimates of correctness unless explicitly calibrated [31], a deficiency that worsens precisely when a model moves to a new hospital [71]. The issue is not a shortage of accurate models but the absence of a governing mechanism deciding, case by case, whether an output is fit to be acted upon. Where specialist review time is the scarcest resource, that mechanism also determines whether AI reduces clinician workload or merely relocates it.

1.2.2 Identified research gaps

No principled policy for when to defer. Clinical AI is deployed either as a fully manual aid, in which every output is reviewed, or as unconditional automation, in which none is. The primitives exist (selective classification [33], learning to defer [34,35], calibration [31], Bayesian and ensemble uncertainty [29,30,70] and conformal guarantees [32,75]), but were built for single models making single decisions, not assembled into a policy accounting for asymmetric error cost, finite expert review time [76], or a chain of dependent sub-decisions [4,28].

Unqualified hand-offs between agents. Agentic systems chain tasks, but auditable hand-offs, principled propagation of uncertainty and guaranteed points of human validation are unsolved for safety-critical medicine [3,5,60]. Errors then compound invisibly: a badly outlined lesion silently becomes a confident but wrong grade, then a wrong recommendation, and the composite decision is attributable to no participant, which is incompatible with clinical governance and with regulatory expectations of meaningful oversight [59].

Fusion that does not survive real data. Diagnosis, grade and treatment depend on integrating imaging and tissue analysis with laboratory markers where available [36], yet the two are handled in isolation. Fusion outperforms single-source models in research settings [47], but robust fusion where one source is missing, delayed or poor remains unsolved [18,19] and largely unexplored for low-resource settings.

Methods that assume abundant, standardised data. State-of-the-art models presuppose large, curated, protocol-standardised datasets [41,42]. Sri Lankan units hold small, heterogeneous collections, so out-of-the-box models under-perform and performance is fragile and scanner-dependent [2,11,54]. Label-efficient formulations exist [45,46,64,68,69] but are unvalidated on South Asian cohorts.

A chicken-and-egg problem in data sharing. Confidentiality, governance and absent data-sharing agreements make centralisation difficult, which starves training. Federated learning offers a route [49,51,52,87] proven at international scale [53], but the regime that matters here, a handful of hospitals holding small cohorts that differ markedly, is exactly where federated aggregation is least well characterised [6,7,8,88,89].

No way to measure the quality of the escalation decision. Clinical AI is reported chiefly through discriminative metrics on retrospective test sets [97]. Standards for early-stage clinical evaluation now require measurement of human factors and safety in situ [57,95,96], and ethical guidance

requires autonomy and accountability to be demonstrable rather than declared [58,59]. No established protocol quantifies whether a system defers on the cases where deferral was warranted.

1.2.3 Problem statement

These gaps identify the central research problem as the absence of a clinically responsible, formally governed multi-agent AI framework that can support the brain-tumour pathway across imaging, tissue analysis, combined diagnosis and pre-operative surgical planning while explicitly managing uncertainty and preserving clinician authority. Such a framework must determine, for each case and each sub-decision, when an agent can provide reliable decision-support information and when the case should be escalated to a qualified clinician. It must remain robust to the small, heterogeneous, non-standardised datasets characteristic of Sri Lankan institutions, and enable privacy-preserving collaboration without centralising raw patient data.

1.3 Aim, Objectives and Research Questions

1.3.1 Purpose and aim

The purpose of this study is to develop and evaluate a clinically responsible human-in-the-loop AI framework in which human oversight is implemented as an explicit, measurable, uncertainty-aware mechanism rather than treated only as a guiding principle, in a resource-constrained setting where datasets are limited, heterogeneous and distributed across institutions. The aim is to design, develop and evaluate an uncertainty-aware, human-in-the-loop multi-agent AI framework for brain-tumour care, integrating imaging, tissue analysis, combined clinical decision support and surgical planning, and to evaluate its feasibility within a national brain-tumour registry framework in Sri Lanka.

1.3.2 Research Objectives

The objectives are deliberately limited to four. The scope is kept controllable at this stage and can be broadened later in response to examiner and reviewer comment.

O1. To design and develop a multi-agent AI architecture for the brain-tumour care pathway (spanning patient intake, MRI analysis, tissue analysis, diagnosis and staging, surgical planning and follow-up), in which an orchestrator routes tasks, propagates uncertainty across agent hand-offs, and maintains an auditable record of every AI output and every clinician validation.

O2. To develop and evaluate a calibrated, uncertainty-aware escalation mechanism that determines when an agent may provide a decision-support output and when a case or sub-decision must be referred to a clinician for validation, based on selective prediction and learning-to-defer principles and on the clinical cost of error.

O3. To develop and evaluate the data-efficient models the agents depend on (for brain-tumour detection, outlining, classification and patient-specific anatomical reconstruction from MRI; for analysis of digitised biopsy slides; and for their combination with available laboratory markers) on small, heterogeneous Sri Lankan data, including training them across institutions by privacy-preserving federated learning without centralising raw patient data, and to benchmark their performance and calibration against established approaches.

O4. To clinically evaluate the proposed human-in-the-loop framework, covering its diagnostic and decision-support performance, the appropriateness of its escalation decisions, clinician

workload, usability, trust and safety, with surgical planning evaluated as a proof-of-concept decision-support component.

1.3.3 Research Questions

The research questions correspond one-to-one with the objectives and are likewise held to four, so that each can be answered to doctoral depth within the study period.

RQ1. How can specialised AI agents be orchestrated across the brain-tumour care pathway so that uncertainty is propagated between them, information is coordinated, and an auditable record of AI outputs and clinician validations is maintained?

RQ2. How can uncertainty estimation and calibration be incorporated into that framework to determine when the system should provide a decision-support output and when it should defer to a clinician?

RQ3. How effectively can data-efficient and multi-source methods detect, outline, classify and grade brain tumours from small and heterogeneous Sri Lankan MRI and biopsy-slide data, including under federated training across hospitals, and how do their performance and calibration compare with established approaches?

RQ4. To what extent can the proposed human-in-the-loop framework improve clinically relevant decision support while maintaining appropriate clinician involvement, acceptable workload, usability, trust and safety, including the feasibility of MRI-based surgical-planning support?

1.4 Scope, Limits of Autonomy and Limitations

1.4.1 Scope of the study

Depth of investigation is deliberately differentiated across three tiers. The first and primary tier is the uncertainty-aware orchestration and escalation mechanism, supported by data-efficient MRI models; this is the central methodological contribution. The second is computational pathology and combined decision support, remaining within the role of decision support rather than autonomous diagnosis. The third is privacy-preserving federated learning. The surgical-planning component is a proof-of-concept module covering patient-specific anatomical reconstruction, outlining of the tumour and of structures at risk, and visualisation of surgical risk; it provides no autonomous surgical decisions. The registry serves as enabling infrastructure for registration, identification, longitudinal monitoring, follow-up and secure management of clinical information. The study is limited to participating Sri Lankan institutions, and clinically to tumours included in the registry under the predefined eligibility criteria.

1.4.2 Limits of AI autonomy

Autonomy is conditional, uncertainty-aware and task-specific rather than absolute. The system is not designed to independently diagnose, determine treatment or perform surgical procedures. Each agent generates a decision-support output with an uncertainty estimate; cases exceeding predefined thresholds are escalated, and the final clinical decision remains the responsibility of the treating professional. On surgery the position is explicit and non-negotiable: the system never performs or controls an operation. It plans, predicts, flags and recommends; the neurosurgeon executes and validates every consequential step. The architecture enforces clinician validation at every decision point rather than relying on procedural convention [58,59]. Three boundaries follow. Bounded decision authority: the AI issues no diagnosis, grade or treatment decision of record, only evidence, a calibrated confidence and a recommendation a clinician accepts, amends or rejects. Conservative default: under uncertainty or novelty the system defers rather than acts. Transparency of basis: every autonomous action carries the evidence and confidence behind it, plus a model-facts style statement of the conditions under which the component was validated [100], so a reviewer can reconstruct not only what the system did but why it did not ask [56].

1.4.3 Limitations and delimitations

Component performance may vary across tumour types, imaging protocols, institutions and patient populations, and the federated component depends on the number and characteristics of participating institutions, computing infrastructure and connectivity. Several adjacent areas lie outside scope and are recorded as future directions: guidance during surgery, robotic actuation and real-time navigation; development of new laboratory assays, existing marker results being consumed as input rather than generated; physics-based simulation of tissue deformation, radiotherapy dose planning and chemotherapy optimisation; paediatric-specific modelling as a separate strand, though paediatric cases are not excluded from the registry; and deployment beyond the participating units and certification as a medical device, though the evidence generated is intended to support a later certification pathway.

Three further limitations should be stated plainly. Cohort size: national incidence means the registry will accrue hundreds rather than thousands of fully linked cases, bounding subgroup analyses and making data-efficient methods a necessity. Reference-standard dependence: clinician-confirmed diagnoses serve as ground truth, so the framework is evaluated against expert consensus rather than an independent biological gold standard. And the study is not a randomised trial of patient outcomes; validation measures diagnostic performance, workload, escalation appropriateness, safety and trust in an assistive workflow [57], so claims about survival benefit are explicitly not made. The work assumes ethics approval and data-sharing agreements can be secured before data access, that units can supply usable MRI and digitisable biopsy material, that clinician time is available albeit scarce, and that units will operate shared governance for federated training; each assumption is paired with a mitigation in the risk register.

1.5 Justification and Significance of the Research

Scientific and clinical. How an AI agent should decide when it is competent to act and when it must defer is a computer-science problem before it is a clinical one, sitting at the intersection of uncertainty quantification [29,30,32,70], calibration [31,71], selective prediction and learning to defer [33,34,35,77] and multi-agent orchestration [3,5,79]. The literature treats these as separate concerns and treats oversight as a principle asserted in prose rather than a mechanism specified, calibrated and measured. Formalising selective autonomy over a chain of dependent sub-decisions, and evaluating it on an explicit workload-versus-safety frontier, is methodologically novel and transferable well beyond this clinical domain. Brain-tumour care is an unusually strong test case: it requires formal integration of several evidence sources [36]; uncertainty must propagate along a sequential chain; the cost of error is severe and asymmetric [38,39]; and specialist review time is the binding constraint, so reduced review load is a directly meaningful outcome.

National and methodological. Sri Lanka has no imaging- and tissue-linked brain-tumour registry, and its absence limits clinical continuity, outcome auditing and research capacity simultaneously [9,10]. Establishing the first such registry delivers durable value to the participating units independently of the research findings, and the associated open, de-identified benchmark addresses the near-total absence of South Asian representation in public brain-tumour datasets [41,42]. The constraints of the setting are also what make the research generalisable: small, heterogeneous data and barriers to centralisation are the normal condition of clinical AI in most of the world, so methods demonstrated under scarcity, heterogeneity and federation reach further than those requiring large curated corpora [50,51,53]. Evaluating on local data guards against the site-specific shortcut learning that has undermined models validated only internally [54,55].

Ethical, regulatory and feasible. International guidance places human autonomy, transparency and accountability among its core principles for AI in health [58], and regulation now requires that high-risk systems be designed so people can oversee them and intervene [59]; explainability alone is insufficient without an account of responsibility [56], and algorithmic bias arises from design choices never examined [98]. A framework making oversight measurable and auditable rather than merely declared addresses a requirement that is becoming mandatory. The enabling components are mature enough to build on but immature enough to contribute to [29,30,32,44,45,84,85,3,5,60,53,6,7,8]. The project combines established computer-science supervision with clinical supervision, giving direct access to the patient pathway, to authentic data and to the clinicians who will validate and use the system; the tiered scope, front-loaded ethics approval and development on public benchmarks while local data accrue make the plan deliverable within 36 months.

2. Literature Review

2.1 Approach to the Review

This chapter reviews the literature the framework builds on and identifies, in each area, the point at which existing work stops. It follows the architecture of the proposed system: the clinical domain, the perception layer, the core contribution of uncertainty and deferral, orchestration, computational pathology, federated learning, surgical planning, and evaluation and governance. Sources were drawn from the major venues for medical image analysis, machine learning and clinical informatics, with recent systematic and scoping reviews [1,2,3,5,11]. Priority was given to work from 2015 onward, to studies reporting external validation, and to sources reporting negative results, since these are more informative about deployment risk.

2.2 Brain Tumours: Burden, Classification and the Care Pathway

Brain and nervous-system tumours are uncommon but carry disproportionate mortality, with poor outcomes concentrated in the most aggressive types [37]. Sri Lanka sees roughly 700 new cases and 587 deaths annually, with a five-year prevalence of about 2,500 [9,10]. The most consequential recent change for computational work is classificatory: the 2021 World Health Organization classification integrates tissue appearance with laboratory marker results to define tumour type and grade [36]. A model trained on a pre-2021 label is therefore not predicting the same target as one aligned to the current classification, and a definitive diagnosis is now multi-source by definition, so single-source systems can only support a diagnosis, never produce one of record. The pathway is a dependent chain in which errors propagate forward and the cost of error is strongly asymmetric [38,39].

2.3 Deep Learning for Brain-Tumour Analysis from MRI

U-Net [61] and its volumetric extension [62] remain the structural basis of most published systems, and shared benchmarks drove the field: BraTS established a common multi-institutional dataset, annotation protocol and evaluation methodology [40,41,42]. Transformer designs such as Swin UNETR followed [63], and nnU-Net showed that careful automated configuration of a conventional architecture matches or exceeds many bespoke designs [43]. Reviews document rapid gains [1,2,11], and strong outlining remains achievable with fewer MRI sequences [12]. The limitation is generalisation: performance depends on large, curated, protocol-standardised data and stays fragile on small single-institution datasets and unseen scanners, where models can learn site-specific artefacts rather than disease [2,11,54,55]. For a Sri Lankan deployment this is the principal technical risk to the perception layer, and it motivates RQ3.

2.4 Data-Efficient, Self-Supervised and Domain-Adapted Learning

Because annotated local data will be scarce, the perception layer depends on methods that reduce label dependence. Contrastive learning [64,65], self-distillation such as DINO [66] and masked autoencoding [67] showed that useful representations can be learned from unlabelled images alone, and these transfer to medicine with measurable benefit, improving classification when labels are few [69] and outlining when pre-trained on unlabelled 3D volumes [68]. The benefit is largest exactly where labels are scarcest, and pre-training can use unlabelled local scans, which carry the scanner characteristics transferred models lack. What remains unestablished is whether the resulting representations yield confidence estimates fit to drive an autonomy decision; most evaluations report only accuracy on curated benchmarks. That intersection is where RQ3 meets RQ2.

2.5 Uncertainty Quantification and Calibration

The premise of the framework is that an agent must know when it does not know, and the foundations are established. Monte-Carlo dropout approximates Bayesian inference in deep networks [29]; deep ensembles are a simple, strong alternative [30]; and separating uncertainty caused by noise from uncertainty caused by limited knowledge clarifies which component more data can reduce [70]. Modern networks are systematically over-confident, and post-hoc calibration such as temperature scaling substantially improves the match between predicted probability and observed correctness [31], while conformal prediction gives distribution-free coverage guarantees [32,75]. Medical imaging has adopted these with mixed results [72,73,74]. The most important negative result is that calibration degrades precisely under dataset shift, exactly where a reliable confidence signal is most needed [71]. The gap is not the absence of estimators but their disconnection from action [28].

2.6 Selective Prediction, Learning to Defer and Human-in-the-Loop AI

Selective classification equips a model with a reject option and characterises the risk-coverage trade-off [33]. Learning to defer generalises this by modelling the downstream human: the system learns a joint policy over predictor and rejector that accounts for the expert's own error profile [34,35]. Related work frames the problem as allocating scarce human effort through triage [76] and as learning to complement rather than imitate human decision-makers [77]. Human-factors work qualifies these results: explanations can increase acceptance of incorrect recommendations [78]. In healthcare, human-in-the-loop configurations outperform both fully automated and clinician-only workflows [25], and control points, escalation and confidence-aware deferral are prerequisites for real use [4]; yet deployed exemplars remain rare [27]. Three limitations define the core gap: these formulations address a single model making a single decision, not a chain in which uncertainty must be composed across agents; they assume a fixed, homogeneous cost of error, whereas clinical costs are asymmetric and vary by decision type; and the expert is modelled as an always-available oracle, whereas specialist capacity is the binding constraint. Addressing these jointly is the substance of RQ1 and RQ2.

2.7 Agentic and Multi-Agent AI in Healthcare

Interest has shifted to agentic systems whose components reason over multiple steps, invoke tools and coordinate with one another, enabled by chain-of-thought prompting [80] and the interleaving of reasoning with tool use [79]. In medicine, large language models encode substantial clinical knowledge [81], the case for generalist medical AI has been argued [82], and conversational diagnostic systems have been evaluated against clinicians [83]. Surveys map the application space while consistently identifying the same unresolved issues: safety, evaluation methodology, traceability of multi-step reasoning, and accountability when a composite decision emerges from several components [3,4,5,60]. The gap is structural: existing work concentrates on capability and end-to-end task success, not on how uncertainty is represented at the interface between agents, how a downstream agent should behave when an upstream input is unreliable, or how an auditable record should be maintained. No published architecture makes uncertainty a first-class element of the contract between agents, which is the contribution proposed under RQ1.

2.8 Computational Pathology and Multi-Source Diagnosis

Clinical-grade performance was demonstrated using weakly supervised learning on digitised slides at scale [44], with attention-based multiple-instance learning letting a single slide-level label supervise attention to individual regions [45,46]. Pathology foundation models now serve as general-purpose encoders [84,85,86], transferring strongly to tasks with few local labels. In brain-tumour work, deep learning supports grading and classification [18,19,20,21], and application during surgery matched conventional interpretation prospectively [48]. Because the 2021 classification is integrated [36], the decisive question is fusion: combining tissue appearance with laboratory features improves diagnosis and prognosis relative to either [47]. What remains unsolved is robustness to real-world incompleteness: published fusion models assume every source present and of consistent quality, whereas in practice tissue analysis lags imaging, slide quality varies, and markers may be unavailable for cost reasons. Fusion that degrades gracefully is not established, and is part of RQ3.

2.9 Federated and Privacy-Preserving Learning

Federated learning enables training across institutions without centralising raw data, using iterative local training and parameter aggregation [49]. Its medical relevance was established early [50,51,52,87], and the decisive demonstration trained a glioblastoma tumour-boundary model across 71 sites on six continents, improving substantially on any single site's model without disclosure of raw patient data [53]. Aggregation across very different sites has been addressed through proximal regularisation [88] and variance reduction correcting for client drift [89], and privacy strengthened through differentially private training [90,91]; recent work adds encryption, reputation-aware aggregation and heterogeneity handling [6,7,8,13,14]. Two gaps persist. The regime of a few hospitals each holding a small, markedly different cohort is under-characterised, and is where aggregation instability and client drift are most severe. And federation is studied as a training-time concern in isolation: how federated updates interact with calibration, and therefore with an escalation policy validated on a previous model version, is essentially unexamined. A model that gains accuracy while losing calibration would silently change deployed safety behaviour, making this a first-order concern for RQ3.

2.10 AI-Assisted Surgical Planning

Pre-operative planning support has developed along two lines: patient-specific anatomical reconstruction from imaging, supported by mature open platforms [94] and digital-twin formulations [15,16,17]; and algorithmic trajectory planning, formulating the approach path as a constrained optimisation over anatomy to be avoided, founded on sampling-based optimal motion planning [92], with clinical instances computing electrode trajectories [93] and risk-aware planning seeking corridors that avoid critical tissue and blood vessels [22,23,24]. This literature establishes feasibility and its own boundaries: reconstruction quality depends on the outlining of structures at risk, itself an uncertain perception task, and evaluation is typically geometric rather than clinical acceptability judged by operating surgeons. Crucially, published systems present a single recommended trajectory without stating confidence in the underlying anatomical delineation, which is why the surgical-planning agent is scoped as a proof-of-concept demonstrator under RQ4.

2.11 Evaluation, Ethics, Governance and Regulation

Methodological guidance distinguishes internal from external validation and warns against inflated claims from inappropriate test-set construction [97]. Reporting standards followed: CONSORT-AI for randomised evaluations [95]; TRIPOD+AI for prediction-model reporting [96]; and DECIDE-AI for the stage most relevant here, small-scale clinical evaluation of AI decision support, which explicitly requires reporting of human factors, of the clinician-system interaction, and of safety in situ [57]. World Health Organization guidance places human autonomy, transparency and accountability among its core principles [58]; interpretability alone cannot discharge accountability without an account of responsibility [56]; algorithmic bias has been shown to arise from an apparently reasonable choice of prediction target [98]; model-facts labelling communicates validation conditions [100]; and maintenance dominates the lifetime cost of deployed systems [99]. Regulation has made this concrete: the European Union's Artificial Intelligence Act classifies medical AI as high-risk and requires, in Article 14, that such systems be designed so people can oversee them, remain aware of automation bias, and intervene or interrupt operation [59]. Yet none of these standards specifies how to measure the quality of an escalation decision itself, and producing such a protocol is part of the contribution claimed here.

2.12 Synthesis: What Is Established and What Remains Open

Four things are established: the individual perception tasks the framework requires are technically solved to a clinically useful standard on curated data, for MRI [1,2,40,42,43] and digitised slides [44,45,84]; the tools for quantifying and calibrating predictive uncertainty are mature [29,30,31,32], with formal mechanisms for acting on it [33,34,35]; federated learning is a proven route to multi-institutional training without data sharing [50,53]; and both ethical guidance and binding regulation now require demonstrable human oversight of high-risk clinical AI [58,59].

Four things remain open, and together they define this research. The primitives for selective autonomy have never been assembled into a calibrated escalation policy for a dependent chain of clinical sub-decisions with asymmetric error costs and finite expert capacity [33,34,76]. Agentic architectures do not treat uncertainty as part of the contract between agents, so hand-offs are unqualified and accountability for composite decisions is undefined [3,5,60]. Multi-source diagnosis is demonstrated on complete, high-quality cohorts rather than under the incompleteness of real practice [47]. And data-efficient, federated and calibrated learning have been studied

separately, leaving unexamined the interaction that matters most: federated updates and domain adaptation can change a model's calibration and therefore silently alter the safety behaviour of any escalation policy built on it [53,71,89]. The framework proposed here closes these four gaps in a single, clinically validated system.

3. Research Design and Methods

3.1 Research Design

This research produces an artefact (a working multi-agent clinical AI framework) and evaluates it against measurable criteria. The design is therefore design-science research, the established paradigm for computing research whose contribution is a constructed system evaluated in its intended environment [101], following the six-activity methodology of problem identification, objective definition, design and development, demonstration, evaluation and communication [102]. It is chosen over a purely experimental design because the central claim, that selective autonomy can be made calibrated and auditable, cannot be demonstrated by benchmarking alone. Methods are selected by fitness to each research question [103], yielding a mixed-methods design. The dominant quantitative strand comprises retrospective and simulated experiments on registry and public data, measuring outlining, classification, calibration, escalation and federation performance. The supporting qualitative strand comprises clinician feedback and interviews measuring trust, perceived usefulness and the acceptability of escalation behaviour. The two combine in an explanatory sequential arrangement in the final validation [103,104]. Work proceeds through iterative build-and-evaluate cycles closed by clinician feedback; because the artefact evolves, every result is tied to a versioned model and escalation policy, and none is carried forward across a change to either without re-evaluation.

3.2 The Agents: What They Are and What Each One Does

3.2.1 What “agent” means in this framework

In this proposal an agent is a self-contained software component with one clearly bounded job, a defined input, a defined output, and one additional obligation that distinguishes it from an ordinary module: it must report how confident it is in what it has just produced, on a scale calibrated so that the number can be believed. An agent is not a person, not a chatbot and not a general-purpose reasoner. It is closer to a specialist on a team who does one job, states how sure they are, and passes the result on.

Three consequences follow and shape the whole architecture. Because each agent does one job, it can be developed, tested and replaced independently of the others. Because each agent emits a confidence alongside its result, the component that coordinates them can reason about reliability rather than merely about outputs. And because confidence travels with the result across every hand-off, a later agent is never misled into treating a doubtful input as a certain one, which is the specific failure this framework is designed to prevent. The system comprises six specialist agents, one orchestrator that coordinates them, and the clinician, who is a participant in the workflow rather than an observer of it.

3.2.2 The orchestrator

The orchestrator is the coordinating component, and is not itself a diagnostic model. Its inputs are the case as it stands and the result and confidence just produced by whichever agent last ran; its outputs are a routing decision (which agent runs next, or whether the case stops here) and a record written to the audit trail. After every step it does four things: it reads the agent's calibrated confidence; it combines that confidence with the clinical cost of getting this particular kind of decision wrong; it applies the escalation rule of Section 3.3, producing one of three actions (proceed, gather more evidence, or escalate to a clinician); and it writes what happened, why, and with what confidence to an auditable record. The orchestrator is therefore where the scientific contribution of this thesis lives: the agents perform recognised tasks, but the policy governing when their outputs may be used without human review is the novel mechanism under study.

3.2.3 The six specialist agents

Each agent is described below by its role, what it receives, what it produces, and what it hands to the next stage. Every output includes a calibrated confidence, and every agent can be interrupted by the orchestrator and its case sent to a clinician.

1. Intake Agent. Role: to turn a messy real-world referral into one clean, standardised, de-identified patient record the rest of the system can rely on. Inputs: patient details, the referral report, the MRI study as it comes from the scanner, and the digitised biopsy slides where these exist. It identifies the patient, checks whether a record already exists so the same person is not registered twice, removes direct identifiers, and applies automated quality checks to the scans and slides. Outputs: a single standardised registry record, plus a quality flag on each item of data. Hands off to: the MRI Perception Agent; and, where a poor-quality or unreadable item is flagged, to the orchestrator, which requests a repeat or refers the case for human attention before any analysis begins.

2. MRI Perception Agent. Role: to find and outline the tumour on the scan and suggest what type it is. Inputs: the standardised multi-sequence MRI study from the Intake Agent. It detects whether a lesion is present, draws its boundary slice by slice, measures its volume, and proposes a tumour type. Because local scans are few and vary between machines, this agent is built with data-efficient methods and is where the calibration work of Objective 3 is concentrated. Outputs: the tumour location and outline, a suggested type, a volume measurement, and a calibrated confidence for each. Hands off to: the Pathology Agent and the Diagnosis and Staging Agent, which receive both the findings and the confidence attached to them; and to the Surgical Planning Agent, which later reuses the same outlines.

3. Pathology Agent. Role: to read the digitised microscope slides of the biopsy and produce supporting evidence about the tumour. Inputs: whole digitised slides of the tissue removed at biopsy. It analyses the slide at high magnification, identifies the regions carrying diagnostic information, and derives evidence for tumour type and severity grade, using methods that need only a label for the slide as a whole, because region-by-region expert annotation is not available at the scale required. Outputs: evidence for tumour type and grade, a map showing which regions of the slide drove that conclusion, and a calibrated confidence. Hands off to: the Diagnosis and Staging Agent. If the slide is of poor quality, or the evidence conflicts with the MRI findings, the conflict itself is a trigger for escalation.

4. Diagnosis and Staging Agent. Role: to bring the separate lines of evidence together into one supported conclusion. Inputs: the MRI findings and confidence from Agent 2, the tissue findings and confidence from Agent 3, and any laboratory marker results recorded for the patient. Because the current international classification defines the diagnosis by combining tissue appearance with laboratory results, this is the agent at which a diagnosis can first be supported at all. It must also cope with the routine case in which one source is missing, delayed or poor; it degrades gracefully rather than failing, and reports how much of its confidence came from each source. Outputs: a supported tumour type, a severity grade, a treatment-related recommendation, a breakdown of which evidence contributed what, and a combined calibrated confidence. Hands off to: the clinician for validation in every case where the decision is consequential, and, once validated, to the Surgical Planning Agent and the Follow-up Agent.

5. Surgical Planning and Risk-Path Agent. Role: to help the surgeon see, before an operation, what the anatomy looks like and which route carries least risk. Inputs: the MRI outlines from Agent 2 and the validated diagnosis from Agent 4. It reconstructs a three-dimensional model of that specific patient's anatomy, outlines the structures that must be avoided (the regions controlling speech, movement and vision, major blood vessels, and the nerve pathways between them) and computes a candidate approach route minimising expected damage. Crucially it also carries forward the uncertainty in the underlying outlines, so a route based on a doubtful boundary is presented as such rather than as a confident recommendation. Outputs: a patient-specific 3D model, a map of structures at risk, a candidate approach route, and an explicit statement of confidence in the anatomy the route was computed against. Hands off to: the neurosurgeon, always. This agent is a proof-of-concept demonstrator; it never performs, controls or guides an operation.

6. Follow-up Agent. Role: to keep track of the patient after treatment. Inputs: the validated record and any later surveillance scans. It schedules surveillance, compares each new scan against the previous one, and measures change in tumour size or appearance over time. Outputs: a surveillance schedule and a report of longitudinal change, with a confidence attached to each measured change. Hands off to: the clinician, where a change exceeds a threshold or the comparison is uncertain, and to the registry, which retains the record as part of the national longitudinal resource.

3.2.4 The clinician-validator

The clinician is treated as a component of the workflow with a defined role, not a reviewer appended to the end of it. Inputs: any case the orchestrator escalates, together with the agent's output, the evidence behind it, and the confidence that triggered the escalation. Outputs: an accept, amend or reject decision, which becomes the decision of record. Two invariants hold throughout: there is no route from any agent to a clinical action of record that does not pass a clinician, and every escalation, every autonomous action and every clinician decision is written to the audit trail.

3.2.5 How the agents work together, step by step

Figure 3.2 traces one case from arrival to final output. The case enters at the Intake Agent and moves down the chain; at each step the agent produces its result and its confidence, the orchestrator applies the escalation rule, and the result is either passed on, returned for more evidence, or sent to a clinician. Nothing crosses a hand-off without its confidence attached, and no step can reach a clinical action without passing the clinician-validator. Figure 3.1 shows the same system as a static architecture, with the agents coordinated by the orchestrator over the shared registry data layer that also holds the audit trail.

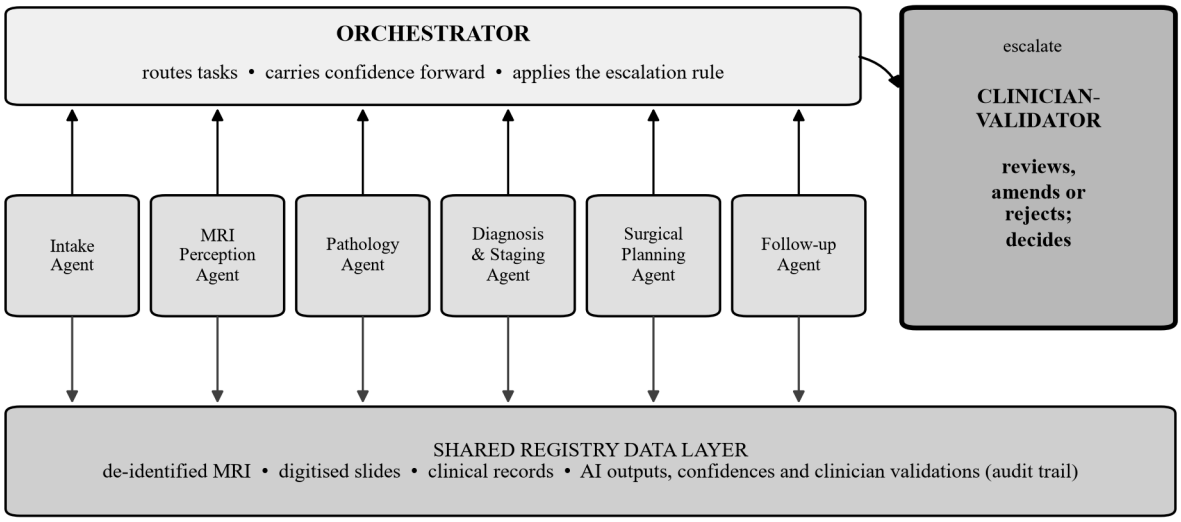


Figure 3.1: System architecture. Specialised agents are coordinated by an orchestrator over a shared registry data layer; every route to a clinical action passes the clinician-validator.

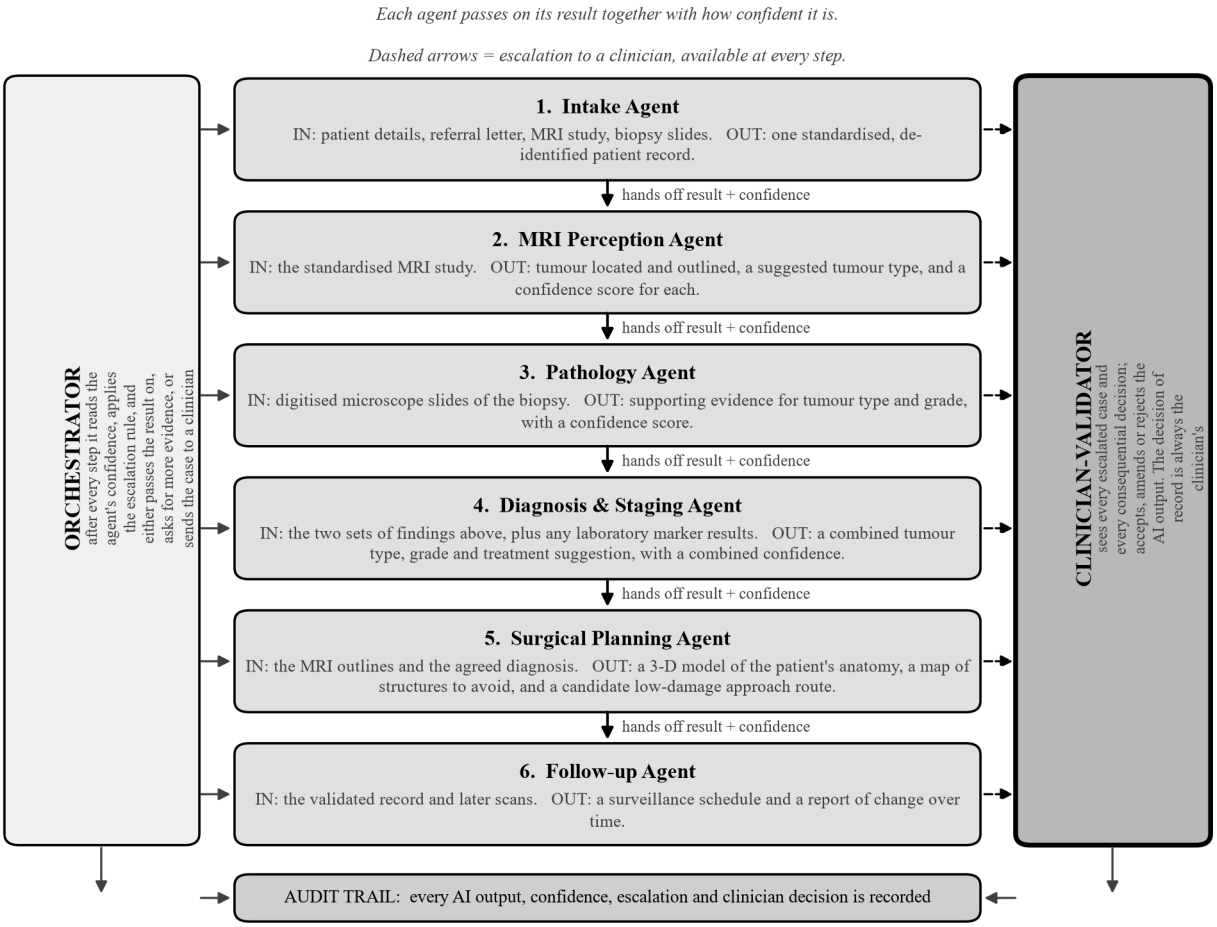


Figure 3.2: The agent workflow from input to final output. Each numbered step names one agent, what it receives, what it produces, and what it hands to the next agent. The orchestrator applies the escalation rule after every step, and escalation to a clinician is available at every step.

3.3 How the Escalation Decision Is Made

The escalation rule is the mechanism under study, and is deliberately simple to state. An agent's raw output score is first calibrated, so a reported probability of 0.8 corresponds to being right about 80% of the time rather than to an arbitrary number produced by the network. The calibrated probability is converted into an uncertainty, then compared not against a threshold chosen by hand but against one derived from the clinical cost of the two kinds of mistake available at that decision, because wrongly reassuring a patient and wrongly alarming one are not equally harmful, and the ratio between those costs is what should set the operating point. The comparison yields exactly one of three actions: proceed, gather more evidence, or escalate to a clinician. The costs are elicited from the clinical supervisor and participating clinicians rather than assumed, and a list of decisions that are always escalated regardless of confidence is fixed in advance.

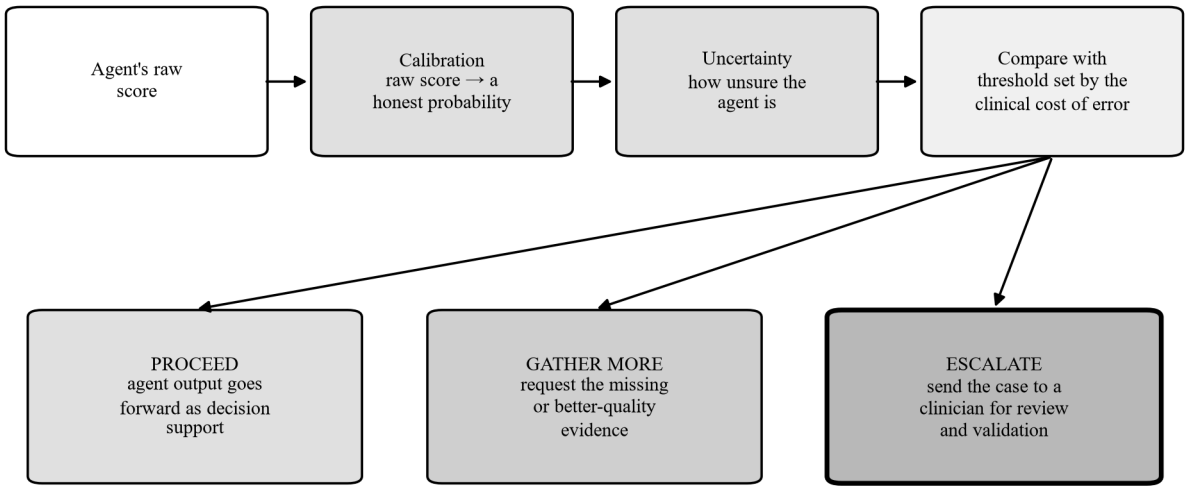


Figure 3.3: How the escalation decision is computed. A raw score is calibrated, converted into an uncertainty, and compared against a threshold derived from the clinical cost of error rather than chosen by hand.

As variables, the independent variables are the design choices under the researcher's control: uncertainty-estimation method, calibration method, escalation policy and its cost parameters, training regime (local, transfer, self-supervised, federated) and fusion strategy. The mediating variable, and the theoretical heart of the framework, is the calibrated uncertainty attached to each agent output, since the thesis claims that the effect of those choices on outcomes is transmitted through the quality of this estimate rather than through raw accuracy. The dependent variables are diagnostic performance, clinician workload, escalation appropriateness, safety and trust. Moderating variables recorded for every case are scanner and acquisition protocol, image and slide quality, tumour type and grade, completeness of evidence, and site of origin.

3.4 Work Packages

The work is organised into seven work packages aligned with the objectives. WP1 delivers the registry platform and data pipeline. WP2 delivers data-efficient MRI perception. WP3, the scientific core, delivers uncertainty-aware orchestration and the escalation policy (RQ1 and RQ2). WP4 delivers computational pathology and combined diagnosis and staging. WP5 delivers federated learning across hospitals. WP6 delivers the surgical-planning agent as a proof of concept. WP7 delivers clinical validation. Each follows the same internal cycle of design, implementation, quantitative evaluation and clinical review. WP2, WP4 and WP5 together answer RQ3; WP6 and WP7 together answer RQ4.

3.5 Population and Study Sample

The target population comprises all patients with primary or secondary brain and nervous-system tumours in Sri Lanka, estimated at approximately 700 new cases annually and a five-year prevalence of about 2,500 [9]. The accessible population comprises patients presenting to the participating tertiary units during the study period and meeting the eligibility criteria, recruited by consecutive sampling. The expected cohort is several hundred patients; the exact number depends on incidence, referral patterns and data availability, and the number with complete linked imaging, tissue, marker, treatment and follow-up information will be smaller. Patients are eligible if they are of any age with a suspected or confirmed brain or nervous-system tumour, managed at a participating institution during the study period, with at least one clinically relevant MRI examination in usable standard scan-file form, sufficient clinical information to establish the reference standard, and appropriate ethical approval and consent or waiver. Records are excluded where imaging is unusable and unrecoverable, where reference-standard information cannot be established, where de-identification cannot be completed, or where data-governance requirements are not met. Patients are not excluded solely for missing tissue, marker or follow-up information when they remain eligible for an individual component; they contribute according to data availability, which is what allows evaluation under the incomplete-data conditions expected in real practice.

A second population comprises the clinicians involved in diagnosis and management at participating institutions: consultant neurosurgeons, radiologists, histopathologists and experienced senior trainees. Because this group is small and highly specialised, clinicians are recruited by purposive sampling based on expertise and are used primarily to evaluate interpretability, trust, usability, workload, escalation decisions and clinical usefulness. For the reader evaluation a multi-reader, multi-case design is used so the available experts are used efficiently: each assesses a common set of cases under assisted and unassisted conditions, with counterbalanced ordering and a washout period. A third population comprises public datasets used for pre-training, benchmarking and external comparison, including BraTS. Public data are not representative of the Sri Lankan population and are therefore not the sole basis for any clinical performance claim; their purpose is to establish reproducible baselines before sufficient local data become available.

3.6 Hypotheses

The hypotheses are testable statements aligned with the objectives and research questions. The corresponding null hypotheses assume no statistically significant difference between the compared methods or conditions.

H1 (RQ2). A calibrated, uncertainty-aware escalation policy will achieve a more favourable workload-safety trade-off than conventional fixed-confidence thresholding, shown by lower decision error at an equivalent escalation rate, or lower escalation at an equivalent error rate.

H2 (RQ1). Propagation of calibrated uncertainty between interacting agents will reduce downstream decision errors compared with conventional hand-offs that do not account for the reliability of upstream outputs.

H3 (RQ3). Self-supervised pre-training on unlabelled local MRI data followed by supervised fine-tuning will improve the performance and calibration of brain-tumour MRI models compared with transfer learning from public benchmark datasets alone.

H4 (RQ3). Combining MRI and biopsy-slide evidence will achieve higher diagnostic and grading concordance with the clinician-established reference standard than the best-performing single-source approach.

H5 (RQ3). Federated training across participating hospitals will improve generalisation compared with single-site training, while maintaining performance within a predefined margin of centrally trained models and preserving calibration within a predefined tolerance, as measured by expected calibration error and Brier score.

H6 (RQ4). The AI-assisted human-in-the-loop workflow will reduce clinician review time and improve perceived usability and trust compared with an unassisted workflow, while maintaining diagnostic performance within a predefined non-inferiority margin.

3.7 Operationalisation and Measures

Outlining quality is measured by Dice similarity coefficient, 95th-percentile Hausdorff distance, sensitivity and specificity, per case with confidence intervals [126]; classification and diagnosis quality by area under the receiver-operating characteristic curve, balanced accuracy, per-class F1 and concordance with the clinician reference standard. Calibration, the mediating variable, is measured by expected and maximum calibration error [135], the Brier score [136], reliability diagrams [31], and empirical coverage against the nominal level [32,75]. Selective autonomy is measured by coverage, the proportion of cases handled autonomously; selective risk, the error rate among those cases; the area under the risk-coverage curve [33]; the appropriate-escalation rate, the proportion of escalated cases in which the clinician changed the AI output; and the missed-escalation rate, the proportion of autonomously handled cases that blinded expert review judged to require human input, the primary safety indicator. Workload is measured as review minutes per case from the audit log and by the NASA Task Load Index [132]; trust and usability by the System Usability Scale [133] and a Likert instrument; inter-clinician agreement by kappa [131] and Krippendorff's alpha [140]. Federated indicators comprise the federated-versus-centralised accuracy difference, communication cost, calibration change after aggregation, the differential-privacy budget [90,91], and an audit-verified indicator that no raw patient data left any institution.

3.8 Sources and Collection of Data

Primary data comprise diagnostic MRI studies in standard scan-file (DICOM) format, digitised biopsy slides, structured clinical information, operative and pathology reports, and follow-up records; secondary data comprise public benchmarks and national incidence statistics [9,10]. MRI is stored in an Orthanc server [108], de-identified, and structural scans defaced to reduce re-identification risk [110,111]; slides are de-identified and managed using OpenSlide [117]; clinical data are held in a PostgreSQL database using HL7 FHIR [109] with terminology based on the 2021 classification [36]. Reference annotations are produced by trained annotators using 3D Slicer [94] and QuPath [118], with clinical review and double annotation of a subset to assess agreement; diagnostic labels come from clinician-confirmed records. Automated quality control assesses scan and slide quality, failed records are quarantined, and access is role-based and fully audited. Collection begins only after ethics approval, and in federated experiments only model parameters are shared.

3.9 Data Analysis

Model evaluation uses patient-level stratified cross-validation for development and a held-out local test set, untouched during development, for final reporting. Splits are stratified by tumour type and site so a model is never evaluated on a patient whose other studies were seen in training; where site-level generalisation is the question, leave-one-site-out evaluation is added, since that reveals the scanner and protocol dependence of Section 2.3 [54]. H1 and H2 compare risk-coverage curves and their areas against each baseline, with paired bootstrap intervals from resampling patients [137]; the primary safety comparison, missed-escalation rate at matched coverage, uses McNemar's test [139]. H3, H4 and H5 use paired predictions over the same test cases: the DeLong method for areas under the curve [129], McNemar's test for paired accuracy [139], and the Wilcoxon signed-rank test for per-case metrics such as Dice. H6 uses mixed-effects models with reader and case as crossed random effects. Agreement between AI- and clinician-derived measures is assessed by Bland-Altman analysis, since correlation measures association and not agreement [130]. Family-wise error is controlled per work package by Holm's procedure [138], with endpoints designated in advance. Interviews are analysed by reflexive thematic analysis [134] and triangulated against the escalation data. Analysis uses Python and scikit-learn [107], reported to the applicable standards [57,96].

Each work package's outputs are defined before data collection rather than chosen afterwards. WP2 produces per-case metrics with confidence intervals, reliability diagrams and calibration error for each uncertainty method, an ablation isolating self-supervised pre-training and domain adaptation, and a failure analysis assigning every case in the worst-performing decile to an error category; the taxonomy is itself a reported result, because it identifies where a competence boundary lies. WP3 produces risk-coverage curves for the policy and each baseline, the workload-safety frontier, its sensitivity to the assumed cost of error, and a decomposition of escalations by trigger (low confidence, distributional novelty, conflicting evidence, mandatory review); a second analysis traces error propagation through the agent chain with and without uncertainty propagation, the direct test of H2. WP4 produces concordance for tissue alone, imaging alone and their combination, with an ablation under simulated missing sources. WP5 produces the federated-versus-centralised-versus-single-site comparison and the calibration change after aggregation. WP6 produces geometric measures of planned trajectories with neurosurgeon ratings, deliberately descriptive and formative. WP7 produces the mixed-effects comparison of assisted and unassisted

workflows, the usability and trust scores, and the qualitative themes. Negative results are committed to in advance and reported in the same detail as positive ones [57].

3.10 Ethical, Privacy and Governance Considerations

Ethical approval will be obtained from the relevant Ethics Review Committee of the University of Sri Jayewardenepura and from each participating hospital before any patient data are accessed, and data use will commence only once all approvals are granted. Informed-consent procedures will follow the requirements of the relevant committees and institutions. At ingestion, robust de-identification is applied: removal of direct identifiers, scrubbing of scan-file headers, defacing of structural MRI, and removal of identifying metadata from digitised slides. Privacy is built into the architecture from the outset: where federated learning is used, raw clinical, imaging and tissue data remain within the respective hospitals and only model updates are shared, with role-based access control, authentication and comprehensive audit logging.

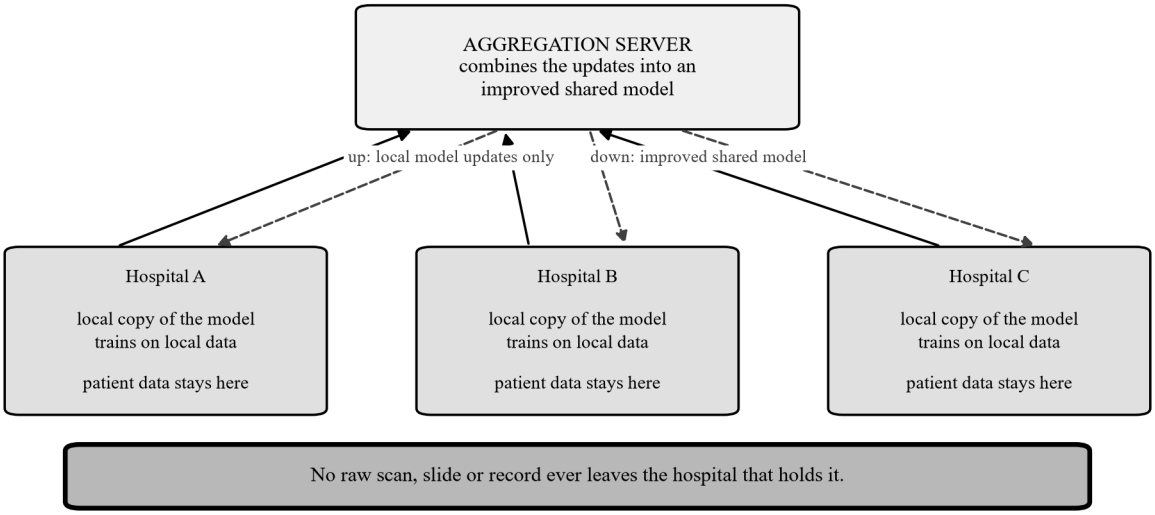


Figure 3.4: Federated training across participating hospitals. Only model parameters are exchanged; raw patient data never leaves an institution.

The system functions strictly as a clinical decision-support tool and will not independently issue or implement unvalidated clinical decisions. All AI outputs require review and validation by appropriately qualified clinicians, and the escalation policy adopts a conservative safety principle whereby cases with insufficient confidence are referred to human experts rather than acted upon automatically. The system supports surgical planning and decision-making only; it will not perform, control or autonomously execute any surgical procedure. All surgical decisions and actions remain under the direct responsibility of the neurosurgeon, who independently reviews and validates the system's outputs before implementation.

3.11 Technology Stack

The stack is chosen for reproducibility, clinical interoperability and mature open-source implementations, so effort goes to the research mechanism rather than infrastructure. Model development uses Python with PyTorch [105] and MONAI [106]; outlining baselines use nnU-Net [43] and Swin UNETR [63]; classical machine learning and metrics use scikit-learn [107]. Image preprocessing uses N4ITK bias-field correction [112], ANTs for registration [113], HD-BET for brain extraction, validated specifically on brain-tumour MRI [114], and FSL and MRtrix3 for nerve-pathway reconstruction [115,116]. Digital pathology uses OpenSlide [117], QuPath [118], stain normalisation following Macenko and Vahadane [119,120], and attention-based multiple-instance learning following CLAM [45,46], with pathology foundation encoders as frozen feature extractors [84,85]. Federated training uses NVIDIA FLARE [121] with Flower and OpenFL as alternatives [122,123], implementing FedAvg, FedProx and SCAFFOLD [49,88,89], and Opacus where differential privacy is required [124,90]. Surgical-path planning uses the Open Motion Planning Library [125,92] with 3D Slicer [94]. The registry is a web application with a PostgreSQL database, an Orthanc imaging server [108], and an HL7 FHIR resource model so it interoperates with national systems rather than becoming another silo [109]. Experiment tracking, dataset versioning and containerisation let every reported figure be regenerated from a recorded model version, data version and seed.

4. Research Plan and Timeline (36 Months)

Table 1: Gantt Chart

Task	Activities	1st Year			2nd Year			3rd Year		
		04	08	12	16	20	24	28	32	36
1.0	**Literature review and proposal refreshment**									
1.1	Literature review and proposal	■	■							
1.2	Ethics application and approval		■	■						
2.0	**WP1: Registry Platform & Data Pipeline**									
2.1	Development of the brain-tumour registry and database and data architecture		■	■	■	■				
2.2	Data ingestion; de-identification;			■	■	■	■			
2.3	Data-quality control; establishment of the research data pipeline.					■	■			
3.0	**Data Collection, Annotation & Quality Control**									
3.1	WP2 MRI perception baselines; calibrated uncertainty; local adaptation.		■	■	■	■	■			
4.0	WP3: uncertainty-aware escalation & orchestration design and experiments.		■	■	■	■	■			
5.0	WP4 computational pathology + multimodal diagnosis & staging.		■	■	■	■	■			
6.0	**Federated Learning Across Hospitals**									
6.1	federated learning across hospitals						■	■	■	■
7.0	Surgical Planning Proof of Concept								■	■
8.0	Thesis writing and publication								■	■

The plan front-loads approvals and the enabling platform, places the core scientific contribution in Year 2, and reserves Year 3 for combined diagnosis, federation, the surgical-planning demonstrator and clinical validation. Third-tier work (federated learning and the surgical-planning demonstrator) is scheduled last so that it can be narrowed without endangering completion.

Table 1: Gantt chart of the 36-month plan.

5. Expected Outcomes and Publications

1. A deployable, doctor-validated, AI-assisted national brain-tumour registry serving the participating units.
2. A validated uncertainty-aware, human-in-the-loop multi-agent framework generalisable beyond this clinical domain.
3. An open, de-identified Sri Lankan brain-tumour imaging and pathology benchmark with an evaluation protocol.
4. At least four peer-reviewed outputs: (1) data-efficient local MRI perception; (2) the core uncertainty-aware escalation and orchestration framework (flagship); (3) computational pathology and combined diagnosis and staging; (4) federated learning across Sri Lankan hospitals. A clinical-validation paper and a surgical-planning proof-of-concept paper are possible further outputs.

6. Evaluation and Success Criteria

Success is judged against measurable criteria rather than a single accuracy figure: (i) perception and pathology models reach clinically useful, well-calibrated performance on held-out local data; (ii) the escalation policy achieves a favourable workload-safety frontier, reducing clinician review load at a controlled and acceptable risk level relative to baselines; (iii) combining imaging and tissue evidence improves diagnosis and staging concordance over single-source baselines; (iv) federated training approaches centralised performance within a quantified margin while sharing no raw data; (v) clinicians rate the assisted workflow, including surgical-planning recommendations, as trustworthy and useful; and (vi) every autonomous action in deployment is traceable and doctor-validated.

References

[1] A review of deep learning for brain tumor analysis in MRI. npj Precision Oncology, 2024. https://www.nature.com/articles/s41698-024-00789-2

[2] Bibi et al. Artificial Intelligence-Based Approaches for Brain Tumor Segmentation in MRI: A Review. NMR in Biomedicine, 2025. https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/nbm.70141

[3] The role of agentic artificial intelligence in healthcare: a scoping review. npj Digital Medicine, 2026. https://www.nature.com/articles/s41746-026-02517-5

[4] Human-in-the-Loop AI Agents: Deploying Agentic AI With Control. Elementum, 2025. https://www.elementum.ai/blog/human-in-the-loop-agentic-ai

[5] Agentic AI in Healthcare: A Comprehensive Survey. TechRxiv, 2025. https://www.techrxiv.org/doi/pdf/10.36227/techrxiv.176238073.31262603/v1

[6] Federated learning with integrated attention multiscale model for brain tumor segmentation. Scientific Reports, 2025. https://www.nature.com/articles/s41598-025-96416-6

[7] Enhancing privacy-preserving brain tumor classification with adaptive reputation-aware federated learning and homomorphic encryption. 2025. https://pmc.ncbi.nlm.nih.gov/articles/PMC12453705/

[8] FedAMM: Federated Learning for Brain Tumor analysis. MICCAI 2025. https://papers.miccai.org/miccai-2025/paper/1764_paper.pdf

[9] GLOBOCAN 2024: Sri Lanka fact sheet. IARC / WHO Global Cancer Observatory. https://gco.iarc.who.int/media/globocan/factsheets/populations/144-sri-lanka-fact-sheet.pdf

[10] National Cancer Incidence and Mortality Data, Sri Lanka. National Cancer Control Programme, Ministry of Health. https://www.nccp.health.gov.lk/

[11] Deep Learning Approaches for Brain Tumor Detection and Classification Using MRI Images (2020-2024): A Systematic Review. 2025. https://pmc.ncbi.nlm.nih.gov/articles/PMC12092918/

[12] Brain tumor segmentation using deep learning: high performance with minimized MRI data. Frontiers in Radiology / PMC, 2025. https://pmc.ncbi.nlm.nih.gov/articles/PMC12281592/

[13] Privacy-preserving federated learning for collaborative medical data mining in multi-institutional settings. Scientific Reports, 2025. https://www.nature.com/articles/s41598-025-97565-4

[14] TwinSegNet: A Digital Twin-Enabled Federated Learning Framework for Brain Tumor Analysis. arXiv:2512.17488, 2025.

[15] Digital twins and brain surgery: revolutionizing neurosurgical procedures through advanced simulation. Elsevier, 2025. https://www.sciencedirect.com/science/article/pii/B9780443342264000216

[16] Virtual reality-augmented differentiable simulations for digital twin applications in surgical planning. Scientific Reports, 2025. https://www.nature.com/articles/s41598-025-99608-2

[17] Current progress of digital twin construction using medical imaging. Journal of Applied Clinical Medical Physics, 2025. https://aapm.onlinelibrary.wiley.com/doi/10.1002/acm2.70226

[18] A multi-class brain tumor grading system based on histopathological images using hybrid YOLO and ResNet networks. Scientific Reports, 2024. https://www.nature.com/articles/s41598-024-54864-6

[19] Weakly supervised deep learning-based classification for histopathology of gliomas. Scientific Reports, 2024. https://www.nature.com/articles/s41598-024-84238-x

[20] Deep learning for digital pathology: a critical overview of the methodological framework. PMC, 2025. https://pmc.ncbi.nlm.nih.gov/articles/PMC12593639/

[21] Performance of deep learning models for automatic histopathological grading of meningiomas: a systematic review and meta-analysis. Frontiers in Neurology, 2025. https://www.frontiersin.org/journals/neurology/articles/10.3389/fneur.2025.1536751/full

[22] Preoperative path planning of a craniotomy surgical robot based on an improved MDP-LQR-RRT* algorithm. Biomedical Signal Processing and Control, 2025. https://www.sciencedirect.com/science/article/abs/pii/S1746809425001582

[23] Machine Learning-Based Surgical Planning for Neurosurgery: AI Approaches to the Cranium. Frontiers in Surgery, 2022. https://www.frontiersin.org/journals/surgery/articles/10.3389/fsurg.2022.863633/full

[24] Integrating AI into neurosurgical decisions: a new frontier in medicine. Egyptian Journal of Neurosurgery, 2025. https://ejns.springeropen.com/articles/10.1186/s41984-025-00412-x

[25] Olawade DB, Plabon SB, Ojo A, Ogunbona MA, Makanjuola BD, Olasilola OR. Human in the loop artificial intelligence in healthcare: applications, outcomes, and implementation challenges. International Journal of Medical Informatics, 213:106362, 2026. https://doi.org/10.1016/j.ijmedinf.2026.106362

[26] Topol EJ. High-performance medicine: the convergence of human and artificial intelligence. Nature Medicine, 25:44-56, 2019. https://doi.org/10.1038/s41591-018-0300-7

[27] De Fauw J, Ledsam JR, Romera-Paredes B, et al. Clinically applicable deep learning for diagnosis and referral in retinal disease. Nature Medicine, 24(9):1342-1350, 2018. https://doi.org/10.1038/s41591-018-0107-6

[28] Kompa B, Snoek J, Beam AL. Second opinion needed: communicating uncertainty in medical machine learning. npj Digital Medicine, 4:4, 2021. https://doi.org/10.1038/s41746-020-00367-3

[29] Gal Y, Ghahramani Z. Dropout as a Bayesian approximation: representing model uncertainty in deep learning. Proceedings of the 33rd International Conference on Machine Learning (ICML), PMLR 48:1050-1059, 2016.

[30] Lakshminarayanan B, Pritzel A, Blundell C. Simple and scalable predictive uncertainty estimation using deep ensembles. Advances in Neural Information Processing Systems (NeurIPS) 30, 2017.

[31] Guo C, Pleiss G, Sun Y, Weinberger KQ. On calibration of modern neural networks. Proceedings of the 34th International Conference on Machine Learning (ICML), PMLR 70:1321-1330, 2017.

[32] Angelopoulos AN, Bates S. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. arXiv:2107.07511, 2021. https://arxiv.org/abs/2107.07511

[33] Geifman Y, El-Yaniv R. Selective classification for deep neural networks. Advances in Neural Information Processing Systems (NeurIPS) 30, 2017.

[34] Mozannar H, Sontag D. Consistent estimators for learning to defer to an expert. Proceedings of the 37th International Conference on Machine Learning (ICML), PMLR 119:7076-7087, 2020.

[35] Madras D, Pitassi T, Zemel R. Predict responsibly: improving fairness and accuracy by learning to defer. Advances in Neural Information Processing Systems (NeurIPS) 31, 2018.

[36] Louis DN, Perry A, Wesseling P, et al. The 2021 WHO Classification of Tumors of the Central Nervous System: a summary. Neuro-Oncology, 23(8):1231-1251, 2021. https://doi.org/10.1093/neuonc/noab106

[37] Price M, Ostrom QT, et al. CBTRUS Statistical Report: Primary Brain and Other Central Nervous System Tumors Diagnosed in the United States in 2017-2021. Neuro-Oncology, 26(Supplement 6):vi1-vi85, 2024. https://doi.org/10.1093/neuonc/noae145

[38] Sanai N, Berger MS. Glioma extent of resection and its impact on patient outcome. Neurosurgery, 62(4):753-766, 2008. https://doi.org/10.1227/01.neu.0000318159.21731.cf

[39] Molinaro AM, Hervey-Jumper S, Morshed RA, et al. Association of maximal extent of resection of contrast-enhanced and non-contrast-enhanced tumor with survival within molecular subgroups of patients with newly diagnosed glioblastoma. JAMA Oncology, 6(4):495-503, 2020. https://doi.org/10.1001/jamaoncol.2019.6143

[40] Menze BH, Jakab A, Bauer S, et al. The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS). IEEE Transactions on Medical Imaging, 34(10):1993-2024, 2015. https://doi.org/10.1109/TMI.2014.2377694

[41] Bakas S, Akbari H, Sotiras A, et al. Advancing The Cancer Genome Atlas glioma MRI collections with expert segmentation labels and radiomic features. Scientific Data, 4:170117, 2017. https://doi.org/10.1038/sdata.2017.117

[42] Baid U, Ghodasara S, Mohan S, et al. The RSNA-ASNR-MICCAI BraTS 2021 benchmark on brain tumor segmentation and radiogenomic classification. arXiv:2107.02314, 2021. https://arxiv.org/abs/2107.02314

[43] Isensee F, Jaeger PF, Kohl SAA, Petersen J, Maier-Hein KH. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. Nature Methods, 18:203-211, 2021. https://doi.org/10.1038/s41592-020-01008-z

[44] Campanella G, Hanna MG, Geneslaw L, et al. Clinical-grade computational pathology using weakly supervised deep learning on whole slide images. Nature Medicine, 25:1301-1309, 2019. https://doi.org/10.1038/s41591-019-0508-1

[45] Lu MY, Williamson DFK, Chen TY, Chen RJ, Barbieri M, Mahmood F. Data-efficient and weakly supervised computational pathology on whole-slide images. Nature Biomedical Engineering, 5(6):555-570, 2021. https://doi.org/10.1038/s41551-020-00682-w

[46] Ilse M, Tomczak JM, Welling M. Attention-based deep multiple instance learning. Proceedings of the 35th International Conference on Machine Learning (ICML), PMLR 80:2127-2136, 2018.

[47] Chen RJ, Lu MY, Wang J, et al. Pathomic fusion: an integrated framework for fusing histopathology and genomic features for cancer diagnosis and prognosis. IEEE Transactions on Medical Imaging, 41(4):757-770, 2022. https://doi.org/10.1109/TMI.2020.3021387

[48] Hollon TC, Pandian B, Adapa AR, et al. Near real-time intraoperative brain tumor diagnosis using stimulated Raman histology and deep neural networks. Nature Medicine, 26(1):52-58, 2020. https://doi.org/10.1038/s41591-019-0715-9

[49] McMahan HB, Moore E, Ramage D, Hampson S, Aguera y Arcas B. Communication-efficient learning of deep networks from decentralized data. Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR 54:1273-1282, 2017.

[50] Sheller MJ, Edwards B, Reina GA, et al. Federated learning in medicine: facilitating multi-institutional collaborations without sharing patient data. Scientific Reports, 10:12598, 2020. https://doi.org/10.1038/s41598-020-69250-1

[51] Rieke N, Hancox J, Li W, et al. The future of digital health with federated learning. npj Digital Medicine, 3:119, 2020. https://doi.org/10.1038/s41746-020-00323-1

[52] Kaissis GA, Makowski MR, Ruckert D, Braren RF. Secure, privacy-preserving and federated machine learning in medical imaging. Nature Machine Intelligence, 2:305-311, 2020. https://doi.org/10.1038/s42256-020-0186-1

[53] Pati S, Baid U, Edwards B, et al. Federated learning enables big data for rare cancer boundary detection. Nature Communications, 13:7346, 2022. https://doi.org/10.1038/s41467-022-33407-5

[54] Zech JR, Badgeley MA, Liu M, Costa AB, Titano JJ, Oermann EK. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: a cross-sectional study. PLOS Medicine, 15(11):e1002683, 2018. https://doi.org/10.1371/journal.pmed.1002683

[55] Finlayson SG, Subbaswamy A, Singh K, et al. The clinician and dataset shift in artificial intelligence. New England Journal of Medicine, 385(3):283-286, 2021. https://doi.org/10.1056/NEJMc2104626

[56] Amann J, Blasimme A, Vayena E, Frey D, Madai VI. Explainability for artificial intelligence in healthcare: a multidisciplinary perspective. BMC Medical Informatics and Decision Making, 20:310, 2020. https://doi.org/10.1186/s12911-020-01332-6

[57] Vasey B, Nagendran M, Campbell B, et al. Reporting guideline for the early-stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI. Nature Medicine, 28(5):924-933, 2022. https://doi.org/10.1038/s41591-022-01772-9

[58] World Health Organization. Ethics and Governance of Artificial Intelligence for Health: WHO Guidance. Geneva: WHO, 2021. https://www.who.int/publications/i/item/9789240029200

[59] European Union. Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act), Article 14 (Human oversight). Official Journal of the European Union, 2024. https://eur-lex.europa.eu/eli/reg/2024/1689/oj

[60] Zhao L, Liu S, Xin T, et al. AI agent in healthcare: applications, evaluations, and future directions. npj Artificial Intelligence, 2:31, 2026. https://doi.org/10.1038/s44387-026-00076-4

[61] Ronneberger O, Fischer P, Brox T. U-Net: convolutional networks for biomedical image segmentation. Medical Image Computing and Computer-Assisted Intervention (MICCAI), LNCS 9351:234-241, 2015. https://doi.org/10.1007/978-3-319-24574-4_28

[62] Cicek O, Abdulkadir A, Lienkamp SS, Brox T, Ronneberger O. 3D U-Net: learning dense volumetric segmentation from sparse annotation. Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2016. https://doi.org/10.1007/978-3-319-46723-8_49

[63] Hatamizadeh A, Nath V, Tang Y, Yang D, Roth HR, Xu D. Swin UNETR: Swin transformers for semantic segmentation of brain tumors in MRI images. MICCAI BrainLes Workshop, 2021. https://arxiv.org/abs/2201.01266

[64] Chen T, Kornblith S, Norouzi M, Hinton G. A simple framework for contrastive learning of visual representations (SimCLR). Proceedings of the 37th International Conference on Machine Learning (ICML), PMLR 119:1597-1607, 2020.

[65] He K, Fan H, Wu Y, Xie S, Girshick R. Momentum contrast for unsupervised visual representation learning (MoCo). IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020.

[66] Caron M, Touvron H, Misra I, Jegou H, Mairal J, Bojanowski P, Joulin A. Emerging properties in self-supervised vision transformers (DINO). IEEE/CVF International Conference on Computer Vision (ICCV), 2021.

[67] He K, Chen X, Xie S, Li Y, Dollar P, Girshick R. Masked autoencoders are scalable vision learners. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022.

[68] Tang Y, Yang D, Li W, et al. Self-supervised pre-training of Swin transformers for 3D medical image analysis. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022.

[69] Azizi S, Mustafa B, Ryan F, et al. Big self-supervised models advance medical image classification. IEEE/CVF International Conference on Computer Vision (ICCV), 2021.

[70] Kendall A, Gal Y. What uncertainties do we need in Bayesian deep learning for computer vision? Advances in Neural Information Processing Systems (NeurIPS) 30, 2017.

[71] Ovadia Y, Fertig E, Ren J, et al. Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift. Advances in Neural Information Processing Systems (NeurIPS) 32, 2019.

[72] Mehrtash A, Wells WM, Tempany CM, Abolmaesumi P, Kapur T. Confidence calibration and predictive uncertainty estimation for deep medical image segmentation. IEEE Transactions on Medical Imaging, 39(12):3868-3878, 2020. https://doi.org/10.1109/TMI.2020.3006437

[73] Jungo A, Reyes M. Assessing reliability and challenges of uncertainty estimations for medical image segmentation. Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2019. https://doi.org/10.1007/978-3-030-32245-8_6

[74] Wang G, Li W, Aertsen M, Deprest J, Ourselin S, Vercauteren T. Aleatoric uncertainty estimation with test-time augmentation for medical image segmentation with convolutional neural networks. Neurocomputing, 338:34-45, 2019. https://doi.org/10.1016/j.neucom.2019.01.103

[75] Vovk V, Gammerman A, Shafer G. Algorithmic Learning in a Random World. Springer, 2005.

[76] Raghu M, Blumer K, Corrado G, Kleinberg J, Obermeyer Z, Mullainathan S. The algorithmic automation problem: prediction, triage, and human effort. arXiv:1903.12220, 2019. https://arxiv.org/abs/1903.12220

[77] Wilder B, Horvitz E, Kamar E. Learning to complement humans. International Joint Conference on Artificial Intelligence (IJCAI), 2020.

[78] Bansal G, Wu T, Zhou J, et al. Does the whole exceed its parts? The effect of AI explanations on complementary team performance. ACM CHI Conference on Human Factors in Computing Systems, 2021. https://doi.org/10.1145/3411764.3445717

[79] Yao S, Zhao J, Yu D, Du N, Shafran I, Narasimhan K, Cao Y. ReAct: synergizing reasoning and acting in language models. International Conference on Learning Representations (ICLR), 2023.

[80] Wei J, Wang X, Schuurmans D, et al. Chain-of-thought prompting elicits reasoning in large language models. Advances in Neural Information Processing Systems (NeurIPS) 35, 2022.

[81] Singhal K, Azizi S, Tu T, et al. Large language models encode clinical knowledge. Nature, 620:172-180, 2023. https://doi.org/10.1038/s41586-023-06291-2

[82] Moor M, Banerjee O, Abad ZSH, et al. Foundation models for generalist medical artificial intelligence. Nature, 616:259-265, 2023. https://doi.org/10.1038/s41586-023-05881-4

[83] Tu T, Palepu A, Schaekermann M, et al. Towards conversational diagnostic AI. arXiv:2401.05654, 2024. https://arxiv.org/abs/2401.05654

[84] Chen RJ, Ding T, Lu MY, et al. Towards a general-purpose foundation model for computational pathology. Nature Medicine, 30:850-862, 2024. https://doi.org/10.1038/s41591-024-02857-3

[85] Vorontsov E, Bozkurt A, Casson A, et al. A foundation model for clinical-grade computational pathology and rare cancers detection. Nature Medicine, 30:2924-2935, 2024. https://doi.org/10.1038/s41591-024-03141-0

[86] Huang Z, Bianchi F, Yuksekgonul M, Montine TJ, Zou J. A visual-language foundation model for pathology image analysis using medical Twitter. Nature Medicine, 29:2307-2316, 2023. https://doi.org/10.1038/s41591-023-02504-3

[87] Li T, Sahu AK, Talwalkar A, Smith V. Federated learning: challenges, methods, and future directions. IEEE Signal Processing Magazine, 37(3):50-60, 2020. https://doi.org/10.1109/MSP.2020.2975749

[88] Li T, Sahu AK, Zaheer M, Sanjabi M, Talwalkar A, Smith V. Federated optimization in heterogeneous networks (FedProx). Proceedings of Machine Learning and Systems (MLSys), 2020.

[89] Karimireddy SP, Kale S, Mohri M, Reddi S, Stich S, Suresh AT. SCAFFOLD: stochastic controlled averaging for federated learning. Proceedings of the 37th International Conference on Machine Learning (ICML), PMLR 119:5132-5143, 2020.

[90] Abadi M, Chu A, Goodfellow I, McMahan HB, Mironov I, Talwar K, Zhang L. Deep learning with differential privacy. ACM SIGSAC Conference on Computer and Communications Security (CCS), 2016. https://doi.org/10.1145/2976749.2978318

[91] Dwork C, Roth A. The algorithmic foundations of differential privacy. Foundations and Trends in Theoretical Computer Science, 9(3-4):211-407, 2014. https://doi.org/10.1561/0400000042

[92] Karaman S, Frazzoli E. Sampling-based algorithms for optimal motion planning. International Journal of Robotics Research, 30(7):846-894, 2011. https://doi.org/10.1177/0278364911406761

[93] Essert C, Haegelen C, Lalys F, Abadie A, Jannin P. Automatic computation of electrode trajectories for deep brain stimulation: a hybrid symbolic and numerical approach. International Journal of Computer Assisted Radiology and Surgery, 7(4):517-532, 2012. https://doi.org/10.1007/s11548-011-0651-8

[94] Fedorov A, Beichel R, Kalpathy-Cramer J, et al. 3D Slicer as an image computing platform for the Quantitative Imaging Network. Magnetic Resonance Imaging, 30(9):1323-1341, 2012. https://doi.org/10.1016/j.mri.2012.05.001

[95] Liu X, Cruz Rivera S, Moher D, Calvert MJ, Denniston AK. Reporting guidelines for clinical trial reports for interventions involving artificial intelligence: the CONSORT-AI extension. Nature Medicine, 26:1364-1374, 2020. https://doi.org/10.1038/s41591-020-1034-x

[96] Collins GS, Moons KGM, Dhiman P, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. BMJ, 385:e078378, 2024. https://doi.org/10.1136/bmj-2023-078378

[97] Park SH, Han K. Methodologic guide for evaluating clinical performance and effect of artificial intelligence technology for medical diagnosis and prediction. Radiology, 286(3):800-809, 2018. https://doi.org/10.1148/radiol.2017171920

[98] Obermeyer Z, Powers B, Vogeli C, Mullainathan S. Dissecting racial bias in an algorithm used to manage the health of populations. Science, 366(6464):447-453, 2019. https://doi.org/10.1126/science.aax2342

[99] Sculley D, Holt G, Golovin D, et al. Hidden technical debt in machine learning systems. Advances in Neural Information Processing Systems (NeurIPS) 28, 2015.

[100] Sendak MP, Gao M, Brajer N, Balu S. Presenting machine learning model information to clinical end users with model facts labels. npj Digital Medicine, 3:41, 2020. https://doi.org/10.1038/s41746-020-0253-3

[101] Hevner AR, March ST, Park J, Ram S. Design science in information systems research. MIS Quarterly, 28(1):75-105, 2004. https://doi.org/10.2307/25148625

[102] Peffers K, Tuunanen T, Rothenberger MA, Chatterjee S. A design science research methodology for information systems research. Journal of Management Information Systems, 24(3):45-77, 2007. https://doi.org/10.2753/MIS0742-1222240302

[103] Creswell JW, Creswell JD. Research Design: Qualitative, Quantitative, and Mixed Methods Approaches. 5th edition, SAGE Publications, 2018.

[104] Saunders M, Lewis P, Thornhill A. Research Methods for Business Students. 8th edition, Pearson, 2019.

[105] Paszke A, Gross S, Massa F, et al. PyTorch: an imperative style, high-performance deep learning library. Advances in Neural Information Processing Systems (NeurIPS) 32, 2019.

[106] Cardoso MJ, Li W, Brown R, et al. MONAI: an open-source framework for deep learning in healthcare. arXiv:2211.02701, 2022. https://arxiv.org/abs/2211.02701

[107] Pedregosa F, Varoquaux G, Gramfort A, et al. Scikit-learn: machine learning in Python. Journal of Machine Learning Research, 12:2825-2830, 2011.

[108] Jodogne S. The Orthanc ecosystem for medical imaging. Journal of Digital Imaging, 31(3):341-352, 2018. https://doi.org/10.1007/s10278-018-0082-y

[109] Bender D, Sartipi K. HL7 FHIR: an agile and RESTful approach to healthcare information exchange. IEEE International Symposium on Computer-Based Medical Systems (CBMS), 2013. https://doi.org/10.1109/CBMS.2013.6627810

[110] Aryanto KYE, Oudkerk M, van Ooijen PMA. Free DICOM de-identification tools in clinical research: functioning and safety of patient privacy. European Radiology, 25(12):3685-3695, 2015. https://doi.org/10.1007/s00330-015-3794-0

[111] Bischoff-Grethe A, Ozyurt IB, Busa E, et al. A technique for the deidentification of structural brain MR images. Human Brain Mapping, 28(9):892-903, 2007. https://doi.org/10.1002/hbm.20312

[112] Tustison NJ, Avants BB, Cook PA, et al. N4ITK: improved N3 bias correction. IEEE Transactions on Medical Imaging, 29(6):1310-1320, 2010. https://doi.org/10.1109/TMI.2010.2046908

[113] Avants BB, Tustison NJ, Song G, Cook PA, Klein A, Gee JC. A reproducible evaluation of ANTs similarity metric performance in brain image registration. NeuroImage, 54(3):2033-2044, 2011. https://doi.org/10.1016/j.neuroimage.2010.09.025

[114] Isensee F, Schell M, Pflueger I, et al. Automated brain extraction of multisequence MRI using artificial neural networks. Human Brain Mapping, 40(17):4952-4964, 2019. https://doi.org/10.1002/hbm.24750

[115] Jenkinson M, Beckmann CF, Behrens TEJ, Woolrich MW, Smith SM. FSL. NeuroImage, 62(2):782-790, 2012. https://doi.org/10.1016/j.neuroimage.2011.09.015

[116] Tournier JD, Smith R, Raffelt D, et al. MRtrix3: a fast, flexible and open software framework for medical image processing and visualisation. NeuroImage, 202:116137, 2019. https://doi.org/10.1016/j.neuroimage.2019.116137

[117] Goode A, Gilbert B, Harkes J, Jukic D, Satyanarayanan M. OpenSlide: a vendor-neutral software foundation for digital pathology. Journal of Pathology Informatics, 4:27, 2013. https://doi.org/10.4103/2153-3539.119005

[118] Bankhead P, Loughrey MB, Fernandez JA, et al. QuPath: open source software for digital pathology image analysis. Scientific Reports, 7:16878, 2017. https://doi.org/10.1038/s41598-017-17204-5

[119] Macenko M, Niethammer M, Marron JS, et al. A method for normalizing histology slides for quantitative analysis. IEEE International Symposium on Biomedical Imaging (ISBI), 2009. https://doi.org/10.1109/ISBI.2009.5193250

[120] Vahadane A, Peng T, Sethi A, et al. Structure-preserving color normalization and sparse stain separation for histological images. IEEE Transactions on Medical Imaging, 35(8):1962-1971, 2016. https://doi.org/10.1109/TMI.2016.2529665

[121] Roth HR, Cheng Y, Wen Y, et al. NVIDIA FLARE: federated learning from simulation to real-world. arXiv:2210.13291, 2022. https://arxiv.org/abs/2210.13291

[122] Beutel DJ, Topal T, Mathur A, et al. Flower: a friendly federated learning research framework. arXiv:2007.14390, 2020. https://arxiv.org/abs/2007.14390

[123] Reina GA, Gruzdev A, Foley P, et al. OpenFL: an open-source framework for federated learning. arXiv:2105.06413, 2021. https://arxiv.org/abs/2105.06413

[124] Yousefpour A, Shilov I, Sablayrolles A, et al. Opacus: user-friendly differential privacy library in PyTorch. arXiv:2109.12298, 2021. https://arxiv.org/abs/2109.12298

[125] Sucan IA, Moll M, Kavraki LE. The Open Motion Planning Library. IEEE Robotics and Automation Magazine, 19(4):72-82, 2012. https://doi.org/10.1109/MRA.2012.2205651

[126] Taha AA, Hanbury A. Metrics for evaluating 3D medical image segmentation: analysis, selection, and tool. BMC Medical Imaging, 15:29, 2015. https://doi.org/10.1186/s12880-015-0068-x

[127] Riley RD, Ensor J, Snell KIE, et al. Calculating the sample size required for developing a clinical prediction model. BMJ, 368:m441, 2020. https://doi.org/10.1136/bmj.m441

[128] Obuchowski NA. Sample size calculations in studies of test accuracy. Statistical Methods in Medical Research, 7(4):371-392, 1998. https://doi.org/10.1177/096228029800700405

[129] DeLong ER, DeLong DM, Clarke-Pearson DL. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. Biometrics, 44(3):837-845, 1988. https://doi.org/10.2307/2531595

[130] Bland JM, Altman DG. Statistical methods for assessing agreement between two methods of clinical measurement. The Lancet, 327(8476):307-310, 1986. https://doi.org/10.1016/S0140-6736(86)90837-8

[131] Landis JR, Koch GG. The measurement of observer agreement for categorical data. Biometrics, 33(1):159-174, 1977. https://doi.org/10.2307/2529310

[132] Hart SG, Staveland LE. Development of NASA-TLX (Task Load Index): results of empirical and theoretical research. Advances in Psychology, 52:139-183, 1988. https://doi.org/10.1016/S0166-4115(08)62386-9

[133] Brooke J. SUS: a quick and dirty usability scale. In: Usability Evaluation in Industry, Taylor and Francis, pp. 189-194, 1996.

[134] Braun V, Clarke V. Using thematic analysis in psychology. Qualitative Research in Psychology, 3(2):77-101, 2006. https://doi.org/10.1191/1478088706qp063oa

[135] Naeini MP, Cooper GF, Hauskrecht M. Obtaining well calibrated probabilities using Bayesian binning. AAAI Conference on Artificial Intelligence, 2015.

[136] Brier GW. Verification of forecasts expressed in terms of probability. Monthly Weather Review, 78(1):1-3, 1950.

[137] Efron B, Tibshirani RJ. An Introduction to the Bootstrap. Chapman and Hall/CRC, 1993.

[138] Holm S. A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics, 6(2):65-70, 1979.

[139] McNemar Q. Note on the sampling error of the difference between correlated proportions or percentages. Psychometrika, 12(2):153-157, 1947. https://doi.org/10.1007/BF02295996

[140] Krippendorff K. Content Analysis: An Introduction to Its Methodology. 2nd edition, SAGE Publications, 2004.

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ROLE IN THIS STUDY

Principal Investigator. Responsible for study design, development of the AI framework, data management of the de-identified registry, the clinician reader study, questionnaires and interviews, analysis and reporting, under the supervision of Dr. WMKS Ilmini, Dr. WGCW Kumara and Dr. AM Deepal K Attanayake.

QUALIFICATIONS

MSc Computer Science (Information Technology Management)
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CAREER AND EMPLOYMENT

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RESEARCH AND PUBLICATIONS

1. Sameera V. An efficient deep learning framework for automated endometrial cancer detection using a ResNet50-ViT hybrid architecture. International Research Conference (IRC), General Sir John Kotelawala Defence University.
2. Sameera V. Voice frequency-based emotion classification: a comparative study of machine learning and deep learning methods. International Conference on Innovations in Information Technology (iCIIT), Informatics Institute of Technology, 2023.

RESEARCH EXPERIENCE RELEVANT TO THIS STUDY

Deep learning for medical images (histopathology image classification with convolutional and transformer models); development of secure, audited software systems and databases; supervision of more than 20 undergraduate research projects.

DECLARATION

I confirm that the information in this curriculum vitae is correct, and I agree to take part in the study "Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings" in the role stated above.

....

Signature (KH Viraj Sameera) Date

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College of Surgeons of Sri Lanka
The Neurosurgeons Association of Sri Lanka
A.O. Spine
A.O. Neurosurgery

HOBBIES

Bridge · Billiards · Chess
Carom · Swimming · Travelling

GENERAL INFORMATION

Date of Birth 19 September, 1963
Gender Male
Marital Status Married
Nationality Sri Lankan
Present Position Consultant Neurosurgeon
Present Station National Hospital of Sri Lanka, Colombo
SLMC Registration No. 10718

PROFESSIONAL QUALIFICATIONS

1992 M.B.B.S., Faculty of Medicine, University of Colombo, Sri Lanka

2003 M.S. (Surgery)

A postgraduate degree in Surgery conducted by the Postgraduate Institute of Medicine, University of Colombo, in affiliation with the Royal College of Surgeons of England — considered equivalent to FRCS / MRCS (UK).

PUBLICATIONS AND PRESENTATIONS

The Annual Session of the College of Sri Lanka, 2004

- Co-author of case report (Asian Journal of Neurosurgery): Removal of a Low Clival Chordoma in a Teenager by Dorsolateral Suboccipital Transcondylar Approach.
- "Hand Rubbing with Bactericidal Agent vs Hand Scrubbing in Surgical Practice" — *M.T.P.R. Perera, H.H.M.K. Herath, J. Masakorala, A.M.D.K. Attanayake, J. Lokuketagoda, S.S.L. Perera, J. Perera — National Hospital of Sri Lanka, Faculty of Medicine, Colombo*
- "Pituitary Adenomas — Sri Lankan Scenario" — *M.P. Gunawardane, A.G.D.N. Gunawansa, J. Lokuketagoda, A.M.D.K. Attanayake, Sunil Perera*

The Annual Session of the College of Surgeons of Sri Lanka, 2005

- Post-surgical outcome in children with posterior cranial fossa tumours. — *N.P. Senanayake, T. Weerasooriya, J. Lokuketagoda, A.M.D.K. Attanayake, S.S.L. Perera*
- A retrospective audit of patients who presented to a neurosurgical unit with an extradural haematoma. — *T.C. Weerasuriya, C.A.H. Liyanage, A.W.P. Manjula, P. Athiththan, M.A.A.N. Marasinghe, D. Attanayake, J. Lokuketagoda, Sunil Perera*

25th Anniversary Academic Sessions, Postgraduate Institute of Medicine, 2005

- A prospective study on objective improvement of clinical features of patients following Carpal Tunnel Decompression. — *T.C. Weerasuriya, C.A.H. Liyanage, A.W.P. Manjula, D. Attanayake, M.A.A.N. Marasinghe, S. Gunasekara, J. Lokuketagoda, Sunil Perera*

The Annual Session of the College of Surgeons of Sri Lanka, 2006

- Intraoperative neuronavigation in cranial surgery: clinical experience in 55 cases. — *Ahiththan, A.M.D.K. Attanayake, W.D.L. Siriwardana, S.S.L. Perera*
- Case load in a single neurosurgical unit at NHSL — An Audit.

WORKSHOPS ATTENDED

2003 Primary Trauma Care — workshop conducted by the World Federation of Societies of Anaesthesiologists, Colombo, Sri Lanka.

2004	RAIN 2004 — International Update on Neuro-intervention, Sri Ramachandra Medical College and Radiology Institution, Chennai, India.
2006	Neurosurgery Workshop: Minimally Invasive Neurosurgery — Keyhole Concept, Singapore.
2010	AO Spine Advanced Trauma Management Course, Davos, Switzerland.
2012	AO Spine Advanced Degenerative Diseases of Spine Course, Davos, Switzerland.

OTHER ACTIVITIES

Participation in educational programmes on neurosurgery on national television:

- "Nuga Sewana" programme, National Television (Rupavahini)
- "Ayubowewa" programme, National Television (Rupavahini)
- "Pahan Dora" programme, Independent Television Network (ITN)
- Resource personnel for newspaper articles on neurosurgery in several newspapers.
- Resource personnel for the Primary Trauma Care workshop held regularly at the Accident & Emergency Department, National Hospital of Sri Lanka — jointly conducted by the College of Surgeons of Sri Lanka and the College of Anaesthesiologists of Sri Lanka.
- Resource personnel for the Advanced Trauma Resuscitation and Initial Management workshop for surgical registrars, held regularly at the College of Surgeons of Sri Lanka — jointly conducted with the College of Anaesthesiologists of Sri Lanka.
- Teaching medical students, Faculty of Medicine, University of Ruhuna.
- Visiting lecturer, Faculty of Medicine, University of Colombo.
- Teaching nursing students, Nurses' Training School, Galle.
- Formerly engaged in teaching nursing students, Nurses' Training School, Colombo.

VOLUNTARY SERVICE

- Participation in voluntary health camps.
- Involved in providing health facilities for tsunami-affected people and war victims in Sri Lanka.

REFEREES

Dr. Ravi Dayasena
MBBS, FRCS Edin (ENT)
Consultant ENT Surgeon (Retired)
National Hospital of Sri Lanka
ravidayasena@hotmail.com
+94 77 747 5338

Dr. Shirani Hapuarachchi
MBBS, MD
Senior Consultant Anaesthetist, Neurosurgical Dept.
National Hospital of Sri Lanka
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Tel (Res) +94 11 267 8781
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DR. WMKS ILMINI

BSC (HONS), PHD
PRINCIPAL SUPERVISOR

Department of Computer Science
Faculty of Applied Sciences
University of Sri Jayewardenepura

Telephone (Mobile)
+94 77 527 8643

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wmksilmini@sjp.ac.lk

CURRENT POSITION

Senior Lecturer (Grade II)
Department of Computer Science, Faculty of Applied Sciences, University of Sri Jayewardenepura

ROLE IN THIS STUDY

Principal Supervisor. Overall academic supervision of the PhD, the AI methodology (uncertainty, calibration and escalation) and the evaluation design.

QUALIFICATIONS

PhD - University of Sri Jayewardenepura - 2017-2023. Thesis: Personality Traits Assessment using Facial Features with Deep Learning Techniques
BSc Honours (Special) in Computer Science (GPA 3.7/4.5) - University of Sri Jayewardenepura - 2008-2014
Gold Medal - Best Postgraduate Student, Physical Science Stream, 50th General Convocation, University of Sri Jayewardenepura - 2024

CAREER AND EMPLOYMENT

Senior Lecturer Grade II, Department of Computer Science, Faculty of Applied Sciences - University of Sri Jayewardenepura - 2025 to present (on contract 2024-2025)
Visiting Lecturer, Advanced Diploma in Data Science - NIBM, Sri Lanka - 2025-2026
Senior Lecturer Grade II (2023-2024) and Head of Department (2024), Department of Computer Science, Faculty of Computing - General Sir John Kotelawala Defence University
Lecturer, Department of Computer Science, Faculty of Computing - General Sir John Kotelawala Defence University - 2017-2023
Temporary Lecturer / Instructor, Department of Computer Science, Faculty of Applied Sciences - University of Sri Jayewardenepura - 2014-2016
Software Developer (Intern) - Solution Infinity (Private) Limited - 2013-2014

SELECTED PUBLICATIONS (UP TO 10)

Athukorala VS, Ilmini WMKS. Explainable AI for critical care: a systematic review of interpretable models for sepsis and ICU mortality prediction. BMC Medical Informatics and Decision Making 2026;26(1):64.
Ilmini W, Fernando T. Detection and explanation of apparent personality using deep learning: a short review of current approaches and future directions. Computing 2024;106(1):275-294.
Ilmini W, Fernando T. Explaining the outputs of CNN-RNN based apparent personality detection models using class activation maps. International Journal of Advanced Computer Science and Applications 2023;14(2).
Ilmini W, Fernando T. Performance analysis of state-of-the-art deep learning models in visual-based apparent personality detection. Vidyodaya Journal of Science 2022;25(2).
Aratchige R, Ilmini W. LLMs working in harmony: a survey on the technological aspects of building effective LLM-based multi-agent systems. Preprint, 2025.
Abeysekara AA, Uwanthika GAI, Ilmini WMKS, Waidyarathna GRNN. Ontology-based knowledge representation for recommending dietary guidelines to non-alcoholic fatty liver disease patients. SLAAI-ICAI 2025, pp. 1-6.
Serasinghe S, Uwanthika G, Ilmini W. Comparative analysis of diagnostic methods for differentiating Hodgkin and non-Hodgkin lymphoma. 4th Student Symposium, Faculty of Computing, KDU, 2024.
Yakandawala Y, Madushanka M, Ilmini W. Explainable AI for transparent phishing email detection. ICATC 2024, IEEE, pp. 1-6.
Nadeeshana T, Ilmini W. Simultaneous detection of COVID-19 and its pneumonia using multitask learning. 15th International Research Conference KDU, 2022.
Vinodi P, Pemarathne W, Ilmini W, Senanayake G. Breast cancer detection using convolutional neural networks. 12th International Research Conference KDU, 2019.

RESEARCH ETHICS TRAINING

[Dr. Ilmini to confirm any GCP / research-ethics training, with year - or "None"]

DECLARATION

I confirm that the information in this curriculum vitae is correct, and I agree to take part in the study "Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings" in the role stated above.

.....	.....
Signature (Dr. WMKS Ilmini)	Date

DR. WGCW KUMARA

BSC ENG (HONS), MENG, PHD
CO-SUPERVISOR

Department of Computer Engineering
Faculty of Engineering
University of Sri Jayewardenepura

Telephone (Mobile)
071 642 5358

Email
chinthakawk@sjp.ac.lk

CURRENT POSITION

Senior Lecturer (Grade I)
Department of Computer Engineering, Faculty of Engineering, University of Sri Jayewardenepura

ROLE IN THIS STUDY

Co-Supervisor. Supervision of the medical-image analysis, federated learning and system-engineering components.

QUALIFICATIONS

PhD in Computer Science and Information Engineering (GPA 3.83) - National Central University, Taiwan - 2018. Thesis: Reconstruction of three-dimensional objects using a network of RGB-D sensors

MEng in Information and Communication Technology (GPA 3.60) - Asian Institute of Technology, Thailand - 2009

BSc Eng (Hons) in Electrical and Information Engineering, Second Class Upper Division (GPA 3.39) - University of Ruhuna, Sri Lanka - 2006

CAREER AND EMPLOYMENT

Senior Lecturer (Grade I), Department of Computer Engineering, Faculty of Engineering - University of Sri Jayewardenepura - 03/2026 to present

Senior Lecturer (Grade II 2019-2025, Grade I 2025-2026), Department of Computer Science and Engineering - South Eastern University of Sri Lanka - 07/2019 to 03/2026

Head, Dept. of Electrical and Telecommunication Engineering (2025-2026); Director, Industrial Training and Career Guidance Centre (2023-2024); Head, Dept. of Mechanical Engineering (2023); Head, Dept. of Computer Science and Engineering (2019-2022) - South Eastern University of Sri Lanka

Lecturer, Department of Mechatronics Engineering - South Asian Institute of Technology and Medicine (SAITM) - 2010-2011 and 2011-2017 (study leave 2012-2017)

Systems Engineer - Networking (Service Provider) - KBSL Information Technologies Limited - 2011

DSL Network Operation Engineer, IP and Broadband Division - Sri Lanka Telecom PLC - 2006-2007

After Sales Engineer - ZTE Corporation, Sri Lanka - 2006

SELECTED PUBLICATIONS (UP TO 10)

Full publication list: Google Scholar https://scholar.google.com/citations?user=j0-9s8cAAAAJ ; ORCID https://orcid.org/0000-0002-4613-275X ; Web of Science AAU-8923-2021.
Research areas: image and depth (RGB-D) processing, 3D reconstruction, robotics and automation, telecommunication engineering.
[Dr. Kumara to add up to 10 selected publications if required]

RESEARCH ETHICS TRAINING

[Dr. Kumara to confirm any GCP / research-ethics training, with year - or "None"]

DECLARATION

I confirm that the information in this curriculum vitae is correct, and I agree to take part in the study "Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings" in the role stated above.

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Signature (Dr. WGCW Kumara)	Date